# Persistence of a certified scalar instability under dynamical interface matter

**Abstract**

We ask whether dynamical conserved interface matter removes a certified scalar instability of a de Sitter two-wall Einstein–scalar background. Under the stated numerical premises, its regular coupled scalar sector has exactly one negative radial mass-squared eigenvalue throughout the nonnegative background-preserving wall-Hessian family. The complete count excludes additional negative radial eigenvalues in that sector, with admissible modes fixed by the physical reconstruction. Centered canonical surface oscillators preserve that count through a monotone boundary response. For a selected completion at zero current, a convergent homogeneous nonlinear growing family has an explicit remainder protecting proper-time separation inequalities and a nonzero wall-scalar departure witness. This is a local constructed-family result in a very small sufficient neighborhood. Nonzero conserved currents admit an actual small-charge analytic branch and a fixed-charge homogeneous growing linearized solution. On each compact time interval, that solution belongs to a compatible classical canonical evolution retaining both interfaces, their phases and the physical width; its momentum form remains coercive for sufficiently small charge. The result supplies a spectrally exhaustive neutral reference problem linked to local neutral nonlinear departure and charged linear persistence in one action. The charged neighborhood is uncomputed, and no nonlinear charged conclusion or endpoint dynamics is asserted.

**Keywords:** Einstein–scalar systems; de Sitter compactification; dynamical interfaces; scalar instability; conserved currents; canonical kinetic form; validated bounds.

## 1 Introduction

Can adding physical matter to the boundaries of an unstable compactification remove its instability? A boundary potential can change the scalar spectrum, but a dynamical material interface also carries conserved currents, propagating fields and a moving embedding. Treating these effects as a freely adjustable constant wall stiffness can miss both the geometric response and additional degrees of freedom.

We examine this question around one certified de Sitter two-wall background. Its pure-potential scalar sector has a complete radial instability count. This gives a precise starting point: one unstable regular scalar radial eigenvalue, rather than an unspecified collection of negative modes. Centered canonical surface oscillators preserve the background and allow a complete negative count in the enlarged neutral problem. Conserved nonzero matter then forces the geometry or interfaces to respond. For a selected action, an actual analytic charged branch and a fixed-charge growing perturbation supply that response without declaring the charged state static.

At zero current, the selected unstable mode also generates an actual convergent nonlinear family. An explicit all-orders remainder protects the sign of its proper-clock separation correction, and a dynamical wall scalar supplies a physical departure witness.

This local result and the nonzero-current linear construction share the declared action but have different quantifiers.

The construction distinguishes three physical questions. The existence of a compatible background is a constraint and junction question. A negative mass or a growing linearized solution concerns instability. Positivity of the physical canonical momentum block concerns the kinetic form and is not a proof of positive restoring stiffness. The results connect these questions in one selected five-dimensional theory while preserving their different domains of validity.

The warped two-brane construction of Randall and Sundrum is the geometric antecedent [18]. Warped black-string models also enter proposals for black-hole seeds [24]; no accretion or horizon-formation scenario is assumed here. Bulk-scalar stabilization and curved-wall instability have substantial precedents [1, 2, 4, 5, 6]. Frolov and Kofman derive the de Sitter shift and a radion instability estimate [3]; Deffayet proves scalar brane-matter well-posedness in a different bulk setting [10]. Finite wall potentials give boundary contributions to scalar products [7]. Established Sturm/form theory supplies the counting machinery [11, 12]; neither instability nor well-posedness is claimed as a new general phenomenon. Regular scalar-turn variables and turning-profile instability arguments are already present in [4, 5]. Matter-supported brane equilibria [26] and recent matter-induced radion tachyonicity [27] are direct precedents; they rule out treating matter dependence itself as novel. Bhattacharyya and SenGupta [28] study bulk-profile and potential conditions for modulus stabilization; those effective-potential criteria are a different diagnostic from the full coupled radial count and growing solution used here. The equation-and-assumption comparisons appear in Appendix I.

Nonlinear radion dynamics itself is established. Martin et al. [29] simulate full five-dimensional two-brane Einstein–scalar evolution, including departure from unstable de Sitter configurations and both signs of separation change. Binétruy, Deffayet and Langlois [30] derive nonlinear homogeneous radion motion from moving-brane junctions, and Langlois and Sorbo [31] derive a nonperturbative homogeneous action in an AdS setting. The present local construction adds selected-action convergence and observable remainder control with dynamical wall oscillators; it does not supersede those nonlinear simulations. Analytic parameterizations and rigorous tail bounds also have established finite-dimensional and parabolic-PDE precedents [32, 33]. Their general strategy is credited, while the constrained Einstein/interface reconstruction is supplied here.

The additional guarantee is completeness and controlled continuation of the instability information. Establishing one negative eigenvalue leaves open whether further negative radial eigenvalues exist. Here the complete count excludes them throughout the declared wall-Hessian family, and the physical reconstruction identifies the admissible modes through turns and limiting wall conditions. The selected dynamical completion connects a growing mode to the coupled constraints and interface equations. At zero current, convergence and an observable remainder control a local nonlinear family; at small nonzero current, the continuation is a separate linear result. These linked guarantees provide a controlled reference problem for this model. Table 1 summarises the relationship to three close precedents; Appendix I gives the equation-level comparisons.

The count concerns radial eigenvalues in the declared scalar sector, not harmonic multiplicities or the dimension of a full Cauchy unstable manifold.

**Table 1. Closely related results and the present contribution. The settings and guarantees differ; no historical-priority claim is implied.**

| Problem and setting | Main result | Relationship to our contribution |
|---|---|---|
| Frolov–Kofman [3]: inflating Einstein–scalar branes | De Sitter shift; lowest-mass variational bound; finite and stiff walls. | Complete negative radial count over the declared wall-Hessian family on one selected background. |
| Konikowska–Olechowski–Schmidt [5]: static/inflating backgrounds, including Gauss–Bonnet interactions | Regular-turn analysis, spectral wall conditions; necessary-and-sufficient static stability criteria. | Curved physical-domain reconstruction and remaining-band exclusion yield the exhaustive Einstein–scalar count. |
| Martin et al. (BraneCode) [29]: nonlinear two-brane evolution | Numerical departure, restructuring and collisions; constraint-compatible initial data. | Convergent local family with wall oscillators and observable remainders; simulations cover larger changes. |
| Present work: selected background and dynamical material completion | Neutral count, oscillator extension, neutral nonlinear family and small-charge linear growth. | Linked reference problem under stated premises; uncomputed charge radius and very small sufficient nonlinear neighborhood. |

The material current is an irrotational stiff fluid, $\rho_{\mathrm{m}} = p_{\mathrm{m}} = c(\Phi)\nu^2$, following Brown's variational formulation [8]. Its timelike domain and zero-charge canonical continuation are distinguished in Section 2.1. Thin-shell matching follows Israel's junction framework [19, 23] and its geometric extensions [9, 21]. The action and its normalization are specified below; no asymptotically flat initial-data or ADM comparison theorem is imported.

Sections 3–4 establish the neutral spectral anchor and its oscillator extension. Section 5 proves a separate local nonlinear theorem at zero current. Section 6 constructs the charged branch and growing direction; Section 7 gives the constrained evolution and momentum form; Section 8 proves their compatibility in Theorem A. The article appendices contain the indispensable analytic estimates, sourced operators, domain arguments and numerical trust boundary; they form part of the article and its requested peer review. The charge neighborhoods are intersected only after the growing solution has been placed in the canonical domain.

## 2 One declared five-dimensional action

### *2.1 Bulk and material variables*

Use signature $(-++++)$, units $\ell = \kappa_5 = 1$, the doubled bulk cover and two reflected timelike interfaces. The action is

$$S = \int_{\widetilde{\mathcal{M}}} \sqrt{-g}\left[\frac{R+12}{2} - \frac{1}{2}(\nabla\Phi)^2 - V(\Phi)\right] d^5x + \sum_{i=a,b} \int_{\Sigma_i} \sqrt{-h_i}\left[-\tau_i(\Phi) - J_i^a D_a\theta_i - c_i(\Phi)\nu_i^2 - \frac{1}{2}(D\chi_i)^2 - W_i(\Phi,\chi_i)\right] d^4x, \tag{1}$$

where $\nu_i^2 = -h_{i,ab}J_i^a J_i^b$. The symbol $\nu_i$ denotes material number density and is distinct from an interface normal. Each current has its own phase $\theta_i$. The two scalars $\chi_i$, both current sectors and both embeddings remain physical variables. The interval formulation includes the Gibbons–Hawking boundary action [20] and the signed Lorentzian joint terms for nonsmooth boundaries [22]. Their coefficients follow the doubled bulk normalization; Appendix E tracks them through reduction and clock exchange.

The centered interaction is

$$W_i = \frac{\omega_i^2\chi_i^2}{2} + \frac{\beta_i}{2}\left[\chi_i - \eta_i(\Phi - \Phi_i^*)\right]^2, \quad \omega_i^2 > 0, \quad \beta_i \geq 0, \quad c_i > 0. \tag{2}$$

Here $\Phi_i^*$ is the exact neutral background trace. Thus $W_i$ and both its first derivatives vanish at $\Phi = \Phi_i^*, \chi_i = 0$. The vacuum energy is already included in $\tau_i$; adding a second independent constant surface tension would change the action.

Varying the auxiliary current and phase gives

$$J_{i,a} = \frac{D_a\theta_i}{2c_i}, \qquad \mathcal{L}_{\theta_i} = -\frac{(D\theta_i)^2}{4c_i}, \qquad D_a J_i^a = 0. \tag{3}$$

The timelike current has $\rho_{\mathrm{m},i} = p_{\mathrm{m},i} = c_i\nu_i^2$, so its equation of state is stiff and its phase potential restricts it to irrotational flow. The phase kinetic coefficient is $1/(2c_i) > 0$. Auxiliary elimination preserves the conserved current; it is not elimination of the phase degree of freedom. The original timelike-fluid interpretation additionally requires a timelike, appropriately oriented current. Its velocity chart degenerates at zero density. The smooth phase continuation through zero charge is useful analytically, but generic phase states there are not automatically timelike-fluid states.

### *2.2 Junctions and energy exchange*

With inward normal orientations $\sigma_a = +1$, $\sigma_b = -1$, the field and interface equations include

$$K_{ab} = -\frac{1}{2}\left(S_{ab} - \frac{S}{3}h_{ab}\right), \qquad 2n^A\partial_A\Phi = \tau_{,\Phi} + W_{,\Phi} + c_{,\Phi}\nu^2, \qquad \Box_h\chi - W_{,\chi} = 0. \tag{4}$$

The Darmois–Israel jump law [19, 23] reduces to the first row of (4) with the stated reflected cover and inward normals. Battye and Carter distinguish this jump condition from the averaged normal-evolution equation [21, Sections II–III]. In the present reflection-symmetric problem, the latter is automatically satisfied: the averaged extrinsic

curvature and jump in normal bulk stress vanish. This does not remove the local junction rows at either wall. The scalar half-factor follows by varying both bulk copies against one wall action, as derived in Appendix J; a one-sided exterior variation has a different coefficient. On the material equations,

$$\begin{aligned} S_{ab} &= 2c\nu^2 u_a u_b + (c\nu^2 - \tau - W)h_{ab} \\ &\quad + D_a\chi D_b\chi - \frac{1}{2}h_{ab}(D\chi)^2, \\ D_a S^a{}_b &= -(\tau_{,\Phi} + W_{,\Phi} + c_{,\Phi}\nu^2)D_b\Phi. \end{aligned} \tag{5}$$

For a homogeneous interface of scale factor $R_i$ and proper time $\tau$,

$$\begin{aligned} &\rho_i = \tau_i + W_i + c_i\nu_i^2 + \dot{\chi}_i^2/2, \qquad p_i = -\tau_i - W_i + c_i\nu_i^2 + \dot{\chi}_i^2/2, \\ &\partial_\tau(R_i^3\nu_i) = 0, \qquad \dot{\rho}_i + 3H_i(\rho_i + p_i) = (\tau_{i,\Phi} + W_{i,\Phi} + c_{i,\Phi}\nu_i^2)\dot{\Phi}_i. \end{aligned} \tag{6}$$

Conserved material number does not require the surface energy to be separately conserved from the bulk scalar. Equation (6) is the exchange law used throughout the continuation.

### *2.3 Why charged matter must change the geometry*

The original neutral walls are umbilic. For any null tangent vector $l^a$, the added canonical material satisfies

$$S_{ab}l^a l^b = 2c\nu^2(u\cdot l)^2 + (l^a D_a\chi)^2. \tag{7}$$

This is strictly positive for nonzero timelike current. The Israel equation requires $K_{ab}l^a l^b = -S_{ab}l^a l^b/2$, whereas an umbilic wall has zero null contraction. Therefore the unchanged umbilic geometry cannot support the nonzero material current. The two interfaces cannot cancel this local condition across the collar. A charged continuation must change the physical embedding or geometry; the analytic branch below does so.

### *2.4 Result and proof strategy*

Adding conserved interface matter forces the background to respond. For the selected action, we construct a nearby time-dependent branch and a fixed-charge homogeneous growing scalar perturbation. On each fixed compact time interval, that history belongs to a compatible canonical evolution whose momentum form remains positive for sufficiently small charge.

The exact neutral background, phase and mass certificates are premises. The charge neighborhood is positive but uncomputed, and the charged conclusion concerns one growing direction. Sections 3–8 give the argument in order. Theorem A at the end of Section 8 states the full common-domain conclusion after its variables have been introduced.

The computer-assisted premises and the scope of prior internal review are stated in Section 9.

Figure 1 summarizes the charged proof dependencies used in Theorem A's simultaneous-applicability statement.

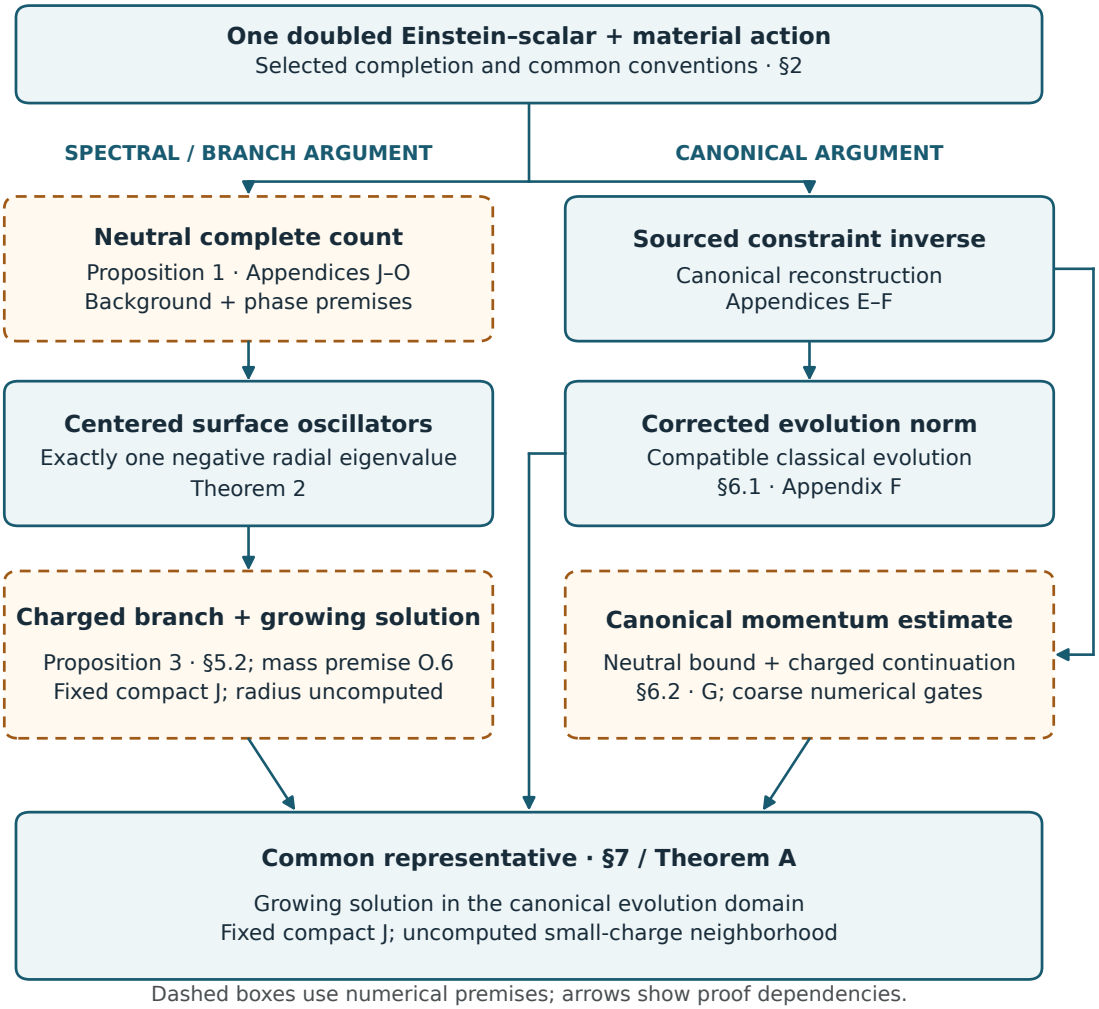



**Figure 1.** Proof dependencies in the common action. Dashed boxes use the numerical premises stated in Appendices H/O; the other analytical assumptions remain those of Theorem A. The corrected evolution norm and canonical momentum estimate are distinct arguments. Their connection to the growing solution is established by the representative and compatibility proof in Section 8. Arrows summarize logical dependencies, not a numerical trajectory or evolution from the neutral state. The charged statements hold on each fixed compact interval, with an uncomputed small-charge neighborhood.

## 3 The certified neutral scalar anchor

### *3.1 Background and regular mode domain*

The neutral notation distinguishes the slice expansion rate $h = \sqrt{k}$, the radial warp gradient $H = a'/a$, and the positive auxiliary $\mathfrak{h}$ used only in the scalar transformation of Appendix K.

The pure-potential neutral background has

$$ds_5^2 = a(s)^2 \gamma_{k,\alpha\beta} dx^\alpha dx^\beta + ds^2, \quad \mathrm{Ric}(\gamma_k) = 3k\gamma_k, \quad h = \sqrt{k} > 0. \tag{8}$$

Set $H = a'/a$, $T = \Phi_0'$, with $a(0) = 1$. Its equations are

$$a' = Ha, \qquad \Phi_0' = T, \qquad H' = -T^2/3 - k/a^2, \qquad T' = V_{,\Phi} - 4HT,$$
$$H^2 - 1 - T^2/12 + V/6 - k/a^2 = 0. \tag{9}$$

For the exact inputs in Appendix A, the validated background has positive warp, positive curvature, $T_a < 0 < T_b$, positive endpoint $T'$, and one simple interior scalar minimum.

Figure 2 illustrates the turn that the regular scalar system must cross.

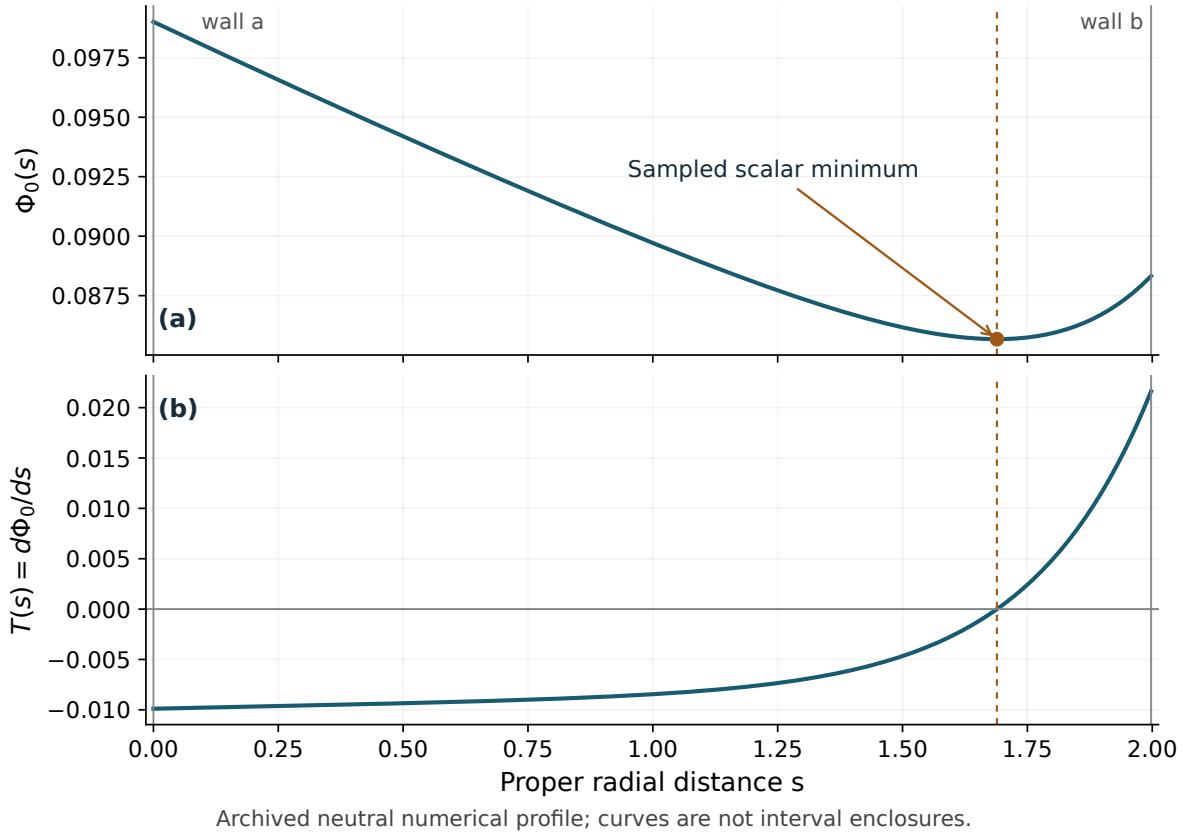



**Figure 2.** Archived numerical illustration of the neutral scalar profile (a) and its proper-distance derivative (b). The walls and minimum of the stored samples are marked; the vertical dashed line denotes that sampled location. The curves are not validated interval enclosures. The exact single-turn conclusion follows from the linked background certificate and the convexity argument in Appendix N.4, rather than from the plot. The vanishing scalar derivative motivates a reconstruction regular through the turn; it is not a geometric throat. No charged trajectory or new numerical integration is shown.

Let $\Box_{\gamma_k} Q = \mu Q$, restricting to the generic scalar harmonics whose traceless Hessian is not identically zero. With $\lambda = \mu + 4k$, the retained longitudinal profiles satisfy

$$\begin{aligned} \psi' &= -2H\psi - Tf/3, \qquad f' = g, \\ g' &= -4Hg + (V_{,\Phi\Phi} - \mu/a^2 + 2T^2)f + (12HT - 4V_{,\Phi})\psi, \\ 0 &= Tg - T'f + (2T^2 - 3\lambda/a^2)\psi. \end{aligned} \tag{10}$$

These equations are regular at $T = 0$. Define $L = g + 2T\psi$. The physical separated domain requires the equations globally in distributions, the physical interface traces and finite original-action kinetic norm

$$Z_A = 2\int_0^D a^2\left(|f|^2 + 6|\psi|^2\right) ds > 0. \tag{11}$$

Global distributional admissibility is important: a glued derivative jump creates a delta source in the scalar equation. Finite norm alone would not exclude it. The retained equations imply the classical regularity needed for the traces. The kinetic pullback assumes vanishing four-dimensional surface flux. No division by $H$ or $T$ is used in the gauge reconstruction.

### *3.2 Complete count and its numerical premise*

**Proposition 1 — certified neutral anchor.** With pure-potential walls and any nonnegative background-preserving Hessian pair $(\kappa_a, \kappa_b)$, including either or both Dirichlet limits, the declared regular generic coupled scalar sector has exactly one simple negative radial eigenvalue and no massless radial mode. Its unique negative eigenvalue $\alpha(\kappa_a, \kappa_b)$ obeys

$$\alpha(\kappa_a, \kappa_b) \le \alpha_{DD} < -4k. \tag{12}$$

The branch is continuous and increases with either stiffness.

**Proof of the restricted count.** The full action and generic Israel reduction are derived in Appendix J. The global distributional equations with finite original-action norm force classical profiles, and Appendix L's entire physical characteristic exhausts their roots, including zero Hessians and Dirichlet faces. In Appendix K the regular map $F = f + (T/\mathfrak{h})\psi$, with $\mathfrak{h} = v'/(2v) > 0$, gives an undivided scalar pencil. Its endpoint product has one negative direction; it cannot be replaced by ordinary positive weighted $L^2$. Its simple artificial zero at $\lambda = 0$ has strictly negative signature and no physical inverse.

Appendix M constructs the closed augmented form explicitly. Its fixed Robin zero mode has exactly the zeros of $T$, so ordinary Sturm theory gives one negative form direction. Positive endpoint squares preserve that index. The polarized kernel is precisely the artificial zero; its orthogonal complement is uniformly positive. Restricting there recovers exactly the nonzero physical roots, with their multiplicities, and leaves one negative $\lambda$. The closed constrained Dirichlet form domains give the same result on either or both faces. Their variational comparison gives stiffness monotonicity and the double-Dirichlet ceiling (12). These are standard spectral consequences after the model-specific physical reconstruction, not a new nodal principle.

To complete the count, the regular zero-Hessian launch at physical $\mu = 0$ is

$$(\psi_a, f_a, g_a) = (1, -12k/T_a', -2T_a), \qquad \theta_a = 3\pi/2. \tag{13}$$

On the same exact background, whole-cell interval flow, a certified continuous phase lift and the entire uncertain endpoint event give

$$\frac{\theta_b(0)}{\pi} \in \left[\frac{577838035}{536870912}, \frac{577858823}{536870912}\right] \subset (1, 3/2). \tag{14}$$

This excludes the first possible additional crossing and a massless root at zero Hessians. Any extra negative root at another wall pair would continue toward that corner with decreasing mass; simplicity and uniform spectral barriers prevent escape or merger. A massless root away from the corner immediately enters the same negative band along that path. Both would contradict (14). Thus one restricted root is the complete negative count.

The existence and input linkage of the exact background are numerical premises. Appendix A fixes the rational inputs. Appendix N proves the phase anchoring, strict spectral monotonicity and soft-corner injection; Appendix O gives the interval-Newton, whole-cell flow, phase-lift and uncertain-wall arguments. Thus the numerical premise has an explicit mathematical interpretation here. Its truth still assumes the identified execution and outward arithmetic, not merely a terminal determinant sign or agreement between approximate integrations.

## 4 A complete negative count with dynamical surface oscillators

### *4.1 The enlarged neutral problem*

At zero charge and $\chi_i = 0$, interaction (2) preserves the exact background. For a generic harmonic, the tracefree Israel consequence removes bending in longitudinal gauge after variation; it is not an independent frozen-embedding assumption. Let $u_i$ be the surface $\chi_i$ amplitude and $\kappa_i = \tau_i'' \geq 0$. The physical undivided boundary equations are

$$\begin{aligned} 2\sigma_i L_i &= (\kappa_i + \beta_i \eta_i^2) f_i - \beta_i \eta_i u_i, \\ \left(\omega_i^2 + \beta_i - \frac{\mu}{a_i^2}\right) u_i &= \beta_i \eta_i f_i. \end{aligned} \tag{15}$$

The oscillator sees $\mu/a_i^2$, not the gravitationally shifted parameter. Its poles at positive $\mu$ must remain in the undivided problem. The phase equation has no negative-mass solution at this anchor. The coupled negative-sector kinetic coefficient is

$$Z_{\text{ext}} = Z_A + \sum_i a_i^2 |u_i|^2 > 0. \tag{16}$$

The corresponding Green identity supplies reality with this same physical norm.

**Theorem 2 — neutral dynamic-wall extension.** Conditional on Proposition 1, arbitrary finite couplings $\omega_i^2 > 0$, $\beta_i \geq 0$, real $\eta_i$ and $\kappa_i \geq 0$ preserve exactly one real simple negative radial eigenvalue in the declared coupled scalar sector. It obeys $\mu_* \leq \alpha_{DD} < -4k$.

**Proof.** For $\mu < 0$, every oscillator denominator in (15) is positive. Eliminating the amplitude for this argument gives the effective Hessian

$$\Gamma_i(\mu) = \kappa_i + \beta_i\eta_i^2 - \frac{\beta_i^2\eta_i^2}{\omega_i^2 + \beta_i - \mu/a_i^2}. \tag{17}$$

It satisfies

$$\begin{gathered} 0 \le \Gamma_{i,0} := \kappa_i + \frac{\beta_i\eta_i^2\omega_i^2}{\omega_i^2 + \beta_i} \le \Gamma_i(\mu) \le \Gamma_{i,\infty} := \kappa_i + \beta_i\eta_i^2, \\ \Gamma_i'(\mu) = -\frac{\beta_i^2\eta_i^2}{a_i^2(\omega_i^2 + \beta_i - \mu/a_i^2)^2} \le 0. \end{gathered} \tag{18}$$

Every negative coupled mode is therefore a fixed point

$$\mu = F(\mu) := \alpha(\Gamma_a(\mu), \Gamma_b(\mu)). \tag{19}$$

The function $F$ is continuous and nonincreasing. Let $\mu_L = \alpha(\Gamma_{a,0}, \Gamma_{b,0})$ and $\mu_U = \alpha(\Gamma_{a,\infty}, \Gamma_{b,\infty})$. Both lie below $-4k$, and $F(\mu_L) \ge \mu_L$, $F(\mu_U) \le \mu_U$. A fixed point exists between them and is unique because $F(\mu) - \mu$ is strictly decreasing. At the root, the norm-slope identity gives

$$\frac{d}{d\mu}[\mu - F(\mu)] = \frac{Z_{\text{ext}}}{Z_A} > 0, \tag{20}$$

so it is simple. No negative pure oscillator/phase mode or negative oscillator pole was omitted. An additional negative bulk-coupled mode would contradict Proposition 1 at its own constant nonnegative Hessian pair. The bound by $\alpha_{DD}$ follows from (12). □

This theorem extends the negative count, not every neutral spectral conclusion. The enlarged phase completion can have massless wall-phase states. Proposition 1's no-massless result applies to its declared coupled sector and does not exclude zero modes of the enlarged matter theory. Likewise, at nonzero charge there is no static separated spectrum to which Theorem 2 can simply be assigned.

## 5 Local nonlinear departure of the neutral growing mode

The neutral oscillator instability also has a local nonlinear realization. Take $\omega_i^2 = \beta_i = \eta_i = 1$ and $\kappa_i = 20$ in action (1), with both conserved currents zero; these are the selected material parameters of Section 6.1. It complements the nonzero-current linear result below; it does not extend that result to charged nonlinear evolution. Earlier work already treats nonlinear homogeneous radion motion and two-brane evolution [29, 30, 31]. The additional claim here is a convergent solution family with an explicit remainder protecting a physical departure witness in this particular Einstein–scalar/interface model.

### *5.1 A convergent family with moving interfaces*

Let $r > h$ be the positive root of $\mu_* = -r(r + 3h)$ for the selected neutral oscillator mode. The strict inequality follows from $\mu_* < -4h^2$. Use the homogeneous metric and

moving embeddings of (22), replacing its redshift variable by the growing amplitude $\eta = \epsilon e^{rt}$. Normalize the first metric coefficient by $p_1(a) = 1$, with first graph coefficients zero; this differs from the scalar-minimum normalization used in Section 6.2. Both wall oscillators remain dynamical. On a fixed affine pullback of the moving interval, write

$$\mathcal{U}(\eta) = \sum_{n\geq 1} \mathcal{U}_n \eta^n, \qquad \mathcal{U} = (p, b, f, \xi_a, \xi_b, \chi_a, \chi_b). \tag{NL1}$$

**Theorem 3 — conditional local neutral nonlinear departure.** Assume the exact background and selected-mode premises, together with the analytic inverse, seven-source domination and observable-tail premises specified in Appendix Q.6. For the selected completion, (NL1) is a real analytic family solving the homogeneous bulk equations and both full moving-interface systems. Its interfaces are timelike and separated. A sufficient amplitude radius and protected real interval are

$$\rho = \frac{1}{10245848708368962467 7120}, \qquad |\eta| \leq \eta_* := \frac{3}{4}\rho. \tag{NL2}$$

The series and the fixed derivatives required by the equations converge on strict inner disks. The radius is a conservative proof bound in the stated normalization, not a measured physical threshold.

**Convergence argument.** At order $n \geq 2$, the temporal rate is $nr$ and the radial mass parameter is $-nr(nr + 3h)$. The regular constrained scalar problem has a uniform inverse in weighted analytic norms; its algebraic constraint denominator is bounded away from zero by a constant times $n^2$. Seven positive source envelopes retain the bulk constraints, graph transport, scalar junctions and oscillator equations. Their derivative and harmonic payments close a scalar analytic majorant. The resonant first mode is bounded separately, rather than passed through that inverse. The resulting series converges with the needed derivatives. An exact moving-wall Codazzi/material identity then recovers the remaining Israel row at each positive harmonic, so the summed fields solve the complete equations. Appendix Q gives these steps and the additional numerical premises. This is a model-specific use of established analytic parameterization and validated-tail ideas [32, 33], not a new general unstable-manifold theorem.

### *5.2 Proper-clock behavior and a scalar witness*

Let $D(\eta)$ be proper separation on the chosen homogeneous slicing, $x = D - D_0$, and let $\tau_a$ be the actual lower-wall proper time. The coefficient and remainder enclosures give, for $0 < |\eta| \leq \eta_*$,

$$\begin{aligned} 25|\eta| \leq |x| \leq 79|\eta|, \qquad \operatorname{sgn} x = -\operatorname{sgn}\eta, \\ -60\eta^2 \leq \frac{dx}{d\tau_a} - rx \leq -11\eta^2. \end{aligned} \tag{NL3}$$

Thus positive amplitude shortens the interval with a nonlinear correction reinforcing the linear law. Negative amplitude opens it, with $0 < dx/d\tau_a < rx$: opening is slowed, not

reversed. Figure 3 displays the protected sign and its remainder margin. The separation remains tied to the specified slicing.

The lower-wall oscillator supplies a separate physical scalar measurement:

$$\operatorname{sgn}\chi_a = \operatorname{sgn}\eta, \qquad |\chi_a| \geq \frac{|\eta|}{40000}, \qquad \operatorname{sgn}\frac{d\chi_a}{d\tau_a} = \operatorname{sgn}\eta. \tag{NL4}$$

Its identically zero background cannot become nonzero through a coordinate change. For any $0 < \epsilon < \eta_*$, restricting the exact family to $\eta(t) = \epsilon e^{rt}$ gives compatible initial data tending to the neutral background, yet reaches the fixed scalar threshold $\eta_*/40000$ at $t = r^{-1}\log(\eta_*/\epsilon)$. The positive bounded lapse makes each elapsed proper time finite. This is a departure witness along a constructed analytic family, not a Cauchy theorem for a neighborhood of arbitrary data. Neither its very small sufficient neighborhood nor its two amplitude signs determine saturation, collision or any other nonlinear endpoint.

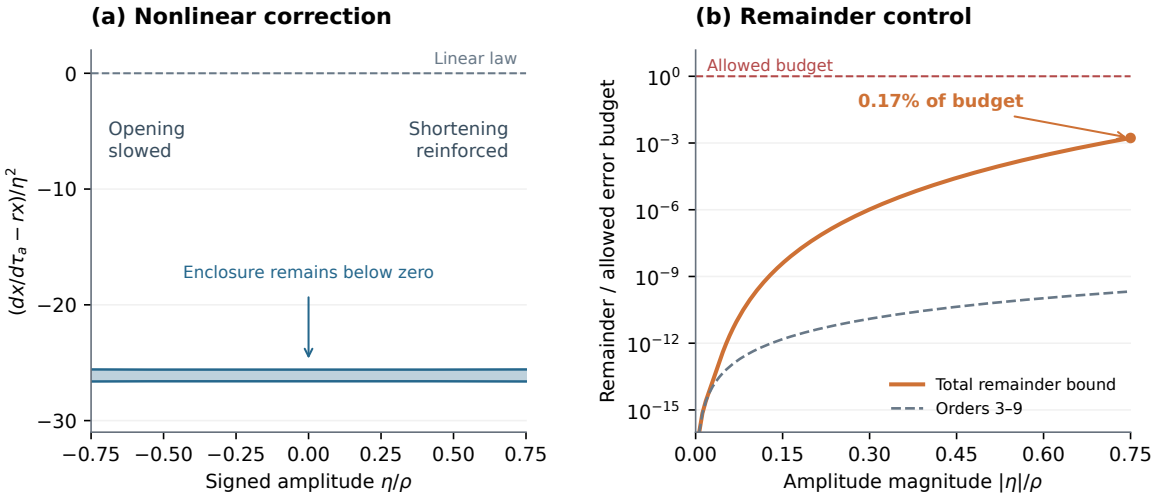



**Figure 3.** Local bounds for the selected neutral, zero-current growing family. (a) Enclosure of $(dx/d\tau_a - rx)/\eta^2$ from the quadratic coefficient and all higher orders; the value at zero denotes the quadratic limit. Its negative sign reinforces shortening and slows opening within the protected neighborhood. (b) The higher-order remainder bound divided by the allowed budget $10\eta^2$, with the acceptance limit at one. At $|\eta|/\rho = 3/4$, the bound uses approximately 0.17% of that budget. The curves render saved certificate bounds, not a physical trajectory. The endpoint is sufficient, not maximal; no nonzero-current nonlinear conclusion is inferred.

## 6 Actual charged continuation and a surviving growing direction

### *6.1 Selected action and analytic branch*

For a concrete continuation, fix the original wall Hessians at 20 and choose

$$\omega_i^2 = \beta_i = \eta_i = 1, \qquad c_i(\Phi) = 1 - \frac{\Phi - \Phi_i^*}{1000}. \tag{21}$$

The coefficient $c_i$ stays positive in a sufficiently small neighborhood. The charges are independent conserved comoving densities in a local spatially flat de Sitter patch. They are not finite total charges on an unspecified compact completion of $\mathbb{R}^3$.

Fix finite density labels $\mathcal{N}_a, \mathcal{N}_b$, introduce a small constant $\delta$, and set $z = \delta^2 e^{-6ht}$. The analytic branch has the form

$$\begin{aligned} ds_5^2 &= -a(s)^2 e^{2p(z,s)} dt^2 + a(s)^2 e^{2ht+2p(z,s)} d\mathbf{x}^2 + e^{2b(z,s)} ds^2, \\ \Phi &= \Phi_0(s) + f(z,s), \qquad S_i(z) = s_i + \sum_{n\geq 1} \xi_{i,n} z^n, \\ \chi_i(z) &= \sum_{n\geq 1} u_{i,n} z^n, \qquad p, b, f = O(z). \end{aligned} \tag{22}$$

The interfaces move, remain disjoint and timelike, and satisfy the full reflected junction equations. The exact current integrals are $R_i^3 \nu_i = \delta \mathcal{N}_i$. The future-directed timelike-fluid branch restricts the signs of the nonzero streams; the canonical phase continuation supplies the neutral endpoint.

**Proposition 3 — conditional charged branch.** Under the exact anchor and selected-mass certificate in Appendix O.6, (22) converges on a common radial collar for sufficiently small $z$. It solves the nonlinear equations in the declared analytic pure-redshift sector. The scalar has exactly one simple radial turn on the selected slices. The positive convergence radius is uncomputed, and uniqueness is within that particular analytic sector and gauge after fixing the two additive phase constants.

**Argument.** The selected-mass certificate excludes a kernel at every temporal rate in the charged recursion. The bulk constraints and moving-interface equations then determine compatible coefficients in a fixed order. Uniform inverse estimates control the radial derivatives, and the nonlinear source budget closes a convergent analytic majorant. The summed fields therefore satisfy the exact equations, rather than only a finite formal expansion. Their closeness to the neutral background preserves its simple scalar turn.

Appendix C gives the coefficient equations, uniform estimates and convergence proof. Appendix C.5 collects the analytic norm and majorant used in Proposition 3. The positive convergence radius remains uncomputed.

### *6.2 Fixed-charge linearized growth*

The neutral mass $\mu_*$ of Theorem 2 defines $\gamma_g > 0$ by

$$\mu_* = -\gamma_g(\gamma_g + 3h), \qquad 1 < \gamma_g/h < 11/10. \tag{24}$$

The selected-mass certificate and its same-root linkage are given in Appendix O.6. The upper ratio uses the selected-action comparison $\mu_* \geq \mu_{20,20} > -33/25000$ and $k > 3/10000$. Thus $(\gamma_g/h)(\gamma_g/h + 3) < 22/5$, which implies (24). It is not a bound for the entire arbitrary-coupling family of Theorem 2. About (22), an actual homogeneous linearized solution exists at fixed $\delta\mathcal{N}_a, \delta\mathcal{N}_b$. Its scalar-minimum readout is

$$\delta\Phi_{\min}(t) = \epsilon e^{\gamma_g t}[1 + O(z(t))]. \tag{25}$$

The pullback includes both scalar and embedding variations. Its shifted temporal rates are $r_n = \gamma_g - 6nh$, with radial parameters $-r_n(r_n + 3h)$; these are not the background

ladder in Proposition 3. Their nonresonance and an offset analytic majorant give convergence of the full linearized bulk/interface solution. Both charges remain fixed, while the wall phases and densities respond consistently.

The same selected-mass comparison excludes every shifted resonance after the leading mode. An offset analytic majorant then sums the perturbation while reconstructing both conserved phases. Appendix D gives the inequalities and summation estimate.

Equation (25) supplies a growing linearized direction on an actual time-dependent charged background. It does not count the full charged perturbation spectrum or prove nonlinear instability. At fixed perturbation amplitude, eventual growth leaves a linear approximation's smallness regime. The exponent and minimum readout use the declared time slicing. Section 8 places this particular smooth solution in the constrained evolution framework.

## 7 Compatible linear dynamics and kinetic control

### *7.1 Classical evolution in the admitted mode class*

The reduced physical scalar action retains both bulk scalar degrees, both independent wall $\chi/\theta$ pairs and, in the actual homogeneous untwisted sector, the width pair $(q, P_q)$. The full sourced reconstruction recovers the original linearized metric, scalar, Israel, current, normal-embedding and width equations. A homogeneous Green identity alone would not establish this converse.

The existence proof uses an auxiliary positive norm equivalent to the energy-space norm. A finite-dimensional correction controls the Hamiltonian's bounded-frequency defect while retaining the wall variables and physical homogeneous width. This construction establishes evolution; the sign of the actual canonical momentum form is checked separately in Section 7.2. Appendix F.3.1 gives the corrected estimate and its domain justification.

For each fixed nonzero Fourier mode, and separately for that physical homogeneous sector, the variational weak construction gives a unique energy-class history on a fixed finite slab. A higher-regularity upgrade uses the time-dependent original conormal $\mathcal{B}_t$. Initial data must satisfy

$$\mathcal{B}_s Z_s = 0, \qquad \dot{\mathcal{B}}_s Z_s + \mathcal{B}_s \mathcal{G}(s) Z_s = 0. \tag{26}$$

The dot term contains the actual moving coefficients and sources; a frozen-domain-square condition is insufficient. Two finite-rank domain lifts, with their time-dependent generator terms retained, yield the compatible classical history

$$Z \in C(J;\mathcal{E}_2) \cap C^1(J;\mathcal{E}_1) \cap C^2(J;\mathcal{E}_0), \qquad \sup_J \|Z\|_{\mathcal{E}_2} \le C_{J,\xi} \|Z_s\|_{\mathcal{E}_2}. \tag{27}$$

Here $\mathcal{E}_m$ uses bulk fields in $H^{m+1}$, true canonical momenta in $H^m$, and all finite wall/width slots. Stable-generator theory supplies the abstract regularity mechanism [13]; the action-specific work is the conormal, source orders and physical reconstruction. The bound is for fixed mode $\xi$, not an original-gauge all-Fourier infrared estimate.

Appendix F.3.4 gives a sufficient local smallness criterion for a linear perturbation to preserve one simple radial turn. Its constants are not numerically supplied.

### *7.2 The canonical momentum block*

The canonical momentum form is computed from the complete reduced action, with configuration fields and the prescribed clock held fixed. Both wall momentum pairs and the physical homogeneous width momentum are retained. Appendix G.1 defines the exact form (C3), including its pressure-boundary and clock terms.

The form separates into a positive square part $\mathscr{S}$ and an adverse remainder $\mathscr{R}$. Solving the constraints gives weighted radial identities that bound this remainder for arbitrary radial momentum inputs, without differentiating those inputs or requesting their endpoint traces. The certified neutral background gates imply the decisive estimate $\mathscr{K}_0 \geq \mathscr{S}_0/2$. Appendix G.2 gives the estimate, its constants and the saved-record premises.

Kernel exclusion and compact-response estimates make $\mathscr{S}_0$ a coercive canonical norm. On the charged branch, bounded-form continuity preserves a positive lower bound on a compact frequency band and the separate physical mean; a uniform large-frequency estimate supplies the complement. The frequency cutoff is fixed before the charge neighborhood is shrunk. Appendix G.3 proves this continuation on each fixed compact time interval. The charged radius and absolute norm-equivalence constants are uncomputed; the neutral relative one-half bound is not asserted unchanged throughout that neighborhood.

The conclusion concerns the actual canonical momentum form, independently of the auxiliary evolution norm. It is compatible with (25): positive kinetic control and a surviving growing direction concern different parts of the dynamics. A pure-momentum form direction need not satisfy the compatibility conditions for classical initial data, and no positivity of restoring energy is asserted.

## 8 The growing solution in the canonical evolution domain

### *8.1 One action and clock dictionary*

The dimensional reduction is $ds_5^2 = \mathfrak{q}_{ab}dx^a dx^b + e^{2\sigma}(dY^2 + dZ^2)$, $g_{ab} = e^{4\sigma}\mathfrak{q}_{ab}$. It is invertible for positive transverse radius. The scalar normal law for $\sigma$ is precisely the internal-fiber Israel equation. Its Gibbons–Hawking normal term [20] cancels the bulk divergence from reduction; the Lorentzian joint measure and boost transform together [22]. Thus stationary reduced histories reconstruct the declared five-dimensional scalar sector.

The branch was constructed in redshift time. Comparing its growth with the canonical momentum estimate requires a representative in the same curvature-clock chart, with the physical width retained. Appendix E.3 derives the canonical one-form (C4) and carries the clock and boundary generators in the chosen orthogonal-cut polarization.

Table 2 summarises the relation between the original fields and the common canonical representation.

**Table 2. Field, clock and boundary correspondence used in the common canonical representation.**

| Original description | Common canonical representative |
|---|---|
| Moving interval and embeddings | Straightened $\rho \in [0,1]$, total pulled fields and metric |
| Two transverse radii equal in the scalar sector | $X = (\sigma, \Phi)$, regular Einstein-frame metric |
| Independent $\chi_i, \theta_i$ and current laws | Independent finite wall canonical pairs; fixed $\delta p_{\theta_i} = 0$ only for the growing solution |
| Homogeneous scalar geometry | Physical $q, P_q$, fixed tangential normalization, no twist |
| Redshift construction clock | Prescribed fixed-$\rho$ curvature profile; explicit normal-cut transformation |
| Original scalar/fiber Israel laws | Time-dependent conormal $\mathcal{B}_t$ and two compatibility conditions |

### *8.2 Explicit homogeneous representative*

The homogeneous representative is obtained by solving the adjoint of the actual coupled metric constraints with their pressure Robin/Dirichlet boundary conditions. The neutral inverse is bounded, and small-charge continuation keeps it invertible on the fixed slab. Thus smooth growing inputs give a smooth gauge transformation without division by spatial frequency or the vanishing scalar gradient. Appendix E.4 displays the system (C5) and its reconstruction.

The width variation is an output of that solve, rather than an extra condition setting it to zero. All fields, densities and wall variables are pulled together, and their canonical momenta are computed from the original action. The full variational reconstruction therefore recovers both width equations and the same fixed-charge growing history in the canonical chart.

### *8.3 Compatibility and simultaneous conclusion*

The analytic growing history and its smooth representative have the regularity required for classical data. Their exact boundary laws and the time derivative of those laws give both conditions (26), including the moving-coefficient term. Uniqueness then identifies this history with the compatible classical propagator. Appendix F.4 gives the differentiated identity (C6); it differentiates the already smooth solution, not an arbitrary energy-class trace.

The minimum-value readout also survives this representation. A radial coordinate change contributes the total scalar radial derivative, which is zero at the branch minimum. Let $\zeta$ denote the proper normal displacement of the time cut and $n_{\mathrm{slice}}$ its unit normal. A change of slice contributes $\zeta n^A_{\mathrm{slice}} \nabla_A \Phi_{\mathrm{bg}}$. The background time derivative is $O_J(z)$, and the bounded gauge representative has size $O_J(\epsilon e^{\gamma_g t})$. Consequently

$$\delta\Phi_{\min}^{\mathrm{clock}} = \epsilon e^{\gamma_g t}[1 + O_J(z)] \tag{C7}$$

on each fixed compact slab. This identifies the same leading physical direction; it does not make a finite-charge slice readout gauge invariant or provide a time-uniform gauge estimate.

The branch, gauge inverse, reconstruction, classical evolution and charged form estimates now refer to the same action and background. Choose a preliminary analytic neighborhood, fix the high-frequency cutoff for the form there, and shrink for its compact band. Intersect the resulting positive neighborhoods with those for the branch, growth and chart on $J$. That proves Theorem A's simultaneous quantifier. The prior review's scope is recorded in Section 9.

**Theorem A — conditional persistence in one canonical domain.** Fix the exact neutral action and background in Appendix A, the selected completion (21), finite independent density labels $\mathcal{N}_a, \mathcal{N}_b$, and a compact interval $J = [t_-, t_+]$ in the local de Sitter patch. Assume the exact background/root linkage and zero-mass phase premise of Proposition 1 and Appendices H/O.1–O.5, the selected-mass certificate in Appendix O.6, and the neutral kinetic coarse gates in Appendices G/H. The neutral physical reconstruction and complete count are proved in Appendices J–N under those premises. Use the sourced constraint and variational reconstructions proved in Appendices E–F, with the original doubled action, its cap/joint terms and the specified new-cut polarization.

There is a positive, uncomputed $\delta_*(J, \mathcal{N}_a, \mathcal{N}_b)$ such that, for $|\delta| < \delta_*$, the following hold on the same actual charged background.

1. The analytic particular branch (22) exists on $J$, with separated timelike interfaces, $c_i > 0$, independent constants $R_i^3 \nu_i = \delta\mathcal{N}_i$, and a simple radial turn on its construction slices. Nonzero future-directed streams use the appropriate signs of $\delta\mathcal{N}_i$; zero charge uses the canonical phase continuation.
2. Its fixed-charge growing solution (25) has a smooth representative in the curvature-clock chart $\kappa = \kappa_{\mathrm{bg}}(t, \rho)$, with new orthogonal cuts, $\rho \in [0, 1]$, fixed tangential normalization and the actual homogeneous untwisted width pair $(q, P_q)$. That representative gives compatible $\mathcal{E}_2$ data and is the unique classical linear history for those data. Both wall phases and both $\chi_i$ are retained. No auxiliary zero-frequency mean replaces $P_q$.
3. The exact canonical Hamiltonian momentum form (C3) is coercive on its full radial $L^2$ momentum form domain for every real Fourier frequency and the separate physical mean fiber. At zero charge it obeys $\mathscr{K}_0 \geq \mathscr{S}_0/2$; charged coercivity has a positive, uncomputed constant depending on $J$. Evolution regularity is asserted mode by mode, not for an infrared-uniform original-gauge Fourier superposition.

The readout (25) uses the original redshift time $t$. On the common curvature-clock slices its first variation has the same leading neutral amplitude, with an $O_J(z)$ correction for the normalized regular gauge representative. This is a small-charge statement on each

fixed $J$; no common all-future bound for the clock map or kinetic norm follows. A pure-momentum form direction need not itself satisfy classical corner compatibility.

**Proof route.** Theorem 2 gives the neutral oscillator count; Proposition 3 constructs the charged branch in the selected action. The shifted construction in Section 6.2 gives a full smooth linearized solution, not just a formal mass substitution. The homogeneous adjoint problem (C5) supplies its gauge representative and its width output; canonical momenta are then computed from that representative. Original stationarity gives both compatibility conditions (26), so uniqueness identifies it with the classical evolution in (27). The canonical one-form and clock calculation in Section 8 identify the momentum coefficient with the same action. The all-mode anchor estimate and bounded-form continuation give assertion 3. Only then take the minimum of the positive branch, inverse, chart, regularity and form neighborhoods on $J$. Appendices C–G give the estimates and the detailed synthesis. □

## 9 Evidence and scope

Table 3 distinguishes the results established in each setting and their scope.

**Table 3. Scientific conclusions and their domains of validity.**

| Setting | Conclusion | Boundary |
|---|---|---|
| Neutral pure-potential sector | Complete negative radial count one; no massless radial mode | Generic coupled sector; exact background and phase premises |
| Neutral oscillator completion | Complete negative radial count one | Positive finite canonical oscillator class; phase zero modes are not excluded |
| Selected neutral nonlinear family | Convergent growing solution; protected separation inequalities and wall-scalar departure | Zero current; selected analytic family and tiny sufficient radius; no endpoint |
| Selected charged branch | Actual solution and a fixed-charge growing direction | Analytic pure-redshift sector; uncomputed radius; no charged spectral count |
| Common canonical domain | Growing history is a compatible classical solution; momentum form is coercive | New cuts, fixed tangential normalization, physical mean retained; finite slabs |

The background-to-phase chain assumes identified source execution and outward Arb arithmetic. The selected-mass certificate retains its separate historical execution premise. The neutral kinetic certificate uses integer fixed-point enclosures; a fresh relocated exact-Fraction check has verified all 16,365 saved Picard/Taylor cells and their endpoint induction, with seven coarse gates and nine rejection controls. This stronger check is distinct from the 822-check record inventory. Neither regenerates interval trajectories or

re-establishes exact-root existence. The pipeline supplement identifies the executable, input closure and new execution records.

OpenAI ChatGPT and Codex were used for research planning, mathematical development and checking, source organisation, manuscript preparation, and symbolic/exact-arithmetic calculations. Internal reviews of earlier versions retain their original scopes. A subsequent bounded independent internal audit reconstructed the seven sharpened nonlinear source bounds, compatible inverse and first-mode seed, and selected majorant; a fresh reviewer challenged the completed chain. The audit supported that chain under the stated exact-equation, numerical-enclosure and execution premises, identifying local forcing-convention and evidence-locator clarifications now incorporated in Appendix Q. It did not independently rederive the full action variation, regenerate the foundational interval calculations or repeat every observable-tail argument. These checks are not external human peer review, arithmetic independence or unconditional verification of every theorem. Appendix H and Q.6 state the retained premises.

Coercive momentum does not imply positive restoring energy or scalar stability. The charged growth result concerns one solution, not a complete charged count. Comoving densities are not finite total charges on the homogeneous infinite patch. Classical linear evolution is not nonlinear evolution. A simple radial turn is not a formation or horizon statement. No original-gauge infrared-uniform Fourier map, old-fixed-cap equivalence, all-future norm bound or quantitative charged amplitude is claimed.

Beyond the protected neutral departure interval, the nonlinear fate of the surviving scalar direction remains a separate question. A concrete comparison is the neutral Einstein-fiber sector: with both currents zero and the oscillators centered, the same radial solution can be combined with a four-dimensional metric satisfying $\mathrm{Ric}(\gamma) = 3k\gamma$, using $g_5 = ds^2 + a(s)^2\gamma$ and $\Phi = \Phi_0(s)$. Appendix P verifies the bulk and interface equations and the transfer of null expansions. This is a model-specific application of the warped-product method used in braneworld black-string geometries [25, Section 3], rather than a new general construction. It gives a setting in which to investigate nonlinear trapping while preserving the radial background. The scalar is fixed in this sector, so it does not describe nonlinear evolution of the growing mode proved here. Connecting that mode to collapse requires constraint-compatible data and a coupled Einstein–interface evolution argument. An event-horizon conclusion would additionally require a specified parent escape destination and global causal control. The present instability and kinetic estimates are not asserted on a collapsing background.

## 10 Conclusion

For one selected five-dimensional action, the certified neutral instability survives canonical oscillators and a small-charge material continuation. At zero current, a convergent nonlinear growing family and its explicit all-orders remainder provide a local departure witness through the physical wall scalar. The neutral oscillator problem retains a complete negative count; the charged time-dependent branch retains a fixed-charge growing direction. Its homogeneous adjoint gauge map puts that direction into the retained-width classical evolution framework. A separately derived and

validated canonical momentum estimate is compatible with this instability.

The useful increment is an exhaustive neutral radial count with a physical reconstruction that connects the counted mode to compatible dynamical-interface solutions. It supplies a controlled reference problem for the selected action, complementing earlier radion instability criteria and nonlinear two-brane simulations. Standard spectral/form theory and analytic parameterization strategies provide the machinery; the physical domain, boundary reconstruction and coupled compatibility arguments establish its application here, under explicit analytical and numerical premises. Quantitative control of the charged neighborhood, nonlinear nonzero-current evolution and behavior beyond the protected neutral family remain open. The explicit neutral radius is a sufficient proof bound, not a physical transition scale. The neutral comparison sector supplies a geometric reference for that investigation without identifying the outcome of the scalar instability.

### Data availability

The exact inputs, numerical enclosures and verification predicates are specified in the integral article appendices. Pipeline source and frozen data, including the neutral nonlinear records identified in Appendix Q, are versioned in the identified research archive. The owner-identifying repository address is reserved for author-facing editorial information. A local referee inspection package includes the inherited certificates, necessary nonlinear source/results, source/inverse ledgers and qualified saved-cell route; external anonymous access must be finalised before submission. The package README distinguishes inspection derivatives and their identities from original execution records. The pipeline supplement separates record inspection, exact-Fraction saved-cell validation, internal mathematical reconstruction and fresh producer regeneration; no portable regeneration of every interval trajectory is claimed.

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## Appendix A. Exact neutral anchor

The central bulk and wall potentials are $V = m^2\Phi^2/2$, $\tau_a = \lambda_a + \gamma(\Phi - v_0)^2/2$, $\tau_b = \lambda_b + \gamma(\Phi - v_1)^2/2$, with exact inputs

Table 4 gives the six exact rational parameters of the neutral anchor.

**Table 4. Exact action parameters; all values are dimensionless in the stated units.**

| Parameter | Exact value |
|---|---|
| $m^2$ | 7385903388887613/18014398509481984 |
| $\gamma$ | 20 |
| $v_0$ | 3602879701896397/36028797018963968 |
| $v_1$ | 6520040733916207/72057594037927936 |
| $\lambda_a$ | 6755388606266369/1125899906842624 |
| $\lambda_b$ | −851862101055633/140737488355328 |

The linked exact root satisfies

$$\Phi_0(0) \in \left[\frac{456607025194779413}{4611686018427387904}, \frac{114151756532473697}{1152921504606846976}\right], \qquad D \in [32727/16384, 32729/16384]. \tag{31}$$

The centered traces $\Phi_i^*$ in (2) belong to this exact solution. The phase calculation propagates the full launch and endpoint uncertainty. Zero perturbative Hessians (13) preserve the background values and slopes, with $\gamma = 20$ unchanged in the shooting map.

The following appendices are integral parts of this article. Equation tags beginning S, E, Q, R and NL retain distinct prefixes to identify the technical constructions. All reference numbers belong to the article bibliography.

The main text states the results and their logical connections; these integral appendices supply the detailed verification. For a first technical reading, use J–N for the neutral foundation, B for the oscillator extension, C–D for charged existence and growth, E–G for canonical reconstruction and evolution, and H/O for the computer-assisted premises. Appendix P is the separate neutral comparison sector. Equation tags are retained when a formula is relocated from the main text.

## Appendix B. Neutral premises and the dynamic-wall count

### *B.1 Exact background and physical spectral problem*

Use the six exact rational parameters, launch interval and wall-distance interval in manuscript Appendix A. In particular, the centered traces in the added interaction are exact traces of the linked root, not independently rounded constants. Let $E = a^{-1}$, $T = \Phi_0'$, $H = a'/a$. The background first integral and regular scalar system are manuscript (9)–(10), with inward signs $\sigma_a = +1$, $\sigma_b = -1$. The metric perturbation $\psi$, scalar $f$, derivative $g$, and $L = g + 2T\psi$ are retained through the turn. The equations hold globally in distributions, with finite original-action Hermitian kinetic norm and the physical wall traces. They are not patched separately on the two sides of $T = 0$.

The neutral reconstruction has four parts, derived in Appendices J–L: the full generic scalar Israel/bending equations are equivalent to the retained characteristic; its artificial simple zero at $\lambda = \mu + 4k = 0$ is removed without deleting finite poles or nonzero roots; the original-action Green identity supplies the positive kinetic weight; compact wall continuation preserves exhaustive simple roots. These results provide the physical input to the subsequent Sturm/phase argument. It is not inferred from an algebraic change of dependent variable alone. Exceptional harmonics, with identically zero traceless Hessian, are outside this count.

Appendices M–N prove the restricted index, wall limits and remaining-band exclusion, using the launch (13) and enclosed phase (14). Appendix O justifies the numerical predicates; their truth remains conditional on the identified outward execution.

### *B.2 Why oscillator elimination is legitimate only on the negative axis*

For each centered canonical surface oscillator the undivided rows are manuscript (15). For $\mu < 0$, $d_i = \omega_i^2 + \beta_i - \mu E_i^2 > 0$, so $u_i = \beta_i \eta_i f_i / d_i$. The effective Hessian $\Gamma_i$ is nonnegative and nonincreasing in $\mu$, while the original unique eigenvalue $\alpha$ is nondecreasing in each constant Hessian. Consequently $F(\mu) = \alpha(\Gamma_a(\mu), \Gamma_b(\mu))$ is nonincreasing. At the two constant-Hessian comparison masses $\alpha(\Gamma_0) \le \alpha(\Gamma_\infty)$, the function $F - \mu$ has opposite weak signs. Its strict decrease yields exactly one root.

Differentiate the original-action Green identity, keeping the surface oscillator equation. The wall inertia adds $\sum_i a_i^2 |u_i|^2$ to the norm. Equivalently, the derivative contribution of $\Gamma_i$ to the mass variation is minus this added term divided by the bulk norm. Therefore, at the fixed point,

$$(\mu - F)' = \frac{Z_A + \sum_i a_i^2 |u_i|^2}{Z_A} > 0. \tag{S1}$$

The undivided equations show that no pure oscillator or phase mode lies at a negative pole. Reality follows from the same Hermitian Green identity before elimination. This completes the negative-sector count conditional on manuscript Proposition 1. At nonnegative mass the oscillator poles must be restored in the characteristic. The no-massless assertion in Proposition 1 does not exclude massless phase states in the enlarged theory.

## Appendix C. Actual analytic charged branch

*C.1 Fixed chart, formal compatibility and analytic scale*

Fix the selected action (21), independent $\mathcal{N}_a, \mathcal{N}_b$, $h = \sqrt{k}$, $z = \delta^2 e^{-6ht}$. Here $s$ is original proper radial coordinate. For this section only, $r$ is a fixed reference radial coordinate, not the later isothermal coordinate. Flatten the moving interval by

$$\begin{gathered} s = r + X(z, r), \quad X = \xi_a(z) + \frac{r-a}{b-a}[\xi_b(z) - \xi_a(z)], \quad J_r = 1 + X_r, \\ D_0 = -6hz\partial_z, \quad W = \frac{D_0 X}{J_r}, \quad D = D_0 - W\partial_r, \quad L = J_r^{-1}\partial_r. \end{gathered} \tag{S2}$$

Background functions are composed with $s = r + X$. Thus $D$ annihilates them exactly, and a new graph coefficient alone creates no spurious bulk source. Let $\mathfrak{m} = m^2$ denote the coefficient in $V = \mathfrak{m}\Phi^2/2$, avoiding confusion with an order index. The full compatible formal recursion follows from the bulk rows (S6), the exact scalar and oscillator wall rows, and Israel. Its compatibility uses the differentiated Einstein/Bianchi and wall/Codazzi identities with the actual current–scalar exchange. It is not a recursion for arbitrary incompatible source triples.

The order-$n$ temporal mass is $\mu_n = -18n(2n-1)k$. At these masses the oscillator elimination has $\Gamma_i > 41/2$. The selected-action comparison excludes a physical scalar kernel for all $\mu \le -18k$. Tracefree bulk and graph divisors have size $n^2$, while phase reconstruction has nonzero divisors $3(2n+1)h$. The known formal sequence is solved in the following order: graphs; the tracefree combination $c = b + 2p$; compatible scalar boundary problem; constrained metric; reconstructed surface oscillators and phases. Bianchi and Codazzi remove redundant residual rows, rather than replacing scalar matching.

Extend the analytic background to a complex stadium $\Omega_\rho$ about the entire real interval. With $d = 1 - \operatorname{dist}(r, I)/\rho$, define $\|v\|_p = \sup d^p|v|$. Then products add indices and

$$\|v'\|_{p+1} \le \frac{e(p+1)}{\rho}\|v\|_p, \qquad \mathcal{U}_n(v) = n^2\|v_n\|_{2n} + n\|v_n'\|_{2n+1} + \|v_n''\|_{2n+2}. \tag{S3}$$

For graphs and surface scalars use $\Xi_n = n^3 \sum_i |\xi_{i,n}|$, $Q_n = n^3 \sum_i |\chi_{i,n}|$. The real endpoints have $d = 1$. The collar is fixed before the recursion; it is not shrunk at each order.

*C.2 Uniform inverse and metric estimate*

Set $D_n = 2T^2 - 3(\mu_n + 4k)E^2$. On the real interval, $D_n \ge 42k(\min E)^2 n^2$. On one prior sufficiently small complex stadium, $D_n/n^2$ stays nonzero uniformly. Reduction gives

$$\begin{gathered} -f_n'' + a_n f_n' + V_n f_n = F_n, \quad \partial_\nu f_n + \beta_{n,i} f_n = g_{n,i}, \\ a_n = -4H - B_0 T/D_n, \quad V_n = \mathfrak{m} - \mu_n E^2 + 2T^2 + B_0 T'/D_n, \quad B_0 = 12HT - 4\mathfrak{m}\Phi_0. \end{gathered} \tag{S4}$$

For large $n$, the real scalar potential is bounded below by $cn^2$ and the outward Robin coefficients by a positive constant. Exponential boundary-layer barriers and the equation

for the derivative give $n^2\|f\|_\infty + n\|f'\|_\infty + \|f''\|_\infty \le C(\|F\|_\infty + n|g|)$. The finitely many lower orders satisfy this after enlarging $C$, by actual nonresonance and regular finite-interval inversion.

For the complex extension integrate the ODE along the normal segment from its closest real point. For weight $p = 2n$, the single/double primitive constants are $\rho/(p+1)$ and $\rho^2/[p(p+1)]$. The linear initial term costs $\rho/(p+1)$. With coefficient bounds obtained on the larger stadium first, choose $\rho$ once so the terms $C(\rho+\rho^2)$ can be absorbed. Work first on smaller closed stadia with the original $d$, then take the supremum limit. The equation supplies the second derivative. This proves

$$\begin{aligned} \mathcal{U}_n(f) &\le C(\|F_n\|_{2n+2} + n|g_n|), \\ \mathcal{U}_n(p) &\le C\left(\|S_{c,n}\|_{2n+2} + \mathcal{U}_n(f) + n\|S_{p,n}\|_{2n+1}\right). \end{aligned} \tag{S5}$$

For the second inequality anchor $p_n = (S_{c,n} - Tf_n' + T'f_n)/D_n$ only on the real interval. Continue by $p_n' = -2Hp_n - Tf_n/3 + S_{p,n}$, whose primitive gains one index, and differentiate once with (S3). Compatibility makes this continuation the actual constrained solution. Multiplication by $D_n^{-1}$ alone has no analytic-index gain.

*C.2.1 Coefficient extraction and the actual compatible bulk sources* Here is a definition of the source maps that fixes their signs as well as their dependence on lower orders. Let $r_n = -6nh$, $c_n = b_n + 2p_n$, and let $\mathcal{R}_T, \mathcal{R}_M, \mathcal{R}_K, \mathcal{R}_R$ be, in order, the four left sides of (S6). For any one of these exact residuals define

$$q_{j,n} = [z^n]\mathcal{R}_j(p_{<n}, b_{<n}, f_{<n}, X_{<n}), \quad v_{<n} = \sum_{1\le j<n} v_j z^j. \tag{Q1}$$

The background in this definition is always composed with $r + X_{<n}$; the differential operators are (S2). Thus (Q1) is a finite coefficient operation on explicitly displayed analytic differential expressions, not a new unspecified forcing. At the first order it includes the explicit current terms in the wall residuals below. New graphs do not generate a pure-background bulk term, since $D$ annihilates the composed background. The new tracefree and mixed equations give

$$c_n = -\frac{q_{T,n}}{r_n(r_n - h)}, \qquad S_{p,n} = Hc_n - \frac{q_{M,n}}{r_n}. \tag{Q2}$$

With $\mathfrak{m}$ as in (S4), the radial and Klein–Gordon source maps are exactly

$$\begin{aligned} S_{c,n} &= 12HS_{p,n} - (12H^2 - T^2)c_n + q_{R,n}, \\ S_{g,n} &= Tc_n' + 2\mathfrak{m}\Phi_0 c_n - 6TS_{p,n} - q_{K,n}, \\ F_n &= -S_{g,n} - B_0 S_{c,n}/D_n. \end{aligned} \tag{Q3}$$

In particular, our sign convention is $f_n'' = -4Hf_n' + (\mathfrak{m} - \mu_n E^2 + 2T^2)f_n + B_0 p_n + S_{g,n}$. The already solved $c_n$ can enter (Q3); it is not a further free order-$n$ unknown. The compatible constraint is $Tf_n' - T'f_n + D_n p_n = S_{c,n}$. Differentiating it and substituting the scalar and momentum rows gives the exact residual identity

$$\begin{aligned} \mathcal{C}_n' + 4H\mathcal{C}_n &= TS_{g,n} + D_n S_{p,n} - S_{c,n}' - 4HS_{c,n}, \\ \mathcal{C}_n &= Tf_n' - T' f_n + D_n p_n - S_{c,n}, \\ S_{c,n}' + 4HS_{c,n} &= TS_{g,n} + D_n S_{p,n}. \end{aligned} \tag{Q4}$$

The last equality is the source compatibility delivered by the exact Bianchi identity, rather than an imposed restriction on arbitrary external loads. To see why no Einstein equation has been suppressed, impose KG, mixed and tracefree bulk rows after all lower residuals vanish. For the remaining mixed-index Einstein residual, $E^s{}_t = 0$ and $E^t{}_t = E^x{}_x$. Its temporal divergence is $\partial_t E^t{}_t + 3(R_t/R)(E^t{}_t - E^x{}_x) + (B_t/B)(E^t{}_t - E^s{}_s) = 0$. The last term is higher order because $B_t/B = O(z)$. At order $n$, $r_n E^t{}_{t,n} = 0$, so the temporal and spatial residuals vanish. The radial divergence then gives (Q4). The scalar and metric background identities used here are (9). Conversely (Q4), the scalar equation and the constraint reconstruct the mixed equation without division by $T$.

*C.2.2 Exact wall source extraction and reduced Robin data* For the selected action let $\mathcal{I}_i$ be (S7) minus $c_i\nu_i^2 + \chi_{i,t}^2/(2\ell_i^2)$. Its new row is $\sigma_i E_i^2 r_n(r_n - h)\xi_{i,n} + q_{I,i,n} = 0$, where $q_I$ is defined by the truncation rule (Q1). This determines the two graph coefficients separately. In particular $\xi_{i,1} = \sigma_i \mathcal{N}_i^2 E_i^4/(42k)$; no density ratio is fixed.

The exact oscillator and normalized scalar residuals used for source extraction are

$$\begin{aligned} \mathcal{O}_i &= \ell_i^{-2}\Big\{\chi_{i,tt} + [3h + 2H(S_i)S_{i,t} + 2p_{i,t}^{\mathrm{wall}} + v_i v_{i,t}/(1 - v_i^2)]\chi_{i,t}\Big\} \\ &\quad + 2\chi_i - [\Phi_0(S_i) - \Phi_i^* + f_i], \\ \mathcal{J}_i &= \frac{e^{-b}(T + Lf) + v_i e^{-p} EDf}{\sqrt{1 - v_i^2}} - \frac{\sigma_i}{2}[\tau_{i,\Phi} + W_{i,\Phi} + c_{i,\Phi}\nu_i^2]. \end{aligned} \tag{Q5}$$

All traces and coefficients in this formula are pulled to the actual wall. Define $o_{i,n} = [z^n]\mathcal{O}_i(U_{<n})$, $j_{i,n} = [z^n]\mathcal{J}_i(U_{<n})$, with the lower graphs included in $U_{<n}$. Put $d_{n,i} = 2 - \mu_n E_i^2 > 0$, $\Gamma_{n,i} = 21 - d_{n,i}^{-1}$. The order-$n$ equations and their known load are

$$\begin{aligned} \chi_{i,n} &= (f_{n,i} + T_i\xi_{i,n} - o_{i,n})/d_{n,i}, \\ 2\sigma_i(f_n' + 2T_i p_n) - \Gamma_{n,i} f_{n,i} &= \mathcal{R}_{i,n}, \\ \mathcal{R}_{i,n} &= 2\sigma_i\left\{T_i c_{n,i} - \left(T_i' - \frac{\sigma_i \Gamma_{n,i} T_i}{2}\right)\xi_{i,n} - j_{i,n} + \frac{\sigma_i o_{i,n}}{2d_{n,i}}\right\}. \end{aligned} \tag{Q6}$$

These formulas retain the displaced background trace and normal, the current force and the oscillator source. Substituting the constraint for $p_n$, with $A_{n,i} = 1 - 2T_i^2/D_{n,i} > 0$, gives the missing coefficients in (S4):

$$\boxed{\beta_{n,i} = \frac{\Gamma_{n,i}}{2A_{n,i}} - \frac{2\sigma_i T_i T_i'}{D_{n,i}A_{n,i}}, \qquad g_{n,i} = -\frac{\mathcal{R}_{i,n}}{2A_{n,i}} + \frac{2\sigma_i T_i S_{c,n,i}}{D_{n,i}A_{n,i}}.} \tag{Q7}$$

The outward derivative here is $-\sigma_i\partial_s$. Endpoint values of $S_c$ really occur; an $L^2$-only bound without these traces would not justify this reduction. For large $n$, $A_{n,i}\to 1$, $\Gamma_{n,i}\to 21$, and the second term of $\beta$ is $O(n^{-2})$. Thus $\beta_{n,i}$ has a uniform positive lower bound at all sufficiently large orders. The finitely many remaining inverses use nonresonance, not a positivity assertion at every low order.

Finally set $\mathcal{J}^a{}_b = K^a{}_b - K\delta^a_b + S^a{}_b/2$. Bulk Codazzi, the independently imposed scalar normal law and the exact material exchange (5) give $D_a\mathcal{J}^a{}_b = 0$. Once tracefree Israel and lower rows vanish, its new residual is a scalar multiple of the identity. Its temporal derivative is $r_n$ times that scalar, hence it vanishes. This is the wall compatibility step. It would not dispose of an additional constant harmonic. Actual reflected geometry makes the averaged normal-force equation redundant; it does not remove either local scalar row.

*C.3 Actual source equations and derivative budgets*

The four pulled bulk rows, with background functions composed as in (S2), are

$$\begin{aligned}
0 &= (D^2 - hD)(b+2p) + (Db)^2 - 2(Dp)(Db) - 2(Dp)^2 + (Df)^2,\\
0 &= L(Dp) - (Db)(H+Lp) + (Df)(T+Lf)/3,\\
0 &= -E^2e^{-2p}[D^2f + (3h + 2Dp + Db)Df]\\
&\quad + e^{-2b}[T' + L^2f + (4H + 4Lp - Lb)(T+Lf)] - \mathfrak{m}(\Phi_0 + f),\\
0 &= e^{-2b}[6(H+Lp)^2 - (T+Lf)^2/2] - 6 + \mathfrak{m}(\Phi_0+f)^2/2\\
&\quad - E^2e^{-2p}[6h^2 + 3(D^2p + 3hDp) + 3(Dp)^2 + (Df)^2/2].
\end{aligned} \tag{S6}$$

The transport identity is $D^2v = D_0^2v - 2WD_0v_r + (WW_r - D_0W)v_r + W^2v_{rr}$. For radial derivative order $a\le 2$ and temporal order $b$, $\|\partial_r^a(z\partial_z)^b v_n\|_{2n+a}\le n^{a+b-2}\mathcal{U}_n(v)$. Graph/oscillator factors have $n^{b-3}$ for $b\le 2$. These factorwise bounds give the following source allocation.

Table 5 records the required source control for the compatible recursion.

**Table 5. Source budgets after the new linear equation is removed.**

| Equation after its new linear row is removed | Required index and control |
|---|---|
| Tracefree bulk | Index $2n$; every transported radial derivative is paid by a positive-order graph factor; divide by order $n^2$, then apply Nagumo twice |
| Mixed Einstein | Index $2n+1$ for $nS_{p,n}$; the exceptional $W_ip_j''$ fits because $2j+2\le 2(i+j)+1$ |
| Klein–Gordon and radial constraint | Index $2n+2$; includes $p_i'p_j'$, $f_i'f_j'$ and already sourced $c_n, c_n'$ |
| Tracefree Israel graph | Outer $n$ source bound; graph divisor order $n^2$ yields cubic weight |

| Equation after its new linear row is removed | Required index and control |
|---|---|
| Scalar normal and oscillator | Outer $n$ wall load; safe negative oscillator inverse supplies two powers of order |

At real endpoints the bulk value, first derivative and second temporal derivative have weights $n^{-2}, n^{-1}, 1$; graph/oscillator value, velocity and acceleration have weights $n^{-3}, n^{-2}, n^{-1}$. For positive factor orders $i_1 + \cdots + i_m = n$, $n/\prod i_j \le m$ and $n^2/\prod i_j^2 \le m^2$. Polynomial factor-count losses are absorbed by an analytic scalar majorant.

For completeness set the actual wall lapse factors $N = a(S)e^p$, $B = e^b$, $v = (B/N)S_t$, $\ell = N\sqrt{1-v^2}$. The exact tracefree Israel left side is

$$K_\tau - K_x = \frac{\sigma}{N\sqrt{1-v^2}}\left[\frac{v_t}{1-v^2} + v(Db - h - Dp)\right]. \tag{S7}$$

Its leading graph row is $\sigma E_i^2(S_{tt} - hS_t)$. Its current source is explicitly $z\mathcal{N}_i^2 c(\Phi_i)/(a(S_i)^6 e^{6p_i})$. The scalar normal is $\sigma(1-v^2)^{-1/2}[e^{-b}(T + Lf) + ve^{-p}EDf]$. These formulas include the actual moving normal, not only a shifted field trace.

The exact oscillator equation contains the friction factor $3h + 2H(S)S_t + 2p_t^{\mathrm{wall}} + vv_t/(1-v^2)$, multiplying $\chi_t$ inside $\ell^{-2}$, plus $\chi_{tt}$, and the potential force $2\chi - [\Phi_0(S) - \Phi_i^* + f_i]$. Terms such as $v_i S_{tt,j}\chi_{t,l}$ have bound proportional to $n/(i^2 j l^2)$; distributing $n = i + j + l$ bounds that ratio. A false bound on the total number of derivatives in a monomial is neither needed nor used. Higher analytic velocity powers obey the same factorwise rule. This accounts for the endpoint terms that a bulk-only estimate would miss.

*C.3.1 A coefficient inequality before the scalar majorant* The preceding source definitions permit a coefficientwise estimate rather than only an analogy with an analytic equation. Set $C_{\mathcal{N}} = |\mathcal{N}_a|^2 + |\mathcal{N}_b|^2$ and $Y_{<n}(z) = \sum_{1\le j<n} Y_j z^j$. There are constants $A \ge 1, C \ge 1$ and one fixed nonnegative integer $q_0$, independent of $n$, such that the actual triangular solve satisfies

$$Y_n \le C[z^n]\left\{C_{\mathcal{N}} z + \sum_{m\ge 2} A^m (m+1)^{q_0} Y_{<n}^m + C_{\mathcal{N}} z \sum_{m\ge 1} A^{m+1}(m+1)^{q_0} Y_{<n}^m\right\}. \tag{Q8}$$

Here $[z^n]Y^m = \sum_{i_1+\cdots+i_m=n} \prod_j Y_{i_j}$. The following lemma makes the separate source estimates behind (Q8) explicit.

**Lemma C.1 — compatible source bounds.** Fix the background, one complex collar as in C.2 and the finite charge labels. For the actual lower-order compatible sequence, define

$$\mathcal{B}_n = [z^n]\left\{ C_{\mathcal{N}} z + \sum_{m\ge 2} A^m (m+1)^{q_0} Y^m_{<n} + C_{\mathcal{N}} z \sum_{m\ge 1} A^{m+1}(m+1)^{q_0} Y^m_{<n} \right\}. \tag{Q9}$$

One can choose $A, q_0$, and constants $C_j$, once, independently of $n$, so that the sources defined in (Q1)–(Q7) and their triangular reconstructions satisfy

$$\begin{aligned}
\|q_{T,n}\|_{2n} + \mathcal{U}_n(c) &\le C_1 \mathcal{B}_n,\\
\|q_{M,n}\|_{2n+1} + n\|S_{p,n}\|_{2n+1} &\le C_2 \mathcal{B}_n,\\
\|q_{K,n}\|_{2n+2} + \|q_{R,n}\|_{2n+2} + \|S_{c,n}\|_{2n+2} + \|S_{g,n}\|_{2n+2} &\le C_3 \mathcal{B}_n,\\
n\sum_i (|q_{I,i,n}| + |j_{i,n}| + |o_{i,n}|) + \Xi_n &\le C_4 \mathcal{B}_n,\\
\|F_n\|_{2n+2} + n\sum_i (|\mathcal{R}_{i,n}| + |g_{n,i}|) &\le C_5 \mathcal{B}_n,\\
\mathcal{U}_n(f) + \mathcal{U}_n(p) + \mathcal{U}_n(b) + Q_n &\le C_6 \mathcal{B}_n.
\end{aligned} \tag{Q10}$$

This is a bound for the compatible recursion, conditional on the inverse (S5); it does not assert solvability for arbitrary independently prescribed residuals.

*Proof.* Analytic products add Nagumo indices, and an index may be increased since $0 < d \le 1$. A bulk factor of order $j$, radial order $a \le 2$ and temporal order $b$ costs $j^{a+b-2}\mathcal{U}_j$ at index $2j + a$. A finite graph or oscillator jet costs $j^{b-3}$ times its cubic norm. Use a slightly larger prior background collar for compositions: its Cauchy bounds give a geometric coefficient bound on the fixed smaller collar. Exponentials, inverse lapse/Jacobian and rapidity are analytic germs about nonzero units and have the same type of bound. Only finitely many differentiated factor slots occur in the exact rows; choosing those slots adds a fixed polynomial in the number of factors, not in the expansion order. Enlarge $A, q_0$ below to cover this polynomial.

For **the tracefree source** (Q1)–(Q2), the nontransport products are products of first temporal jets. The transported terms are $WD_0c_r$, $(WW_r - D_0W)c_r$, and $W^2c_{rr}$. Every added radial index is paid by a positive-order graph: for example $2j + 1 \le 2(i + j)$, and $2j + 2 \le 2(i + l + j)$ for $i, l \ge 1$. Their temporal weights are bounded by the jet rules above. No new pure-background graph source occurs because $D$ annihilates the composed background. Thus $\|q_T\|_{2n} \le C\mathcal{B}_n$. The divisor $r_n(r_n - h) = 6h^2n(6n + 1)$ is bounded below by $36h^2n^2$; one and two Nagumo derivatives cost at most $Cn$ and $Cn^2$. This proves the first row of (Q10), including both derivatives of $c_n$.

For **the mixed source**, expand $L(Dp)$ as $J_r^{-1}D_0p_r - J_r^{-1}(W_rp_r + Wp_{rr})$. After its linear term is removed, $(J_r^{-1} - 1)D_0p_r$, the two transport terms, $(Db)Lp$ and $(Df)Lf$ fit index $2n + 1$. In particular $W_ip_j''$ fits since $2j + 2 \le 2(i + j) + 1$; its time weight introduces no positive power of an unpaired order. Composed background coefficients have the reserved analytic bound. Hence $\|q_M\|_{2n+1} \le C\mathcal{B}_n$. Since $|r_n| = 6hn$, (Q2) and the first row give the second row.

For **the Klein–Gordon and radial sources**, all nontransport radial products in (S6), including $p_i' p_j'$ and $f_i' f_j'$, fit index $2n+2$. A term consisting of a positive-order analytic coefficient times $D_0^2 v_j$ has no order loss: $j^2$ cancels the bulk value weight $j^{-2}$. Transport terms have the same positive-graph index payment as above. Pure background residuals cancel by (9). Therefore both raw sources have the third-row bound. In (Q3), $c_n'$ and $S_{p,n}$ already satisfy stronger indices, giving the stated bounds on $S_c, S_g$. The differentiated identity (Q4) enforces compatibility; the inversion does not require estimating $S_c'$ at the undifferentiated source index.

For **the three wall residuals**, evaluation at the real endpoints costs no analytic weight, since $d = 1$. In (S7), the only linear graph acceleration has been removed. Every remaining graph acceleration carries a positive-order coefficient; typical outer-order ratios are $n/(i^2 j)$. The oscillator stress gives $n/(i^2 j^2)$. In the scalar normal, a positive-order coefficient multiplying a first bulk jet has the ratio $n/(i^2 j)$; velocity times $Df$ has at least as much decay. The potential compositions are less differentiated. In (Q5), a coefficient times $\chi_{tt}$ or its first derivative times $\chi_t$ gives respectively $n/(i^2 j)$ or $n/(ij^2)$. The rapidity/friction term also includes $n/(i^2 j l^2)$, as noted above. Distribute $n = i_1 + \cdots + i_m$ among the positive-order factors; these ratios are bounded by a fixed polynomial in $m$. Extra analytic velocity or inverse-lapse factors improve the order denominators. The explicit current source is $z$ times an analytic germ: here distribute $n = 1 + \sum i_j$, giving the same bound, with its constant term covered by $C_{\mathcal{N}} z$. This proves the raw wall bound. The graph divisor is of order $n^2$, so multiplying the graph solution by $n^3$ costs exactly the outer $n$ already present. This proves the fourth row.

For **the reduced scalar/Robin load** (Q6)–(Q7), uniformly $D_n, d_{n,i} \geq cn^2$, $A_{n,i} \geq c > 0$, and $\Gamma_{n,i}$ is bounded. In particular $n|c_{n,i}| \leq \mathcal{U}_n(c)/n$, $n|\xi_{i,n}| \leq \Xi_n/n^2$, and $n|S_{c,n,i}/D_{n,i}| \leq C\|S_{c,n}\|_{2n+2}/n$. The endpoint $S_c$ is an actual trace controlled by the analytic norm. Substitution gives the fifth row, including $F = -S_g - B_0 S_c/D_n$.

Finally **the scalar, metric and oscillator reconstructions** use (S5), $b_n = c_n - 2p_n$, and $n^3|\chi_{i,n}| \leq Cn(|f_{n,i}| + |\xi_{i,n}| + |o_{i,n}|)$. The preceding rows control all these terms. This proves the final row. The source bounds use finitely many analytic products and a fixed number of reconstructions, so the constants can be enlarged once for the entire chain. Summing its bulk, graph and oscillator outputs gives (Q8). □

For an integer $M > q_0$, the binomial coefficients of $(1-x)^{-M}$ dominate a fixed positive multiple of $(m+1)^{q_0}$. Increasing one fixed $K$ therefore majorizes the right side of (Q8) by the coefficient of $K[zC_{\mathcal{N}} + Y^2 + zC_{\mathcal{N}}Y](1-KY)^{-M}$. Induction gives equation (C2). This is an existential bound on the explicitly defined source maps; it supplies neither numerical inverse constants nor a computed charge radius.

### *C.4 Summation, current reconstruction and turn*

Let $Y_n$ sum the three bulk $\mathcal{U}_n$ norms and both finite cubic weights. The triangular solve and the bounds above dominate the actual compatible coefficients by equation (C2), with fixed positive $K, M$. All remaining sources are quadratic in positive-order unknowns or an explicit current $z$ times an analytic germ. At the origin the majorant's right-hand derivative in $Y$ vanishes. The analytic implicit-function theorem gives a

convergent nonnegative coefficient series. On one inner stadium, $d \geq d_{\min} > 0$; restricting $|z|$ by the factor $d_{\min}^2$ absorbs every $d^{-2n}$. All fixed derivatives converge on compact subsets. Vanishing exact-equation coefficients then imply vanishing exact residuals.

Reconstruct $\nu_i = \delta\mathcal{N}_i/R_i^3$ and $\theta_{i,t} = -2c_i\delta\mathcal{N}_i\ell_i/R_i^3$. With the signed variable $w_c = \delta e^{-3ht}$, $w_c^2 = z$, the phase is $\theta_{i,0} + w_c\Theta_i(z)$. Its nonzero divisors $3(2j+1)h$ give convergence and the exact Noether law. The algebra also bounds absolute charge-monomial sums, so independent density directions and mixed terms are retained; no ratio is tuned. For each fixed finite pair there is a positive small-$\delta$ domain.

Uniform $C^2$ closeness preserves positive warp/lapse, positive $c_i$, timelike separated walls and the one simple radial turn, using its endpoint signs. Uniqueness is only in this analytic pure-redshift particular sector after fixing the two additive phase constants, or equivalently modulo independent constant phase shifts. These constants change neither the geometry nor the currents or growing solution. The radius is existential and uncomputed.

### *C.5 Estimates used in Proposition 3*

The scale (S3) is also recorded in the original normalization of Proposition 3:

$$\mathcal{U}_n(v) = n^2\|v_n\|_{2n} + n\|v_n'\|_{2n+1} + \|v_n''\|_{2n+2}, \tag{23}$$

The uniform scalar and constrained-metric estimates are (S5), equivalently

$$\begin{aligned} &\mathcal{U}_n(f) \leq C(\|F_n\|_{2n+2} + n|g_n|), \\ &\mathcal{U}_n(p) \leq C\Big[\|S_{c,n}\|_{2n+2} + \mathcal{U}_n(f) + n\|S_{p,n}\|_{2n+1}\Big]. \end{aligned} \tag{C1}$$

Here the compatible source classes and graph/surface weights are those defined in C.1–C.3. Their coefficient budget (Q8) closes the scalar majorant

$$Y = \frac{K[zC_N + Y^2 + zC_NY]}{(1-KY)^M}, \qquad C_N = |\mathcal{N}_a|^2 + |\mathcal{N}_b|^2. \tag{C2}$$

Its derivative with respect to the unknown majorant variable vanishes at the origin. The analytic implicit-function theorem and coefficient induction give convergence on one fixed collar. The exact equations follow from the vanishing Taylor coefficients, and C.4 reconstructs the currents and simple turn.

## Appendix D. Shifted nonresonance and actual growing linearized solution

Let $\mu_*$ be the selected completed negative mass, $\mu_* = -\gamma_g(\gamma_g + 3h)$. The same-action estimates $\mu_* \geq \mu_{20,20} > -33/25000$, $k > 3/10000$, and $\mu_* < -4k$ give $1 < \gamma_g/h < 11/10$. The principal scalar auxiliary form is positive far to the left of its first physical kernel. With only one negative physical mass, continuity implies its first kernel at $\mu_*$ is its lowest auxiliary eigenfunction. ODE uniqueness and finite Robin laws make that eigenfunction strictly positive. Normalize $f_0(s_*) = 1$ at the neutral turn. Its surface oscillators are

genuine, $u_{i,0} = f_0(s_i)/(2 - \mu_* E_i^2)$; its graph amplitude is zero by the nonzero tracefree Israel factor $\gamma_g(\gamma_g - h)$.

For a linearized coefficient multiplied by $\epsilon e^{\gamma_g t} z^n$, the actual rate is $r_n = \gamma_g - 6nh$, and its radial parameter is $\nu_n = -r_n(r_n + 3h)$. For $n \geq 1$,

$$\nu_n \leq \nu_1 < -\frac{931}{100}k < -\frac{2793}{1000000} < -\frac{33}{25000} < \mu_{20,20}. \tag{S8}$$

A kernel there would reduce to constant Hessians $\Gamma_i(\nu_n) > 20$, implying mass at least $\mu_{20,20}$, a contradiction. The other divisors are $r_n$, $h - r_n$, $r_n(r_n - h)$, $2T^2 - 3(\nu_n + 4k)E^2$, $2 - \nu_n E_i^2$, and $\gamma_g - (6n + 3)h$; each is nonzero, with the required order-$n$ or order-$n^2$ growth. The leading scalar kernel is selected, never inverted. In particular, (S8) is a new comparison for the shifted ladder, not an application of the background $-18n(2n - 1)k$ ladder.

Linearize every row in (S6)–(S7) and the full material equations about the actual analytic branch. At order $n \geq 1$, charged coefficients multiply lower perturbation orders. The new graph row is solved first with coefficient $\sigma_i E_i^2 r_n(r_n - h)$; current stress variation is $O(z)$, and background oscillator velocity is $O(z)$. Tracefree bulk, scalar/momentum reconstruction and safe oscillator reconstruction follow in the same compatible order as Appendix C. Linearized Bianchi and Codazzi use $r_n \neq 0$ to remove the remaining redundant temporal and Israel residuals. Reflection symmetry makes the averaged normal-evolution condition automatic [21, Section III]; the local jump and scalar junction rows remain in the recursion.

Use the offset norms

$$\widehat{\mathcal{U}}_n(v) = (n + 1)^2 \|v_n\|_{2n+2} + (n + 1)\|v_n'\|_{2n+3} + \|v_n''\|_{2n+4}. \tag{S9}$$

The uniform inverse proof in C.2 applies with this offset; finite low orders are absorbed into its constant. A product of background total order $i \geq 1$ and perturbation order $j$ has index $2i + (2j + 2) = 2n + 2$. Endpoint budgets use $j + 1$ for the perturbation factor. With $T_0$ finite and a nonnegative analytic $B(z)$, $B(0) = 0$, the full norm satisfies $T_n \leq K \sum_{i=1}^n B_i T_{n-i}$. The convergent majorant $T_0/(1 - KB(z))$ gives an actual smooth linearized solution on one inner collar.

Fixed-charge reconstruction is $\delta\nu_i = -3\nu_i \delta \log R_i$, $\delta(R_i^3 \nu_i) = 0$, and $\delta\theta_{i,t} = \theta_{i,t}[(c_{i,\Phi}/c_i)\delta\Phi_i + \delta \log \ell_i - 3\delta \log R_i]$. Its phase quadratures have the nonzero divisors above. The total scalar pull is $\widehat{f} + T\widehat{X}$. At the branch minimum the total background radial derivative vanishes, so the variation of its location contributes zero to the first variation of the minimum value. Since $T = O(z)$ there and $\widehat{X}_0 = 0$, manuscript (25) follows. The exponent is in the construction clock. This is a linearized solution, not an exact neighboring nonlinear family.

## Appendix E. Regular action, constraints and the common chart

*E.1 Invertible reduction and complete boundary laws*

Write $ds_5^2 = \mathfrak{q}_{ab}dx^a dx^b + e^{2\sigma}(dY^2 + dZ^2)$, $g = e^{4\sigma}\mathfrak{q}$. The warped-product identities give $R_5 = R[\mathfrak{q}] - 4\Box\sigma - 6(\nabla\sigma)^2$ and $K_5 = K_{\mathfrak{q}} + 2n\sigma$. Integration of the bulk divergence cancels its normal partner in doubled GHY. The subsequent Einstein-frame divergence cancels the transformed GHY normal term. Per transverse coordinate area the exact action is

$$\begin{aligned} S_E = &\int \sqrt{-g}\,[R - \tfrac{1}{2}G_{AB}\partial X^A \partial X^B - B(X)] + 2\sum_i \int \sqrt{-h}\,K_{\text{out}} \\ &- \sum_i \int \sqrt{-h}\,[P_i + \tfrac{1}{2}\mathsf{H}_{i,rs}D\psi_i^r D\psi_i^s], \\ X = (\sigma, \Phi), \quad G = \operatorname{diag}(12, 2), \quad B = 2e^{-4\sigma}(V - 6), \\ P_i = e^{-2\sigma}(\tau_i + W_i), \quad \mathsf{H}_i = e^{2\sigma}\operatorname{diag}(1, 1/(2c_i)). \end{aligned} \tag{S10}$$

The boundary term in (S10) follows the Gibbons–Hawking prescription [20], with outward normals and the coefficient fixed by the bulk action. Caps carry their corresponding causal signs. At a cap–wall joint the signed area-times-boost term is the Lorentzian nonsmooth-boundary term [22]. Under this positive conformal reduction its measure acquires the lower-dimensional area factor and its boost is invariant. The sign follows the chosen normals and ordering of caps, rather than an unsigned angle convention. Joint and normal work are retained. The scalar/fiber normal forces are

$$G_{AB}n_{\text{in}}X^B = F_A, \qquad F_A = P_{,A} + \tfrac{1}{2}\mathsf{H}_{rs,A}D\psi^r D\psi^s. \tag{S11}$$

The $\sigma$ row is the original internal-fiber Israel law. The base Israel equations and intrinsic $\chi, \theta$ equations follow from the same action. The Ward identity is $D^\alpha S_{\alpha\beta} = E_{\psi,r}D_\beta\psi^r - F_A D_\beta X^A$; its off-shell $E_\psi$ term is needed in the Hessian identity. The densitized phase current agrees exactly with the original one per transverse area. These transformations use no inverse scalar gradient and remain regular at the turn.

*E.2 Sourced constraint inverse and physical mean*

The background chart can be chosen near the construction slices on each compact slab. For example, near a timelike graph $s = S_i(t)$, set $\tau = t + A_i(t)[s - S_i(t)]$. The orthogonality condition $n_i\tau = 0$ fixes $A_i = -n_i^t/(n_i^s - n_i^t\dot{S}_i)$; the denominator is nonzero near the neutral wall, and $A_i = O_J(z)$. Use disjoint smooth radial cutoffs to extend these two first jets into the collar. The resulting gradient remains timelike for sufficiently small charge, because it is a small smooth perturbation of the neutral time function. Its cuts are new and orthogonal at both walls. Straighten the walls along this foliation and use the positive slice-metric ratio to define an isothermal radial coordinate, retaining its width. This gives a smooth chart tending to the neutral one, with the background scalar's clock derivative still $O_J(z)$. It imposes no fixed-old-cap condition.

In the spatial chart $\gamma = e^{2\lambda}\bar{\gamma}$, set $\bar{\gamma} = \operatorname{diag}(q^2, 1)$ in the mean, $\bar{p} = \Pi/q$, $\bar{A}^{\rho\rho} = A/q^2$,

$\bar{A}^{xx} = -A$. True $\Pi$ is a density per $d\rho$. At a fixed time let $r = q_{\mathrm{bg}}\rho$, so $\partial_r$ holds that time fixed. The divided scalar operator is

$$\begin{aligned} \bar{L}_\xi &= -\partial_r^2 + \xi^2 + \bar{W}, \\ \bar{W} &= e^{-2\lambda}\left(2A^2 + \tfrac{1}{2}G^{AB}\bar{p}_A\bar{p}_B\right) + e^{2\lambda}(\kappa^2/2 - B). \end{aligned} \tag{S12a}$$

All coefficients are background fields on the chosen slice; $B$ is the reduced potential in (S10), and $\bar{p}_A = \Pi_A/q$. Thus the first bracket equals $Q$ in (S13), and this $\bar{W}$ is the potential used in Appendix G.1. The divided metric block is

$$\mathcal{A}(w,\zeta) = (\bar{L}_\xi w - C\zeta',\ -\zeta'' + \xi^2\zeta + D_c w), \quad C = 2e^{-2\lambda}A, \quad D_c = e^{2\lambda}\kappa'. \tag{S12}$$

Its domain is pressure Robin for $w$ and Dirichlet for $\zeta$. Matter, clock-profile, width, density and boost variations are loads, not deleted fields. In the physical mean, write $a_m = \mathrm{avg}_r(a_m) + \zeta'$, with $2\,\mathrm{avg}_r(a_m) = \delta P_q$. This is the unique Dirichlet primitive at the supplied canonical mean. It supplies the actual homogeneous block; auxiliary continuation to $\xi = 0$ alone would not fix it.

For reference, the exact mean scalar and momentum constraints are

$$\begin{aligned} \mathcal{F} &= -2\lambda_{\rho\rho}/q^2 - G_{AB}X^A_\rho X^B_\rho/(2q^2) - e^{-2\lambda}Q + e^{2\lambda}(\kappa^2/2 - B), \\ \mathcal{M} &= A_\rho - e^{2\lambda}\kappa_\rho/2 - \Pi_A X^A_\rho/(2q), \\ Q &= 2A^2 + G^{AB}\Pi_A\Pi_B/(2q^2), \qquad P_q = 2\int_0^1 A\,d\rho. \end{aligned} \tag{S13}$$

Their full differentials, together with the actual wall energy row, define the source map. In particular $\delta\bar{p} = \delta\Pi/q - \bar{p}\delta q/q$; width transport is not discarded. The wall sources use $\delta Z_i = 2p_i^T\mathsf{H}_i^{-1}\delta p_i - p_i^T\mathsf{H}_i^{-1}\mathsf{H}_{i,A}u_i^A\mathsf{H}_i^{-1}p_i$, $\delta P_i = P_{i,A}u_i^A + P_{i,r}v_i^r$. The wall energy load is $e^{\lambda_i}\delta P_i/2 + e^{-\lambda_i}\delta Z_i/4$, with the explicit width and boost terms from differentiating the actual normal. Thus no trace of an input $L^2$ momentum is required.

Here is the anchor inverse proof. Write $w = \alpha y$, $\alpha = a^3e^{2ht}$, $B_c = \alpha^2$, $M_c = 10h^2e^{2ht}$, $K_c = \xi^2 + M_c$. The homogeneous block becomes

$$\begin{aligned} -(B_c y')' + B_c K_c y + h\zeta' = 0, \qquad -\zeta'' + \xi^2\zeta - \frac{M_c}{4h}B_c' y = 0, \\ y'|_{a,b} = 0, \qquad \zeta|_{a,b} = 0. \end{aligned} \tag{S14}$$

The Neumann row follows from the actual lapse pressure Robin law. Define $V_c = B_c y' - h\zeta$, $a_c = M_c/(4h)$, $\eta_c = \zeta + a_c V_c/K_c$, $q_c = -hK_c\eta_c/d_1$, with $d_1 = \xi^2 + 3M_c/4$, $d_2 = \xi^2 + M_c/4$, $d_3 = 3M_c^2/(16d_1)$. Differentiating the first row gives $V_c' = K_c B_c y$, and the transformed Dirichlet system is

$$-V_c'' + (B_c'/B_c)V_c' + d_1(V_c - q_c) = 0, \qquad -q_c'' + d_2 q_c - d_3 V_c = 0. \tag{S15}$$

Since $d_2 - d_3 = \xi^2 K_c/d_1 \geq 0$, take the common positive barrier $F = 1 + \cosh(c\ell/2) - \cosh(c(r - \ell/2))$, $c > \|B_c'/B_c\|_\infty$. Both row residuals are strictly

positive. The largest positive maximum of $V_c/F, q_c/F$, or of their negatives, contradicts (S15); hence the kernel is zero. The regular mixed boundary problem has Fredholm index zero, so this is also existence and bounded invertibility. It includes auxiliary zero frequency. The mean primitive and supplied $P_q$ convert it into the physical mean inverse.

For small charge use the actual positive lapse $\alpha_\delta$, so $w = \alpha_\delta y$ has the same homogeneous Neumann/Dirichlet domain after continuous source lifts. On a preliminary neighborhood choose a high-frequency cutoff from the coupled weak-form cross bounds. On the remaining compact $(\xi, t)$ band the anchor inverse has a uniform bound. Shrink charge once so its operator-norm perturbation times that bound is less than one half. This proves the full source inverse without an omitted intermediate band or a circular cutoff choice.

*E.3 Adjoint, clock and width equations*

On the Robin/Dirichlet domains the adjoint of (S12) is $\mathcal{A}^*(f, Z) = (\bar{L}_\xi f + D_c Z, -Z'' + \xi^2 Z + (Cf)')$. The cross boundary pairing vanishes because $\zeta = Z = 0$; the scalar Green pairing uses the same pressure Robin row. At homogeneous frequency this is exactly equation (C5), with $Z = -\xi^r$, and the retained width output. The boost transformation is geometric: $\delta_g \eta = -(\partial_\nu + p_{\text{wall}}/2)f$, because the actual orthogonal wall acceleration is $-p_{\text{wall}}/2$. Arbitrary Neumann time shifts would use the wrong domain.

Before clock exchange the metric one-form is $-\int \kappa \delta v + 2\int A \delta q$. The exact identity $-\kappa\delta v = v\delta\kappa - \delta(\kappa v)$ gives equation (C4), with $\kappa_{\text{bg}}(t, \rho)$ prescribed independently of variable $q$. The Hamiltonian clock term is $-\int v\dot{\kappa}_{\text{bg}}$. If instead the profile were held at $r = q\rho$, an additional width transport term would be necessary; that convention is not mixed into this proof.

The post-clock canonical one-form and generator are

$$\begin{aligned} \Theta &= P_q\,\delta q + \int v\,\delta\kappa + \int \Pi_A\,\delta X^A + \sum_i p_i\,\delta\psi_i - \delta F_{\text{clock}}, \\ v &= qe^{2\lambda}, \quad P_q = 2\int_0^1 A\,d\rho, \quad F_{\text{clock}} = \int \kappa v. \end{aligned} \tag{C4}$$

On $\delta\kappa = 0$, new orthogonal cuts have $\delta\eta_i = 0$. The corner symplectic pair then has zero pullback in this polarization, while $\delta P_q \wedge \delta q$ remains. The Hamiltonian includes $-\int v\partial_t\kappa_{\text{bg}}$; the endpoint generator is retained. This is the chart in which (C3) was computed. It is not a fixed-old-cap equivalence theorem.

Retain the mean augmentation $\tau(2\int A - P_q)$ in the canonical action. Varying $P_q$ gives $\tau = \dot{q}$. The $A$ row gives $\tau = q[2\alpha e^{-2\lambda}A + \partial_\rho\beta^\rho]$. Its mean is the physical width-velocity equation, and varying $q$ gives $\dot{P}_q = -\partial_q H_{\text{new}}$ at fixed true densities, fixed canonical fields, multipliers and prescribed clock. Both are retained. The augmentation differs from the original action by $(P_q - 2\int A)(\dot{q} - \tau)$, so it is exact once its own algebraic and canonical rows hold.

*E.4 Smooth growing solution: the actual domain correspondence*

Let $F, G, d, \eta_i^{\mathrm{in}}$ be the spatial tracefree metric, off-diagonal metric, curvature and boost variations of the smooth growing solution, after the invertible field reduction and wall straightening. At each time let $\zeta$ be a proper normal cut deformation and $\xi^r$ the spatial generator, with $\xi^r = 0$ at both ends. For $C = 2e^{-2\lambda}A$, $D_c = e^{2\lambda}\kappa'$, the simultaneous gauge problem is

$$\begin{aligned} \bar{L}_0\zeta - D_c\xi^r &= e^{2\lambda}d, & -\partial_r^2\xi^r - (C\zeta)' &= -F', \\ \mathcal{B}_R\zeta|_i &= \eta_i^{\mathrm{in}}, & \xi^r|_i &= 0, \\ \frac{\delta q}{q} &= \mathrm{avg}_r(C\zeta - F), & (\xi^x)' &= G. \end{aligned} \tag{C5}$$

Here $\bar{L}_0$ is the zero-frequency operator defined in (S12a), with the potential and fixed-time coordinate conventions stated there. The boundary operator is $\mathcal{B}_R = \partial_\nu + p_{\mathrm{wall}}/2$ in the physical metric, equivalently $\partial_{\bar{\nu}} + e^{\lambda}p_{\mathrm{wall}}/2$ after conformal rescaling. The last integration constant fixes the tangential origin. Crucially, $\delta q$ is an output, not an additional condition $\delta q = 0$.

The differential pair in (C5) is the adjoint of the actual coupled metric constraint block, with its pressure Robin/Dirichlet domain. Appendix E proves that block is invertible at the neutral anchor by a cooperative barrier and continues it on a common domain for sufficiently small charge. The physical mean and its sourced adjoint include the retained canonical mean row. Therefore smooth inputs give a unique smooth $\zeta, \xi^r$ on a fixed slab, with no division by Fourier frequency or scalar gradient. At zero charge and in a sufficiently small neighborhood the inverse is bounded; a normalized growing input gives $\zeta, \xi^r = O_J(\epsilon e^{\gamma_g t})$.

Pull all fields, densities and wall variables with this transformation. Compute $\Pi, p_i, P_q$ by their original velocity–momentum relations, including (C4); the width is not chosen independently afterward. Covariance and the full boundary action imply that the transformed history satisfies the original linear equations, and the converse variational reconstruction gives the canonical equations. The exact mean augmentation differs from the unaugmented action by $(P_q - 2\int A)(\dot{q} - \tau)$, so it adds no residual equation once the mean identity and $\tau = \dot{q}$ hold. Both width equations are recovered.

Compute $\Pi, p_i, P_q$ from the original canonical relations in this representative. In particular do not impose $P_q = 0$ or freeze $\bar{p}$ when the width changes. Equations (S10)–(S13), the carried joint/cap terms and the Hessian Noether identity imply stationarity of the complete constrained action. The history is in the homogeneous untwisted class because neither its input nor its radial/time gauge map creates a tangential twist. Both original charge variations are zero and remain so by Noether conservation between the new cuts. At zero charge the neutral phase continuation is used; for nonzero streams choose a sufficiently small perturbation on the finite slab to preserve the timelike orientation.

The representation is a local smooth map on each fixed slab. It does not impose old physical caps in addition to the new orthogonal cuts. Its near-anchor slice readout

satisfies manuscript (C7), because a radial coordinate term vanishes at the minimum and a time-cut change multiplies the branch derivative $O_J(z)$. The bounded neutral-limit inverse excludes a singular $1/z$ clock shift. This proves the common-chart part of Theorem A as a deduction from the actual homogeneous adjoint and full action correspondence, rather than by intersecting unrelated existence assertions.

## Appendix F. Full variational reconstruction and two compatibility levels

### *F.1 Necessity and sufficiency of the sourced construction*

Let $y$ be eliminated metric data, $x$ retained scalar/wall/width data, and $g$ the fixed gauge coordinates. The full linear constraints are $C_y y + C_x x = 0$, so $y = Tx = -C_y^{-1} C_x x$. A stationary section history gives $R_x + T^* R_y = 0$, not $R_y = 0$ by itself. Solve the full Green-adjoint multiplier problem $\Lambda = -C_y^{-*} R_y$. Its action contribution cancels $R_y$, and section stationarity cancels $R_x$. The physical traces of these multipliers are the same wall lapse and shift traces fixed by GHY cancellation; they are not independent forces. The mean includes its finite multiplier $\tau$ and both width equations.

The complete Hessian satisfies $B_2(Y, \mathcal{P}W) = 0$ off shell in $Y$, since the background solves all bulk, wall and endpoint equations and the original moving-domain action is diffeomorphism invariant. The sourced gauge adjoint spans the remaining curvature, spatial representative and boost tests. Hence their residuals vanish as well. A smooth normal graph test can be absorbed into a bulk vector field with the opposite wall embedding pull; the same identity then proves the normal-embedding equation. In the mean, width is a retained test, not a gauge slot. This is the converse that a homogeneous Green identity by itself would miss.

For energy histories define the original weak functional from the full quadratic canonical action, moving time derivatives to smooth compact-time tests and leaving spatial gradients in Green form. Instantaneous reconstruction has $w \in H^2$, tracefree momenta in $H^1$, and lapse/shift in $H^2$ from bulk $H^1$ fields, $L^2$ momenta and finite wall data. Pressure loads need no bulk momentum trace. A smooth full test splits into geometric gauge, multiplier, raw constraint-normal and section-tangent tests. The corresponding identity of full and section weak functionals extends by density on these unknown spaces. Normal-graph work is defined by the same smooth-test Noether identity. This proves weak necessity and sufficiency in the declared reconstructed class, not unrestricted gauge-space uniqueness.

### *F.2 Exact original conormal and graph domain*

Let $\mathcal{E}_m$ have bulk configurations $u \in H^{m+1}(0,1)$, true momenta $p = \delta\Pi \in H^m$, and every independent finite wall/mean slot. Set $M = \alpha e^{-2\lambda} G^{-1}/q$, $C_b = \alpha G/q$, $\beta = \beta^\rho$. The actual Hamiltonian rows take the form

$$u_t = Mp + \beta u_\rho + R_0 Z, \qquad p_t = \partial_\rho (C_b u_\rho + \beta p) + R_1 Z, \tag{S16}$$

where the full reconstructed remainders satisfy $R_0 : \mathcal{E}_m \to H^{m+1}$, $R_1 : \mathcal{E}_m \to H^m$, smoothly in time. These contain the actual metric, width, source, clock, shift and

coefficient terms. They are not a discarded lower-order model. On the small-charge finite slab, $\alpha^2 e^{-2\lambda}/q^2 - \beta^2$ has a positive lower bound; at orthogonal wall cuts $\beta_i = 0$.

With wall $b = e^\lambda$, define $L_A = \mathsf{H}^{-1}\mathsf{H}_{,A}\mathsf{H}^{-1}$. The force in canonical variables is $F_A = P_A - p_i^T L_A p_i/(2b^2)$. Its variation is

$$\delta F_A = P_{AB}u_i^B + P_{A,r}v_i^r - b^{-2}p_i^T L_A \widehat{p}_i + b^{-2}(p_i^T L_A p_i)w_i - \frac{1}{2b^2}p_i^T (L_A)_{,B}p_i u_i^B. \tag{S17}$$

For outward endpoint signs $\epsilon_a = -1, \epsilon_b = +1$, the original scalar/fiber Israel conormal is

$$\mathcal{B}_t Z|_i = \epsilon_i u_{\rho,i} - \epsilon_i(w_i + \delta q/q)X_{\rho,i} + qe^{\lambda_i}G^{-1}\delta F_i = 0. \tag{S18}$$

The spatial kinetic wall term has zero first variation about the homogeneous background, not zero Hessian. After wall velocity–momentum restoration, (S17) needs no bulk velocity or momentum endpoint trace. Thus $\mathcal{B}_t = T_u + K_t$, $T_u Z = (\epsilon_i u_{\rho,i})_i$, and $K_t : \mathcal{E}_0 \to \mathbb{R}^4$ is bounded.

The maximal energy-valued generator domain is exactly $D_t = \{Z \in \mathcal{E}_1 : \mathcal{B}_t Z = 0\}$. Indeed differentiating the first interior row of (S16) and combining with the second yields a system for $u_{\rho\rho}, p_\rho$ with principal matrix $\left(\begin{smallmatrix} \beta I & M \\ C_b & \beta I \end{smallmatrix}\right)$. Subluminality makes it invertible; the remaining terms are $L^2$. Hence these derivatives are $L^2$. The remaining endpoint functional cannot be a bulk $L^2$ output, so (S18) vanishes. The converse follows by integration by parts. This also gives graph-norm equivalence to $\mathcal{E}_1$.

*F.2.1 The reconstructed interior remainders* For clarity write $X_0, \Pi_0$ for the canonical background in this subsection, $w = \delta\lambda$, $h_q = \delta q/q$, $a_\alpha = \delta\alpha/\alpha$, $b^\rho = \delta\beta^\rho$, $b^x = \delta\beta^x$. The last three quantities are outputs of the sourced multiplier problem, not independent perturbations. The background is homogeneous in $x$, with zero tangential shift. Linearizing the full scalar Hamilton equations of (S10) in true per-$\rho$ momenta gives the following explicit operators in (S16):

$$\begin{aligned} R_0 Z &= M\Pi_0(a_\alpha - 2w - h_q) + b^\rho X_{0,\rho}, \\ R_1 Z &= \partial_\rho[C_b(a_\alpha - h_q)X_{0,\rho} + b^\rho \Pi_0] - \xi^2 \alpha q G u + i\xi b^x \Pi_0 \\ &\quad - \alpha q e^{2\lambda}\left[B_{,AB}u^B + (a_\alpha + h_q + 2w)B_{,A}\right]. \end{aligned} \tag{R1}$$

The target-space matrix $G$ is constant. In the physical mean $\xi = 0$, $b^x = 0$ and $h_q$ is retained; at nonzero frequency $h_q = 0$. All quantities multiplying the linear responses in (R1) are background coefficients. In particular no $\dot{q}\,p/q$ term appears: such a term would belong to time differentiation of $\Pi/q$, not the true canonical density used in (S16).

Here are explicit volume and endpoint operators defining the responses in (R1). Use $r = q_{\mathrm{bg}}\rho$ at fixed time only, a prime for $\partial_r$, $\bar{p}_0 = \Pi_0/q$, and $\widehat{p} = p/q$ at nonzero frequency. For $\xi \neq 0$, the metric outputs $w, a, b$ satisfy

$$
\begin{aligned}
2\bar{L}_\xi w - 4e^{-2\lambda}Aa &= f_H,\\
a' + i\xi b - D_c w &= f_r, \quad b' - i\xi a = f_x,\\
f_H &= GX_0' \cdot u' + e^{-2\lambda}G^{-1}\bar{p}_0 \cdot \widehat{p} + e^{2\lambda}B_{,A}u^A,\\
f_r &= (\widehat{p}_A X_0^{A\prime} + \bar{p}_{0,A}u^{A\prime})/2,\\
f_x &= i\xi\bar{p}_{0,A}u^A/2,\\
\bar{B}_i w &= g_i, \quad b_i = -\epsilon_i i\xi p_i \cdot v_i/2,\\
\bar{B}_i &= \epsilon_i\partial_r + e^{\lambda_i}p_{\mathrm{wall},i}/2,\\
g_i &= e^{\lambda_i}\delta P_i/2 + e^{-\lambda_i}\delta Z_i/4.
\end{aligned}
\tag{R2}
$$

The contractions in the first line defining $f_H$ use $G_{AB}$ and $G^{AB}$, respectively. The wall differentials $\delta P_i, \delta Z_i$ are those stated after (S13). The exact mean rows (S13), differentiated at fixed true density, replace (R2) when $\xi = 0$; they include the finite condition $2\int a = \delta P_q$, not an additional boundary condition on $a$.

The lapse and shift outputs are fixed by pressure and gauge preservation. At nonzero frequency they are

$$
\begin{aligned}
\bar{L}_\xi\delta\alpha - e^{2\lambda}\kappa' b^r &= -(\delta\bar{L})\alpha + 2w\bar{L}\alpha,\\
(b^r)' - i\xi b^x + C\delta\alpha &= -2\alpha e^{-2\lambda}(a - 2Aw),\\
(b^x)' + i\xi b^r &= -2\alpha e^{-2\lambda}b, \qquad b_i^r = 0,\\
\bar{B}_i\delta\alpha &= \alpha_i[e^{\lambda_i}\delta P_i/2 - e^{-\lambda_i}\delta Z_i/4 + e^{\lambda_i}\rho_{\mathrm{wall},i}w_i/2],\\
\delta\bar{L} &= e^{-2\lambda}[4Aa + G^{-1}\bar{p}_0 \cdot \widehat{p} - 2Qw]\\
&\quad + e^{2\lambda}[2(\kappa^2/2 - B)w - B_{,A}u^A].
\end{aligned}
\tag{R3}
$$

Here $b^r = q_{\mathrm{bg}}b^\rho$. The exact fixed-$\rho$ curvature equation is $\bar{L}\alpha = -e^{2\lambda}\dot{\kappa} + e^{2\lambda}\beta^\rho\kappa_\rho$; it fixes the time/shift convention in (R3). For the physical mean use $\widehat{p} = p/q - h_q\bar{p}_0$, add $2h_q\alpha_{\rho\rho}/q^2$ to $(\delta\bar{L})\alpha$, and add $-\alpha_i e^{\lambda_i}p_{\mathrm{wall},i}h_q/2$ to its boundary source. The first shear row is then $\partial_\rho b^\rho + C\delta\alpha - \dot{h}_q = -2\alpha e^{-2\lambda}(a - 2Aw)$. Its integral determines $\dot{h}_q$, as in E.3; the second width equation is retained there. These formulas are evaluated at new orthogonal cuts, so boost variations and the normal shift trace vanish after full variation.

The independent finite wall rows are equally explicit. Write $v_i = \delta\psi_i$, $\pi_i = \delta p_i$, $p_{0,i}$ for the background wall momentum. Their linear equations are

$$
\begin{aligned}
\dot{v}_i &= \alpha_i e^{-\lambda_i}\Big[\mathsf{H}_i^{-1}\pi_i - \mathsf{H}_i^{-1}\mathsf{H}_{i,A}\mathsf{H}_i^{-1}p_{0,i}u_i^A + \mathsf{H}_i^{-1}p_{0,i}(a_{\alpha,i} - w_i)\Big],\\
\dot{\pi}_{i,r} &= -\xi^2\alpha_i e^{-\lambda_i}\mathsf{H}_{i,rs}v_i^s + i\xi b_i^x p_{0,i,r}\\
&\quad - \alpha_i e^{\lambda_i}[P_{i,rs}v_i^s + P_{i,rA}u_i^A + P_{i,r}(a_{\alpha,i} + w_i)].
\end{aligned}
\tag{R6}
$$

They use only configuration traces, finite data and reconstructed multiplier traces, all bounded on $\mathcal{E}_0$. Each phase momentum remains independent. For the physical mean, the two finite width equations in Appendix E.3 are generated by the following full post-clock Hamiltonian at fixed true density and prescribed $\kappa(t,\rho)$:

$$
\begin{aligned}
H_{\mathrm{new}} = \int_0^1 d\rho \Big[ & -\frac{2\alpha_\rho \lambda_\rho}{q} + \frac{\alpha G_{AB} X_\rho^A X_\rho^B}{2q} + 2\alpha q e^{-2\lambda} A^2 + \frac{\alpha e^{-2\lambda} G^{AB} \Pi_A \Pi_B}{2q} \\
& - \alpha q e^{2\lambda}(\kappa^2/2 - B) + \beta^\rho(-2qA_\rho + q e^{2\lambda}\kappa_\rho + \Pi_A X_\rho^A) - q e^{2\lambda}\dot{\kappa} \Big] \\
& + \sum_i \alpha_i [e^{\lambda_i} P_i + e^{-\lambda_i} Z_i/2].
\end{aligned}
\tag{R7}
$$

Together with $2\int A = P_q$ and the multiplier augmentation in Appendix E.3, (R7) fixes both finite linear width rows by ordinary differentiation; the wall Hamiltonian has no direct $q$ dependence at fixed tangential normalization. In particular $\dot{P}_q = -\partial_q H_{\mathrm{new}}$ holds before eliminating the response and at fixed multipliers, $\lambda, A, \Pi, X$ and wall pairs. Indirect metric response is retained in its full linear variation. No partial derivative at fixed $qe^{2\lambda}$ is substituted.

For completeness the direct mean source map, with $\bar{L}_0 = -q^{-2}\partial_\rho^2 + \bar{W}$ and $a = \delta A$, is

$$
\begin{aligned}
2\bar{L}_0 w - 4e^{-2\lambda} A a &= q^{-2} G_{AB} X_\rho^A u_\rho^B + q^{-2} e^{-2\lambda} G^{AB} \Pi_A p_B + e^{2\lambda} B_{,A} u^A \\
&\quad - h_q[4\lambda_{\rho\rho}/q^2 + G_{AB} X_\rho^A X_\rho^B/q^2 + e^{-2\lambda} G^{AB} \Pi_A \Pi_B/q^2], \\
a_\rho - e^{2\lambda}\kappa_\rho w &= [(p_A/q - h_q \bar{p}_{0,A}) X_\rho^A + \bar{p}_{0,A} u_\rho^A]/2, \\
2\int_0^1 a\, d\rho &= \delta P_q, \\
\bar{B}_i w &= \epsilon_i h_q \lambda_{\rho,i}/q + e^{\lambda_i}\delta P_i/2 + e^{-\lambda_i}\delta Z_i/4.
\end{aligned}
\tag{R8}
$$

The clock variation is zero on the chosen section. Formula (R8) displays all direct width and true-density loads and supplies the finite mean input that the auxiliary zero-frequency limit does not determine.

For fixed frequency, (R2) has volume loads in $H^m$, and its endpoint loads require only $H^1$ field traces and the finite canonical data. Elimination can initially give a distributional derivative of $f_x/(i\xi)$; recovering the original first-order rows yields $a, b \in H^{m+1}$ and then $w \in H^{m+2}$. In (R3) these responses give volume loads in $H^m$, pressure loads with no momentum endpoint trace, and consequently $\delta\alpha, b^\rho, b^x \in H^{m+2}$. The mean primitive and its finite loads have the same orders. The inverse of E.2 and its Green adjoint are bounded on these common mixed Sobolev spaces. Differentiating their defining equations in the time parameter inserts only smooth coefficient derivatives and the same bounded inverses. Thus these bounds hold uniformly on the selected compact slab, with every fixed finite number of explicit time derivatives. Substitution in the actual formulas (R1) proves the two asserted mapping properties after (S16). No general evolution theorem is used for this step.

### *F.3 Two domain lifts and classical propagation*

*F.3.1 Corrected energy and the finite momentum defect* The auxiliary norm is constructed from the same full quadratic Hamiltonian. The configurations $Q$, momenta $P$ and spaces $\mathcal{Q}_0, \mathcal{V}$ are defined in (E3) below. The projection retains all finite wall/width momenta:

$$\begin{aligned} H_2 &\geq c(\|u'\|_2^2 + \|P\|^2) - C(\|Q\|_{\mathcal{Q}_0}^2 + \|P_N P\|^2), \\ \mathcal{E}_{\mathrm{aug}} &= H_2 + A\|Q\|_{\mathcal{Q}_0}^2 + B\|P_N P\|^2 \asymp \|Q\|_{\mathcal{V}}^2 + \|P\|^2. \end{aligned} \tag{C8}$$

Its range lies in the configuration form space, so differentiating its contribution uses smooth tests rather than momentum traces. The weak construction and both graph-domain lifts use this corrected norm. Canonical momentum positivity is established separately in Appendix G.

**Lemma — positive auxiliary evolution norm.** On a fixed compact slab and a sufficiently small charged neighborhood, the complete quadratic section Hamiltonian $H_2(t,\xi)$ has a positive equivalent evolution norm. It controls bulk $H^1$ configurations, true $L^2$ momenta, both independent wall pairs and, separately at physical zero frequency, the retained $q, P_q$. Constants may depend on the slab. This lemma supplies the norm used by the weak construction and domain lifts; it does not assert positivity of $H_2$ or replace its dynamics.

Write $u = \delta X$, $v_i = \delta\psi_i$, $p = \delta\Pi/q_{\mathrm{bg}}$ at a fixed time, and $\pi_i = \delta p_i$. Time derivatives always use true $\Pi$ at fixed rho. The positive principal form from the complete action is

$$\begin{aligned} E_{\mathrm{pr}} = \frac{1}{2}\int \sqrt{\bar\gamma}\,\alpha \left[e^{-2\lambda} G^{AB} p_A \overline{p_B} + G_{AB}\bar D u^A \cdot \overline{\bar D u^B}\right] \\ + \sum_i \frac{\alpha_i e^{-\lambda_i}}{2}\left[\pi_i^* \mathsf{H}_i^{-1}\pi_i + \xi^2 v_i^* \mathsf{H}_i v_i\right]. \end{aligned} \tag{E1}$$

The two-dimensional identity $\sqrt{\gamma}\gamma^{mn} = \sqrt{\bar\gamma}\bar\gamma^{mn}$ makes the bulk gradient exactly independent of conformal response $w$. Its expansion creates no uncontrolled $w|u'|^2$. The homogeneous background has $\psi_{i,x} = 0$, so the quadratic wall spatial principal term is positive, with wall fields independent of bulk traces.

For large $|\xi|$, the actual sourced constraints and parameter trace inequality give

$$\begin{gathered} \|w'\|_2 + |\xi|\|w\|_2 + \|\delta\bar A\|_2 \leq C|\xi|^{-1/2} E_{\mathrm{pr}}^{1/2}, \\ \|w\|_2 \leq C|\xi|^{-3/2} E_{\mathrm{pr}}^{1/2}, \qquad \sum_i |w_i|^2 \leq C|\xi|^{-2} E_{\mathrm{pr}}, \qquad \sum_i |u_i|^2 \leq C|\xi|^{-1} E_{\mathrm{pr}}. \end{gathered} \tag{E2}$$

Robin lifts cost $C|g|/\sqrt{|\xi|}$ in parameter energy, while Dirichlet traction lifts cost $C\sqrt{|\xi|}|B|$. Actual constraint loads use field traces and finite wall data, with no normal derivative or momentum endpoint trace. These give the worst $|\xi|^{-1/2}$ response in (E2).

Besides (E1) and the nonnegative tracefree-momentum square, the full Hamiltonian has bounded bulk $u^2, uw, w^2, wp, w\delta\bar A$, actual transport $\beta^\rho \delta\Pi\, u_\rho$, and bounded wall $u_i^2, v_i^2, u_i v_i, u_i w_i, v_i w_i, w_i^2, \pi_i u_i, \pi_i w_i$. This exhaustive list follows by expanding the scalar constraint, conformally invariant gradient, wall Hamiltonian and momentum row. Lapse/shift variations multiply solved linear constraints; clock, endpoint and joint terms remain. The worst nontransport term $\pi_i u_i$ is $O(E_{\mathrm{pr}}/\sqrt{|\xi|})$. Transport is absorbed using the actual small-charge bound $|\beta|_\gamma/\alpha < 1/4$ on a preliminary neighborhood; frequency alone does not make it small. One cutoff $K_J$ therefore gives $cE_{\mathrm{pr}} \leq H_2 \leq CE_{\mathrm{pr}}$ above it. Explicit time differentiation at fixed true canonical inputs differentiates smooth

coefficients, common-domain inverses and load lifts. The same parameter bounds give $|\partial_t H_2| \le C E_{\mathrm{pr}}$, retaining the background width derivative.

On the bounded band use fixed reference spaces

$$\begin{aligned} \mathcal{Q}_0 &= L^2(\text{bulk}) \oplus \{\text{finite configurations}\}, \\ \mathcal{V} &= H^1(\text{bulk}) \oplus \{\text{finite configurations}\}, \\ \mathcal{P} &= L^2(\text{true momenta}) \oplus \{\text{finite momenta}\}. \end{aligned} \tag{E3}$$

Positive smooth density congruences have bounded inverses. The auxiliary nonzero-frequency limit and physical mean are distinct families: only the latter adds $q, P_q$ and its actual mean loads. Include both families in the following compact-parameter argument.

At bounded frequency, derivative/momentum loads give $H^2$ scalar and $H^1$ tracefree responses; their embeddings into the $L^2$ and trace spaces occurring in the form are compact. Configuration endpoint traces are finite-rank maps on $H^1$, not on $L^2$. Uniform Ehrling/Young estimates absorb arbitrary small derivative/momentum portions, leaving a $\mathcal{Q}_0$ configuration defect. The remaining pure momentum term is a continuous compact self-adjoint family with finite wall/width blocks.

Choose one fixed smooth finite-rank orthogonal momentum projection $P_N$, identity on every finite wall/width momentum and with bulk range spanned by smooth compactly supported radial functions. Strong approximation by such projections converges uniformly in operator norm on a compact family of compact operators: choose finitely many operator-norm balls and the finite union of their approximating ranges. Include the separate physical-mean family in this union. Consequently

$$\begin{aligned} |\langle K_{\mathrm{c}} P, P\rangle| &\le \varepsilon \|P\|_{\mathcal{P}}^2 + C_\varepsilon \|P_N P\|_{\mathcal{P}}^2, \\ H_2 &\ge c(\|u'\|_2^2 + \|P\|_{\mathcal{P}}^2) - C(\|Q\|_{\mathcal{Q}_0}^2 + \|P_N P\|_{\mathcal{P}}^2). \end{aligned} \tag{E4}$$

This Gårding estimate makes no assertion about the canonical defect's sign. Fixed sufficiently large $A, B > 0$ give

$$E_{\mathrm{aug}} = H_2 + A\|Q\|_{\mathcal{Q}_0}^2 + B\|P_N P\|_{\mathcal{P}}^2 \asymp \|Q\|_{\mathcal{V}}^2 + \|P\|_{\mathcal{P}}^2. \tag{E5}$$

Dynamics remain those of $H_2$. For compatible smooth solutions, the full natural-boundary variation gives $\dot{Q} = (H_2)_P \in \mathcal{Q}_0$, $\dot{P} = -(H_2)_Q \in \mathcal{V}^*$, $dH_2/dt = \partial_t H_2$, and $\|(H_2)_P\|_{\mathcal{Q}_0} + \|(H_2)_Q\|_{\mathcal{V}^*} \le C(\|Q\|_{\mathcal{V}} + \|P\|_{\mathcal{P}})$. Fixed positive component identifications put the projection's range in $\mathcal{V}$. Hence

$$\frac{d}{dt}\|P_N P\|^2 = -2\,\mathrm{Re}\langle (H_2)_Q, P_N^* P_N P\rangle_{\mathcal{V}^*,\mathcal{V}}, \qquad |\dot{E}_{\mathrm{aug}}| \le C E_{\mathrm{aug}}. \tag{E6}$$

No bulk $\dot{P}$ in $L^2$, endpoint momentum derivative or Robin operator-domain condition on the projection basis is used. Finite wall/width components use independent tests. The configuration correction uses only $\dot{Q} \in \mathcal{Q}_0$.

A time-independent smooth cutoff equal to one below $K_J$ and zero above $2K_J$ joins the corrections to the high-frequency $H_2$. In the transition band $H_2$ is already coercive. The joined reference norm includes $(1+\xi^2)\|u\|_2^2$ and $(1+\xi^2)|v_i|^2$; physical mean variables remain separate. This is a section-variable norm statement, not infrared-uniform reconstruction of the original coordinate gauge.

For existence, use finite canonical Galerkin systems whose configuration spaces contain the fixed range of $P_N$ and all finite wall/width slots. Their symmetric Hamiltonian gives the same energy identity and all correction tests remain in the trial space. Uniform bounds yield a weak variational limit; backward adjoint/symplectic uniqueness identifies its energy history and strong continuity. The corrected norm is thus available during construction, rather than inferred by assuming smooth existence. The frozen autonomous construction supplies the group bounds used below. The actual conormal and this smoothly varying norm support both graph-domain steps under the stable-generator regularity theorem [13].

*F.3.2 First domain lift* Choose a smooth configuration-only endpoint derivative lift $L$, with zero endpoint values, $T_u L = I$, and arbitrarily small $\mathcal{E}_0$ norm. Fix it once so $\|K_t L\| < 1/2$ on $J$. For reference time $s$, set

$$V_t = I - L(I + K_t L)^{-1}(K_t - K_s), \qquad G_t = V_t^{-1}\mathcal{G}(t)V_t - V_t^{-1}\dot{V}_t. \tag{S19}$$

The identity $\mathcal{B}_t V_t = \mathcal{B}_s$ fixes the first domain. These maps change no finite wall or width slot. Their time-generator terms are retained. Frozen autonomous weak flow and uniqueness identify its maximal generator with $D_t$. In a smoothly varying corrected energy norm it has two-sided exponential bounds. Norm-transfer factors at ordered times telescope to one slab-dependent factor, rather than one fixed equivalence constant raised to the number of factors. This gives the stable-family hypothesis in Dorroh's form of Kato's theorem [13]. The graph resolvent $\lambda - G_t : D_s \to \mathcal{E}_0$ supplies its differentiable isomorphism on the proper operator domains. The first application yields $C(\mathcal{E}_1) \cap C^1(\mathcal{E}_0)$.

The second boundary remainder can now be displayed. Put $\ell_t = V_t - I$, $\bar{\ell}_t = V_t^{-1} - I$, and let $\mathcal{G}(t)$ denote the full original generator including its finite rows. Both lifts have configuration-only smooth finite-dimensional range. The configuration row of $G_t$ is $Mp + \beta u_\rho + \mathcal{F}_t Y$, with

$$\mathcal{F}_t Y = \beta\partial_\rho(\ell_t Y)_u + R_0(V_t Y) + \{\bar{\ell}_t[\mathcal{G}(t)V_t Y - \dot{V}_t Y]\}_u - (\dot{V}_t Y)_u. \tag{R4}$$

All terms in this formula are bounded $D_s \to H^2$: (R1)–(R3) apply with $m = 1$, and every other range is smooth and finite dimensional with coefficient functionals bounded by the first graph norm. Normalize the second boundary row by the bare endpoint multiplier $M_i^{-1}$. Its exact expression is

$$\begin{aligned}\operatorname{diag}(M_i^{-1})\mathcal{B}_s G_t Y &= T_p Y + K_t^{(2)} Y,\\ (K_t^{(2)}Y)_i &= \epsilon_i M_i^{-1}[M_{\rho,i}p_i + \beta_{\rho,i}u_{\rho,i} + \partial_\rho(\mathcal{F}_t Y)_i] + M_i^{-1}(K_s G_t Y)_i,\\ T_p Y &= (\epsilon_i p_{\rho,i})_i.\end{aligned} \tag{R5}$$

Only in (R5), $p_i$ inside the square bracket denotes the bulk momentum trace at that endpoint, available on $D_s \subset \mathcal{E}_1$, not the independent wall momentum. The latter remains in the finite slots. The $\beta_i u_{\rho\rho,i}$ term vanishes by orthogonality. Each remaining trace in the square bracket is bounded by $\mathcal{E}_1$, while $K_s G_t$ is bounded there because $G_t : D_s \to \mathcal{E}_0$. Thus (R5) displays a bounded $D_s \to \mathbb{R}^4$ operator, with no hidden endpoint $p_\rho$ or $u_{\rho\rho}$ in its remainder. This normalization uses positive bare $M_i$, not an inverse of the full momentum form. Start with a small smooth momentum derivative lift $J_\eta$, and correct its first boundary row:

$$L_2 = J_\eta - L(I + K_s L)^{-1}\mathcal{B}_s J_\eta, \qquad R_t = I - L_2(I + K_t^{(2)}L_2)^{-1}(K_t^{(2)} - K_s^{(2)}). \tag{S20}$$

Then $\mathcal{B}_s L_2 = 0$, $T_p L_2 = I$. Choose $\eta$ once so the second finite matrix is invertible. This lift is bounded on the first graph space; boundedness on $\mathcal{E}_0$ is not required or asserted. Repeat the stable-generator argument there, retaining $-R_t^{-1}\dot{R}_t$. The graph norms identify the second domain with $\mathcal{E}_2$. Undoing both lifts gives manuscript (27) for precisely $\mathcal{B}_s Z_s = 0$, $\dot{\mathcal{B}}_s Z_s + \mathcal{B}_s \mathcal{G}(s) Z_s = 0$. The dot term cannot be replaced by the square of a frozen domain. Both transformations leave finite material/width freedoms intact. A compatible smooth datum can be obtained by successive derivative corrections, with no finite-slot restriction.

*F.3.3 Hypotheses at the two stable-generator applications* For the first application of [13], take base space $X = \mathcal{E}_0$, fixed dense domain $Y = D_s$ with its $\mathcal{E}_1$ norm, and the sign convention $A(t) = -G_t$ in that paper. Density follows by smooth approximation and arbitrarily small energy-cost derivative lifts; finite wall/width variables are independent. The maximal-domain calculation in F.2 and (R1)–(R3) gives the uniform graph-norm equivalence and $C^1(J; \mathcal{L}(Y, X))$ dependence. The corrected energy (E5)–(E6), frozen autonomous construction and the bounded $-V_t^{-1}\dot{V}_t$ term give a uniform two-sided exponential group bound in $|Y|_t^2 = E_{\mathrm{aug,t}}(V_t Y)$. The logarithmic derivative of this norm at fixed $Y$ is bounded uniformly on the slab. Ordered transfers therefore give a single overall norm-equivalence constant and an exponential in the slab length, rather than a constant raised to the number of factors. Laplace transformation yields the ordered resolvent stability condition (i) of [13]. For one real $\Lambda$ above that growth bound, $S_t = \Lambda - G_t : Y \to X$ is a differentiable isomorphism. It commutes with the generator on the requisite domain; its conjugation remainder in condition (ii″) is zero. This verifies the hypotheses, rather than deriving source regularity from them.

For the second application the base is $X = D_s$, the fixed domain is $Y = D_2(s)$, and the generator is $\widehat{G}_t = R_t^{-1} G_t R_t - R_t^{-1}\dot{R}_t$. The part of frozen $G_t$ on $D_s$ generates its restricted strongly continuous group, with norm $|Y|_{D,t}^2 = |Y|_t^2 + |G_t Y|_t^2$. Commutation

with its frozen group gives the same growth bound for both entries. The graph estimate makes this norm equivalent to $\mathcal{E}_1$ and smoothly time dependent on the fixed space. Formula (R5) and (S20) give smooth common-domain dependence $D_2(s) \to D_s$, with $D_2(s)$ carrying its $\mathcal{E}_2$ norm. The range of $L_2$ is smooth, $\mathcal{B}_s L_2 = 0$, and its norm into $D_s$ tends to zero with the support width. Thus the finite inverse and its derivatives are bounded on the compact slab. The $\dot{R}_t$ term is bounded on this base $D_s$; boundedness on $\mathcal{E}_0$ is not needed. The same ordered-norm argument and graph resolvent $\Lambda - \widehat{G}_t$ verify conditions (i), (ii″) and (iii) at this level. Density follows by momentum derivative corrections within the first domain. This gives $C(\mathcal{E}_2) \cap C^1(\mathcal{E}_1)$, with both compatibility conditions including $\dot{\mathcal{B}}_s$, for the same weak-unique history. No additional arbitrary-data or infrared-uniform assertion is inferred.

*F.3.4 A sufficient linear criterion for the radial turn* For a compact interior turn tube of radius $d$, with background second radial derivative of fixed sign and magnitude at least $m_T > 0$, the sufficient linear criterion

$$\sup |u_\rho| < m_T d/2, \qquad \sup |u_{\rho\rho}| < m_T/2 \tag{28}$$

gives one shifted simple radial turn in that tube. The displacement is at most $2 \sup |u_\rho|/m_T$. The evolution constant and tube constants are not numerically supplied; this is a local smallness criterion for background plus linear perturbation, not nonlinear turn persistence.

*F.4 The growing datum and simultaneous quantifiers*

The analytic growing solution and its smooth gauge representative have the radial derivatives required by $\mathcal{E}_2$. Its scalar/fiber Israel laws give $\mathcal{B}_t Z(t) = 0$. Differentiating that identity along this already smooth solution gives

$$\dot{\mathcal{B}}_t Z(t) + \mathcal{B}_t \mathcal{G}(t) Z(t) = 0. \tag{C6}$$

Thus its data satisfy both conditions (26). This does not differentiate a trace of a general energy solution. The uniqueness theorem of Section 7 identifies the transformed growing history with the compatible classical propagator. Homogeneous phase momenta remain independently conserved; the chosen history has zero variation of each charge, while the propagator and form domain retain their other allowed directions.

Fix $J$, the charge labels and a preliminary branch neighborhood. The inverse, gauge, regularity, growth and form arguments give positive neighborhoods of the neutral anchor in that same action and chart. For the form choose its high-frequency cutoff on the preliminary neighborhood before shrinking its compact band. Their finite intersection is positive. Timelike orientation for nonzero currents and the linear amplitude condition are imposed on this fixed slab. This establishes Theorem A's simultaneous applicability and states exactly why separate review receipts alone would not establish it. It does not produce an all-future common radius, a quantitative charge radius or a nonlinear instability theorem.

## Appendix G. Exact canonical momentum form and its bound

*G.1 Square completion with clock, shift and wall pressure*

In the three-dimensional Einstein frame, $X = (\sigma, \Phi)$, $G = \mathrm{diag}(12, 2)$, $\psi_i = (\chi_i, \theta_i)$, $\mathsf{H}_i = e^{2\sigma_i}\,\mathrm{diag}(1, 1/(2c_i))$. Write the spatial metric as $\gamma = e^{2\lambda}\bar{\gamma}$; in the mean $\bar{\gamma} = \mathrm{diag}(q^2, 1)$. The true canonical bulk density is $\Pi_A$ per $d\rho$, so $\bar{p}_A = \Pi_A/q$. These variables follow from the invertible reduction in Appendix E, not from adding a second model.

For a pure-momentum direction hold $X, \psi_i, q, \kappa, \eta_i$ fixed, where $\eta_i$ is the cut boost. Solve the actual sourced constraints for $w = \delta\lambda$ and $\delta\bar{A}$, retaining the prescribed $\delta P_q$ in the mean. Put

$$\begin{gathered}
\mathsf{T} = \delta\bar{A} - 2w\bar{A}, \qquad \mathsf{P} = \delta\bar{p} - 2w\bar{p}, \qquad \mathsf{R}_i = \delta p_i - w_i p_i, \\
\mathscr{K} = \mathscr{S} + \mathscr{R}, \\
\mathscr{S} = \int \sqrt{\bar{\gamma}}\,\alpha e^{-2\lambda}\left(|\mathsf{T}|^2 + \tfrac{1}{2}G^{AB}\mathsf{P}_A\overline{\mathsf{P}_B}\right) + \sum_i \frac{\alpha_i e^{-\lambda_i}}{2}\mathsf{R}_i^*\mathsf{H}_i^{-1}\mathsf{R}_i, \qquad \text{(C3)} \\
\mathscr{R} = -2\int \sqrt{\bar{\gamma}}\,\bar{\Delta}\alpha\,|w|^2 - \sum_i \frac{\alpha_i e^{\lambda_i}}{2}p_{\mathrm{wall},i}|w_i|^2.
\end{gathered}$$

Integrals use the fixed reference strip measure. The endpoint term also equals $\sum_i \partial_{\bar{\nu}}\alpha_i|w_i|^2$ by the actual pressure Robin law. Both orientations and both wall momentum vectors occur. In the mean, $\delta\bar{A} = \mathrm{diag}(a_m, -a_m)$ in an orthonormal reference basis, and $2\,\mathrm{avg}_r\, a_m = \delta P_q$. The term $\mathscr{R}$ is adverse in part; positivity comes from its bound on the solved response, not from dropping it. The clock and shift cancellation is derived below.

Use the complete post-clock action in E.3 and a pure-momentum path with $X, \psi_i, q, \kappa, \eta_i$ fixed. The original lapse/shift variations multiply the linear constraints and vanish only after those constraints are imposed. Completing squares in the scalar constraint Hamiltonian gives the bulk square in equation (C3) and $-2\alpha\bar{W}|w|^2$, where $\bar{W} = e^{-2\lambda}Q + e^{2\lambda}(\kappa^2/2 - B)$. The lapse equation is $\bar{L}\alpha = -e^{2\lambda}(\partial_t - \beta\cdot D)\kappa$. The clock term contributes $-2v\dot{\kappa}|w|^2$, and shift contributes $+2v\beta\cdot D\kappa|w|^2$. Their combination leaves exactly $-2\int\sqrt{\bar{\gamma}}\bar{\Delta}\alpha|w|^2$. Dropping clock or shift would change the form.

At a homogeneous wall let $Z_i = p_i^T\mathsf{H}_i^{-1}p_i$, $\rho_i = P_i + e^{-2\lambda_i}Z_i/2$, $p_{\mathrm{wall},i} = -P_i + e^{-2\lambda_i}Z_i/2$. Its Hamiltonian is $\alpha_i[e^{\lambda_i}P_i + e^{-\lambda_i}Z_i/2]$. Its coefficient of the path parameter squared is

$$\frac{\alpha_i e^{-\lambda_i}}{2}(\delta p_i - w_i p_i)^*\mathsf{H}_i^{-1}(\delta p_i - w_i p_i) - \frac{\alpha_i e^{\lambda_i}}{2}p_{\mathrm{wall},i}|w_i|^2. \qquad \text{(S21)}$$

This gives precisely $\mathscr{S} + \mathscr{R}$. Both independent momentum components at both walls and the physical mean $\delta P_q$ remain. The time-dependent canonical section's additional pure-momentum coefficient vanishes because all extended paired configuration rows are fixed; this is not permission to discard $q, P_q$ from the action. Corners have zero section pullback at $\delta\eta = 0$ with their generator carried. Thus the form is the physical coefficient in the specified polarization, not a volume-only Hessian or the corrected evolution norm.

*G.2 Weighted all-radial estimate at the exact anchor*

At the exact neutral anchor, put $dr = ds/a$, $\alpha = a^3$, $g = \alpha_r/\alpha$, $Q = \partial_r - g$. The solved pure-momentum response $w$ obeys both physical Robin laws, so $Qw$ is Dirichlet at the two ends. For $u = \sqrt{\alpha}Qw$, $E = \|u\|_2^2$, the full constraint/response identities give

$$\begin{aligned} \mathscr{A} &= \int \left[ |u_r|^2 + \left( \xi^2 + \frac{15}{4} a^2 H^2 + a^2 T^2 \right) |u|^2 \right] dr, \\ \mathscr{K} \geq \mathscr{S} - \frac{3}{2} E, \qquad & E \leq \frac{13}{2} x \left( \frac{21}{5} + \frac{9}{2} x \right) \mathscr{S}, \quad x = \frac{h^2}{\lambda_0}. \end{aligned} \tag{29}$$

The differentiated response is an $H^{-1}$ identity with $L^2$ forcing. It requests neither a derivative nor an endpoint trace of an input momentum and does not invert the scalar ground crossing.

The exact-background gates are $0 < h^2 < 1/3000$, $0 < a^{-1} < 8$, $m^2\Phi_0^2 < 1$. The first integral gives $H^2 > 11/12$ and hence $\mathscr{A} \geq \lambda_0 E$ with $\lambda_0 = 55/1024$. Consequently

$$\mathscr{K} \geq (1 - \Delta)\mathscr{S} \geq \tfrac{1}{2}\mathscr{S}, \qquad \Delta = \frac{39}{4} x \left( \frac{21}{5} + \frac{9}{2} x \right) < \tfrac{1}{2}. \tag{30}$$

This estimate covers arbitrary radial momenta and all real frequencies, with the physical mean separate. Appendices H/O identify the inherited background/root premises and the exact-Fraction saved-cell verification.

At neutral time zero use proper $s$, isothermal $dr = ds/a$, $\alpha = a^3$, $g = \alpha_r/\alpha = 3aH$, $Q_r = \partial_r - g$. The background identity is $g_r = g^2/3 - a^2T^2 - 3h^2$. Both pressure Robin rows give $Q_r w = 0$ at both ends. In the square variables $t = a_m + h\alpha w$, $P_\sigma = \delta\bar{p}_\sigma - 24h\alpha w$, the coupled constraint solution obeys

$$(Q_r^\dagger Q_r + \xi^2 - 3h^2)w = f_c, \qquad f_c = \frac{h}{\alpha}(-t + P_\sigma/2). \tag{S22}$$

This is an identity of the coupled solved response, including the actual physical mean. It never inverts the scalar ground crossing $\xi^2 = 3h^2$. Let $v = Q_r w$, $u = \sqrt{\alpha}v$, $E_u = \|u\|_2^2$. Thus $E_u$ is the quantity denoted $E$ in (29). The differentiated equation lies in $H^{-1}$ with $L^2$ forcing. Testing against $\alpha v$, with Dirichlet $v$, gives

$$\begin{aligned} \mathscr{A} &= \int [|u_r|^2 + (\xi^2 + 15a^2H^2/4 + a^2T^2)|u|^2] \\ &= -\operatorname{Re} \int \sqrt{\alpha} f_c \, \overline{(u_r + 3gu/2)}, \\ \|u_r + 3gu/2\|_2^2 \leq \tfrac{21}{5}\mathscr{A} + \tfrac{9}{2}h^2 E_u, \qquad & \|\sqrt{\alpha} f_c\|_2^2 \leq \tfrac{13}{2} h^2 \mathscr{S}, \\ \mathscr{R} \geq -\tfrac{3}{2} E_u. & \end{aligned} \tag{S23}$$

The remainder bound can be checked directly. Put $y = w/\alpha$, so $E_u = \int \alpha^3 |y'|^2$. Retaining both oriented pressure endpoints and using $g' = g^2/3 - a^2T^2 - 3h^2$ gives

$$\begin{aligned}\mathscr{R} &= -2\int \alpha''|w|^2 + [\alpha'|w|^2]_a^b \\ &= -\tfrac{3}{2}E_u + \int \alpha^3\left[\tfrac{3}{2}|y' + \tfrac{2}{3}gy|^2 + (a^2T^2 + 3h^2)|y|^2\right] \geq -\tfrac{3}{2}E_u.\end{aligned} \tag{F1}$$

This proves the adverse-remainder estimate with both wall terms. No derivative or endpoint value of an input momentum appears. If $\mathscr{A} \geq \lambda_0 E_u$, Cauchy–Schwarz gives $E_u \leq (13/2)x(21/5 + 9x/2)\mathscr{S}$, $x = h^2/\lambda_0$. The certified gates $h^2 < 1/3000$, $a^{-1} < 8$, $\mathfrak{m}\Phi_0^2 < 1$ and the first integral imply $H^2 > 11/12$, so $\lambda_0 = 55/1024$ works. Since $x < 1024/165000 < 1/100$,

$$\Delta = \frac{39}{4}x(\tfrac{21}{5} + \tfrac{9}{2}x) < \frac{33111}{80000} < \frac{1}{2}, \qquad \mathscr{K}_0 \geq \mathscr{S}_0/2. \tag{S24}$$

The proof includes arbitrary radial $L^2$ inputs, complex Hermitian Fourier amplitudes and the separate physical mean. A finite matrix positivity check could not replace it.

*G.2.1 Neutral-time covariance in true canonical densities* The bound just proved at neutral time zero transports to every finite neutral construction time. Put $\Lambda = e^{h\tau} > 0$. At time $\tau$, keep the same proper coordinate $s$ and reference strip $\rho$, but use $r_\tau = \Lambda^{-1}r_0$, $q_\tau = \Lambda^{-1}q_0$. The neutral reduced background satisfies

$$\begin{gathered}\sigma_\tau = \sigma_0 + \log\Lambda, \quad \alpha_\tau = \Lambda^2\alpha_0, \quad e^{2\lambda_\tau} = \Lambda^6 e^{2\lambda_0}, \quad \mathsf{H}_{i,\tau} = \Lambda^2\mathsf{H}_{i,0}, \\ B_\tau = \Lambda^{-4}B_0, \quad \kappa_\tau = \Lambda^{-2}\kappa_0, \quad \bar{A}_\tau = \Lambda^4\bar{A}_0, \quad \bar{p}_\tau = \Lambda^4\bar{p}_0, \quad p_{\mathrm{wall},i,\tau} = \Lambda^{-2}p_{\mathrm{wall},i,0}.\end{gathered} \tag{G1}$$

Here $B$ is the reduced potential and $\bar{A}$ the tracefree momentum tensor, not the radial quadratic form $\mathscr{A}$. The neutral wall background momenta vanish. The time shift is compared in the construction clock; it is not a change in the action or in the charge labels.

For pure-momentum inputs at fixed $\rho$, define the positive congruence $\mathcal{D}_\tau$ by

$$\begin{gathered}\delta\Pi_\tau = \Lambda^3\delta\Pi_0, \qquad \delta p_{i,\tau} = \Lambda^3\delta p_{i,0}, \qquad \delta P_{q,\tau} = \Lambda^4\delta P_{q,0}, \\ \widehat{\xi} = \xi/\Lambda, \qquad w_\tau(\xi) = w_0(\widehat{\xi}), \qquad \delta\bar{A}_\tau(\xi) = \Lambda^4\delta\bar{A}_0(\widehat{\xi}).\end{gathered} \tag{G2}$$

Both components of each independent wall vector scale as stated. The width momentum occurs only in the separate physical mean. All paired configuration variations, including $\delta q$, remain zero. Indeed $\delta\bar{p}_\tau = \delta\Pi_\tau/q_\tau = \Lambda^4\delta\bar{p}_0$, and substitution in the coupled constraint rows and the homogeneous pressure Robin rows gives the response in (G2). The mean condition $2\int_0^1 \delta a_m\, d\rho = \delta P_q$ is preserved. At nonzero frequency $\partial_{r_\tau} = \Lambda\partial_{r_0}$ gives the frequency relabelling; the auxiliary limiting zero is still distinct from the physical mean. Holding $\bar{p}$ as though it were the true density, or freezing $P_q$, would not give this transformation.

Substitution in the exact form (C3), with both pressure endpoints retained, yields

$$\begin{aligned}\mathscr{S}_\tau(\xi;\mathcal{D}_\tau P) &= \Lambda^3\mathscr{S}_0(\xi/\Lambda;P),\\ \mathscr{R}_\tau(\xi;\mathcal{D}_\tau P) &= \Lambda^3\mathscr{R}_0(\xi/\Lambda;P),\\ \mathscr{K}_\tau(\xi;\mathcal{D}_\tau P) &= \Lambda^3\mathscr{K}_0(\xi/\Lambda;P) \geq \tfrac{1}{2}\mathscr{S}_\tau(\xi;\mathcal{D}_\tau P).\end{aligned} \tag{G3}$$

For example, the bulk square prefactor $dr_\tau\alpha_\tau e^{-2\lambda_\tau}$ scales as $\Lambda^{-5}$, while its squared inputs scale as $\Lambda^8$. The wall prefactor $\alpha_i e^{-\lambda_i}\mathsf{H}_i^{-1}$ scales as $\Lambda^{-3}$ and its squared momentum as $\Lambda^6$. In the remainder, $dr_\tau\bar{\Delta}_\tau\alpha_\tau$ and $\alpha_i e^{\lambda_i}p_{\mathrm{wall},i}$ both scale as $\Lambda^3$. Thus the two parts have the same weight.

Equivalently $u_\tau = \sqrt{\alpha_\tau}(\partial_{r_\tau} - g_\tau)w_\tau = \Lambda^2 u_0$, so $E_{u,\tau} = \Lambda^3 E_{u,0}$ and $\mathscr{A}_\tau = \Lambda^5\mathscr{A}_0$ at the relabelled frequency. The spectral lower bound scales by $\Lambda^2$, as does the $h^2$ coefficient in the differentiated response identity; their ratio $x$ and the remainder fraction $\Delta$ are unchanged. On a fixed compact time slab, $\Lambda$, its inverse and both congruence norms are bounded, and a compact time–frequency band maps into a compact scaled-frequency band. This supplies the neutral-time input to G.3. It supplies no uniform all-future unweighted norm or quantitative charge radius.

*G.3 Norm equivalence and charged continuation*

At fixed frequency the sourced response sends bulk $L^2$ momenta and finite wall/mean loads to $H^2 \times H^1$; compact embedding makes $\mathscr{S}_0$ positive bare multiplication plus compact response. Its kernel vanishes: at nonzero frequency the off-diagonal square forces the momentum primitive to vanish, then the diagonal square and constraints give $w = 0$ and all inputs zero. For the physical mean its zero squares give $a_m = -h\alpha w$, $\delta\bar{p}_\sigma = 24h\alpha w$, $\delta\bar{p}_\Phi = 0$. The momentum row forces $w_r = 0$. The constant response in (S22) then has coefficient $g^2 + g' - 3h^2 = a^2(12 - \mathfrak{m}\Phi_0^2) + 6h^2 > 0$ under the same coarse gate; hence $w = 0$ and $P_q = 0$, with no auxiliary limiting-mean assumption. The separate finite wall squares remain positive. Compact-band inverse continuity gives a positive common lower norm bound, and the large-frequency parameter estimate supplies the complement.

For the charged branch retain the true-density fixed-$\rho$ form domain. Wall loads now contain background momenta times independent $\delta p_i$; these are finite loads, not frozen density conditions. Continuous Robin lifts and the coupled inverse give response continuity in $\mathcal{B}(\mathcal{H}_{\mathrm{mom}}, H^2 \times H^1)$, including traces, on compact frequency bands. The inverse identity $L_\delta^{-1} - L_0^{-1} = L_\delta^{-1}(L_0 - L_\delta)L_0^{-1}$ makes this operator-norm continuity. Thus the exact forms converge in bounded-form norm, not merely coefficientwise.

Choose the ultraviolet cutoff first on a preliminary background neighborhood, using the actual pressure trace loads and parameter estimates. Above it $|\mathscr{R}_\delta| \leq C\mathscr{S}_\delta/\xi^2$, with uniform comparison to canonical inputs. Keep that cutoff fixed. The neutral form has a positive minimum on the remaining compact band and the physical mean; shrink the charge neighborhood until the bounded-form difference is smaller than half that minimum. Both independent charge directions remain. This proves finite-slab charged coercivity with uncomputed radius and constants. It does not assert the neutral relative

half-$\mathscr{S}$ bound throughout the charged low band, positive restoring energy, or a nonlinear/quantum no-ghost theorem.

## Appendix H. Computer-assisted premises and local evidence

The numerical premises enter at three distinct places. Their source executions are inherited; none is regenerated here.

Table 6 summarises the three numerical premises.

**Table 6. Numerical inputs, certified outputs and trust boundaries.**

| Premise | Exact input and checked output | Consequence and trust boundary |
|---|---|---|
| Original background and phase | Six rational parameters and linked root/launch/distance intervals; whole launch and endpoint propagated; rigorously lifted physical soft-wall $\mu = 0$ phase, equation (S25) | Complete original coupled negative count and no massless root, conditional on physical reconstruction, faithful identified execution and outward Arb arithmetic |
| Baseline selected mass | Same six rational action parameters and entire launch/event intervals; physical wall Hessians fixed at 20; completed opposite-sign endpoint enclosures at $-33/25000$ and $-131/100000$, both implementations covering the full uncertain wall | Continuity and the restricted count give the open bracket in (N11), hence the branch and shifted-ladder nonresonance; historical identified source/runtime and outward-Arb execution remain premises. See Appendix O.6 and the mass-source map. |
| Neutral kinetic coarse gates | Same root/launch linkage; enclosure over the full collar through the upper endpoint; 16,365 integer fixed-point cells with Picard/Taylor and remainder tests; independent exact-Fraction reconstruction of saved cells | $0 < h^2 < 1/3000$, $0 < a^{-1} < 8$, $\mathfrak{m}\Phi_0^2 < 1$, hence (S24); root existence remains inherited |

The exact phase enclosure used at this interface is

$$\theta_b(0)/\pi \in \left[\frac{577838035}{536870912}, \frac{577858823}{536870912}\right] \subset (1, 3/2). \tag{S25}$$

The phase calculation uses the background wall parameter 20 while varying only the

perturbative Hessians to zero. Root uniqueness links the launch box to that exact action. A terminal phase sign, a producer pass flag or overlapping compact intervals alone is insufficient: the trajectory must remain in its whole-cell enclosure, the phase must be continuously lifted through chart events, and the full uncertain endpoint must be covered. Appendix O supplies the predicates and their mathematical justification; the identified execution records supply the numerical premises. Agreement between its two implementations shares Arb and chart mathematics; arithmetic independence is not claimed.

The baseline mass certificate is a separate numerical dependency from the soft-wall zero-mass phase calculation: counting one root does not locate it. Appendix O.6 supplies the determinant normalization, exact endpoint signs, same-root linkage and inference needed for the selected material completion.

For the kinetic gates the cell verifier reconstructs strict saved-cell inclusions with exact rational arithmetic. This is additional arithmetic diversity for those coarse bounds; it does not recertify the complete phase or root proof. No broad global cover or simultaneous-wall Jacobian is needed here. The analytical charge radius, response constants and finite-slab energy constants are uncomputed.

The numerical inputs and mathematical validity predicates are stated here and in Appendix O; the kinetic implication is proved in Appendix G. Exact source/result identities and execution records accompany the local reading package. They establish provenance rather than arithmetic validity by themselves. The proof exposition uses the recorded executions without repeating them. Public portable qualification and an approved evidence-access route remain separate from the proof exposition.

### Appendix I. Closest literature: equations and assumptions

Frolov–Kofman [3] already exhibit the de Sitter shift in their equations 8–9. Their scalar variable and the weight in their equation 17 divide by the background scalar derivative; their equations 14 and 16 give finite-potential matching and its stiff-wall limit. Their Section III treats bending and its matter dependence. This paper does not claim the curvature shift, radion instability or wall-stiffness dependence as new. The physical reconstruction, exceptional-root removal and certified complete application must instead be assessed together.

Lesgourgues–Sorbo [4] do not simply stop at that singular variable. Their equations 12–15 introduce a Mukhanov-type variable regular through a scalar-derivative zero, provided the warp logarithmic derivative is nonzero. Their equations 16–17 include finite wall Hessians. Their setting has flat four-dimensional slices. Thus regular passage through a scalar turn is an explicit precedent. The present retained system (10), its original-action domain, curved-slice characteristic and complete certified count are the more specific objects requiring comparison.

Konikowska–Olechowski–Schmidt [5] include inflating backgrounds and the shifted spectral parameter. Their equations 33–35 use a regular Mukhanov-type variable at a scalar turn, and their equations 45–47 give a turning-profile instability argument in the

static stability analysis. Turning profiles as a source of instability are therefore established. Our proposed increment is an exhaustive count, including the remaining band on one linked background, followed by the specified dynamical matter completion; no new general turning-point criterion is proposed.

Grinstein–Nolte–Skiba [26] construct static two-brane solutions with matter density and pressure as well as tension. Their Section III analyzes perturbations about the matter-supported equilibrium; stability depends on the equation of state. Their bulk contains a cosmological constant and their static metric allows different temporal and spatial warp factors. The present timelike current is an irrotational stiff fluid, with $p_{\mathrm{m}}/\rho_{\mathrm{m}} = 1$, together with independent canonical oscillators and a nonconstant bulk scalar. Their stabilization result prevents any general inference that adding brane matter must preserve instability. What is claimed here is persistence on the selected analytic time-dependent branch about the certified turned scalar background, using its full scalar/metric/interface problem rather than transferring a static matter-equilibrium criterion.

Karmakar–SenGupta [27], published in September 2026, study matter-induced radion tachyonicity in the low-energy four-dimensional effective theories of tuned, detuned and Goldberger–Wise-stabilized Randall–Sundrum models. Their equations 50–51 describe the matter-dependent effective potential and their equation 58 gives the tuned-case mass response to the stress trace. Matter can therefore alter radion stability, and its trace is directly relevant: our four-dimensional stiff current has $T_{\mathrm{m}} = -\rho_{\mathrm{m}} + 3p_{\mathrm{m}} = 2\rho_{\mathrm{m}}$. This overlap is substantive. Their effective-potential analysis does not supply an exhaustive radial count or the actual five-dimensional moving-interface charged branch and canonical-domain reconstruction used here. Conversely our result neither tests their parameter families nor establishes a universal response to positive-trace matter.

Deffayet [10], Section 2.1, uses an empty maximally symmetric bulk without a stabilization scalar. His master equation and characteristic boundary reduction establish well-posedness with adiabatic perfect-fluid or scalar brane matter. Our nonconstant bulk scalar and two interacting dynamical walls require the constraint, width and conormal constructions below. Compatible initial data and dynamical brane boundary equations remain established precedents; no first general brane-matter well-posedness theorem is claimed.

Bhattacharyya–SenGupta [28] analyze how bulk scalar profiles and potentials affect modulus stabilization, including singular-perturbation approximations and effective radion potentials. Their criterion concerns stabilization in those models, whereas the present propositions count the complete declared coupled negative radial sector of one exact background and then reconstruct selected dynamical solutions. Neither conclusion is transferred between the two problems.

Martin et al. [29], Section V A and Figure 5, already give nonlinear numerical departure from unstable static de Sitter two-brane configurations with a bulk scalar, including increasing and decreasing separation; Section VI examines collisions. Their scalar-potential boundary model differs from this article's independent canonical wall oscillators and current sectors. Theorem 3 is a small analytic family with an explicit tail

and scalar witness, not the first nonlinear radion result, and its extremely small sufficient neighborhood does not match the physical reach of those simulations. In the inspected arXiv version (hep-th/0309001v1), Section 3.3 describes initial configurations, Appendix B.1 constructs constrained perturbations, and Table 1 gives simulation parameters.

Binétruy–Deffayet–Langlois [30] derive nonlinear homogeneous separation dynamics directly from the moving-wall junctions and distinguish the two wall proper times in their equations 4–10. Langlois–Sorbo [31] derive a nonperturbative homogeneous radion action for an AdS two-brane bulk and compare the reduced dynamics with the full five-dimensional system. These are precedents for nonlinear radion motion, moving interfaces and proper-clock conversion. The turned bulk scalar, dynamical wall oscillators, full constraint reconstruction and remainder-protected witness are the selected objects of Theorem 3.

Van den Berg–Mireles James–Reinhardt [32] validate analytic invariant-manifold parameterizations with truncation-error bounds for analytic vector fields. Reinhardt–Mireles James [33] develop a Fourier–Taylor construction, nonresonant coefficient solves, convergence and validated unstable manifolds for parabolic PDEs. These methods prevent treating the general spectral-to-nonlinear-series strategy as new. Here the task-specific requirements are the Einstein constraints, source derivative payments, both moving timelike interface systems and the actual proper-clock observable. No full unstable-manifold or general-data flow theorem for this Einstein/interface problem is inferred from the cited theories.

Finite-wall norms and standard spectral/form methods [7, 11, 12] are also credited. The candidate contribution is the model-specific combination of complete count, oscillator extension, local neutral nonlinear departure with a controlled remainder, actual charged branch and compatible canonical linear evolution with kinetic control. This targeted primary-source comparison does not establish historical priority or replace specialist assessment of whether that combination is useful.

## Appendix J. Neutral action reconstruction and kinetic norm

This section treats the pure-potential wall action obtained by omitting the material terms, before applying the oscillator extension of manuscript Section 4. It does not set a coupled oscillator perturbation to zero in the selected completed theory. Here $\nu_{\mathrm{met}}$ is the metric lapse perturbation, not material density; $\mathcal{X}$ is a gauge compensator, not a surface oscillator. In these neutral appendices, $Z = Z_A$ is the original-action norm of manuscript (11). The lower/upper signs and background are those of manuscript (4), (8)–(10).

### *J.1 Variation and constraints*

The neutral matching uses the Darmois–Israel framework [19, 23]; the reflected-wall specialization is described in manuscript Section 2.2. The neutral pure-potential interval action is twice the bulk integral, with twice the usual Einstein Gibbons–Hawking boundary term (coefficient one per copy when the Einstein bulk coefficient is one half), and each wall potential once. The Gibbons–Hawking term uses the outward normal; the

junction formulas of manuscript (4) use inward normals. Vary the lapse, scalar shift and tangential metric before fixing longitudinal gauge. This retains the Hamiltonian and momentum constraints that would otherwise be lost by premature gauge fixing. The bulk equations are Einstein's equation with cosmological term -6 and the canonical scalar stress, together with $\nabla^2\Phi - V_{,\Phi} = 0$. Their background components give manuscript (9); the boundary variations give manuscript (4) with the material terms omitted.

For a scalar harmonic $Q$, define

$$\mathcal{H}_{\alpha\beta}[Q] = D_\alpha D_\beta Q - \tfrac{1}{4}\gamma_{\alpha\beta}\Box_\gamma Q. \tag{I1}$$

Before fixing longitudinal gauge, write

$$\delta g_{ss} = 2\nu_{\rm met} Q, \quad \delta g_{s\alpha} = bD_\alpha Q, \quad \delta g_{\alpha\beta} = 2a^2\{\psi\gamma_{\alpha\beta}Q + ED_\alpha D_\beta Q\}.$$

For $X^s = \alpha Q$, $X^\alpha = \beta D^\alpha Q$, with $\delta g \mapsto \delta g - \mathcal{L}_X g$, the transformations are

$$E \mapsto E - \beta, \quad b \mapsto b - \alpha - a^2\beta', \quad \psi \mapsto \psi - H\alpha, \quad \nu_{\rm met} \mapsto \nu_{\rm met} - \alpha', \quad f \mapsto f - T\alpha. \tag{I2}$$

Choosing $\beta = E$, $\alpha = b - a^2E'$ gives $E = b = 0$ without an inverse warp gradient, scalar gradient or spectral parameter. These are scalar diffeomorphisms of the general perturbation; the lapse, shift and traceless equations have already been varied. Equivalently, $\mathcal{X} = b - a^2E'$ transforms as $\mathcal{X} \mapsto \mathcal{X} - \alpha$, so

$$\widehat{\psi} = \psi - H\mathcal{X}, \quad \widehat{f} = f - T\mathcal{X}, \quad \widehat{\nu}_{\rm met} = \nu_{\rm met} - \mathcal{X}', \quad \widehat{\xi}_i = \xi_i + \mathcal{X}_i Q \tag{I3}$$

are gauge invariant within this regular scalar sector. They equal the longitudinal profiles and wall graph after the stated gauge choice. Thus the finite kinetic norm in Appendix L may be specified on $(\widehat{\psi}, \widehat{f})$; no inverse gradient is hidden in that identification. No assertion is made that bare finite signed action alone supplies all required metric/gauge regularity. The generic sector has a nonvanishing traceless Hessian and a nonconstant harmonic. The traceless tangential equation gives $\nu_{\rm met} = -2\psi$. The mixed equation is

$$\psi' - H\nu_{\rm met} = -Tf/3. \tag{I4}$$

In proper distance the perturbed Hamiltonian equation has

$$\begin{aligned} \delta R[h]/Q &= -6a^{-2}(\mu + 4k)\psi, \\ \delta G_{nn}/Q &= 3a^{-2}(\mu + 4k)\psi + 12H(\psi' - H\nu_{\rm met}), \\ \delta T_{nn}/Q &= T(g - \nu_{\rm met}T) - V_{,\Phi}f. \end{aligned} \tag{I5}$$

Equating the last two expressions, using (I4), $T' = V_{,\Phi} - 4HT$, and $\nu_{\rm met} = -2\psi$, yields the constraint in manuscript (10). Direct variation of the scalar equation gives

$$f'' + 4Hf' + (\mu/a^2 - V_{,\Phi\Phi})f + (4\psi' - \nu'_{\rm met})T - 2\nu_{\rm met}V_{,\Phi} = 0. \tag{I6}$$

Substitution of (I4) gives the scalar row of manuscript (10). The background identities and manuscript (10) imply $C' = -4HC$, so $a^4C$ is conserved. At a scalar turn $T = 0$, the constraint is the finite relation $T'f + 3(\mu + 4k)\psi/a^2 = 0$; the scalar equation (I6) is still imposed. There is no new interior boundary or freely chosen matching datum.

For explicit sufficiency, put $q = \psi' - H\nu_{\rm met}$. The unreduced tangential Einstein variation in longitudinal gauge is

$$\delta G_\alpha{}^\beta = -\frac{2\psi + \nu_{\rm met}}{a^2} D_\alpha D^\beta Q + \left[\frac{(2\mu + 6k)\psi + \mu\nu_{\rm met}}{a^2} + 3q' - 3\nu_{\rm met}H' + 12Hq\right] Q\delta_\alpha{}^\beta, \tag{I7}$$

while $\delta T_\alpha{}^\beta = (-Tg + \nu_{\rm met}T^2 - V_{,\Phi}f)Q\delta_\alpha{}^\beta$. The cosmological term has zero mixed-index variation. One way to derive (I7) is to use

$$R_\alpha{}^\beta = R[h]_\alpha{}^\beta - N^{-1}\partial_s K_\alpha{}^\beta - KK_\alpha{}^\beta - N^{-1}D_\alpha D^\beta N,$$

$$R_{nn} = -N^{-1}\partial_s K - K_\alpha{}^\beta K_\beta{}^\alpha - N^{-1}\Box_h N, \qquad K_\alpha{}^\beta = (H + qQ)\delta_\alpha{}^\beta,$$

with $N = 1 + \nu_{\rm met}Q$ and the intrinsic conformal-curvature variation. These identities fix the signs of both normal and tangential components.

Define the momentum residual $M = q + Tf/3$. After the generic traceless equation is imposed, the full remaining tangential residual $\delta(G - 6g - T)_\alpha{}^\beta = \mathcal{E}_{\rm tan}Q\delta_\alpha{}^\beta$ obeys

$$\mathcal{E}_{\rm tan} = 3q' + 12Hq + Tg + V_{,\Phi}f + (6k/a^2 + 6H' + 2T^2)\psi = 3M' + 12HM. \tag{I8}$$

The last bracket vanishes by the background $H'$ equation; the remaining equality uses $T' + 4HT = V_{,\Phi}$. Thus momentum and traceless equations imply the omitted tangential equation through both $T = 0$ and $H = 0$. The normal residual is $-C$, and direct differentiation gives

$$C' = -4H\{Tg - T'f + (2T^2 - 3\lambda/a^2)\psi\} = -4HC. \tag{I9}$$

Necessity of the retained system follows from the varied equations; (I7)–(I9) establish its bulk sufficiency in the classical generic scalar sector. This is a scalar-sector reconstruction, not a classification of vector or tensor perturbations.

### *J.2 Finite walls and bending*

Before setting the bending to zero, take a wall graph $s = s_i + \xi_i(x)$. With $\sigma_a = +1$, $\sigma_b = -1$, its scalar boundary variation is

$$\sigma_i[g - \nu_{\rm met}T + T'\xi_i] = \tfrac{1}{2}\kappa_i(f + T\xi_i). \tag{I10}$$

Harmonic factors are understood for the profile terms. Its mixed extrinsic curvature varies as

$$\delta K_\alpha{}^\beta = \sigma_i\{[\psi' - H\nu_{\rm met} + H'\xi_i]\delta_\alpha{}^\beta - a_i^{-2}D_\alpha D^\beta\xi_i\}. \tag{I11}$$

In the full Israel condition the $f$ terms cancel by (I4). The identity $H' + T^2/3 = -k/a^2$ leaves

$$D_\alpha D_\beta \xi_i + k\gamma_{\alpha\beta}\xi_i = 0. \tag{I12}$$

For clarity, before this last reduction the entire Israel condition, after its inward orientation is divided out, is

$$MQ\delta_\alpha{}^\beta + (H' + T^2/3)\xi_i\delta_\alpha{}^\beta - a_i^{-2} D_\alpha D^\beta \xi_i = 0. \tag{I13}$$

Here the background relation $\tau_{i,\Phi} = 2\sigma_i T_i$ cancels the scalar terms. With $M = 0$, it is exactly (I12). Under (I2) the graph changes by $\xi_i \mapsto \xi_i + \alpha_i Q$. Therefore zero bending in longitudinal gauge is a physical consequence of the generic Israel equation, not a second independent gauge choice imposed at both ends.

For generic harmonics the bending amplitude must vanish. Equation (I10) then gives the neutral wall law $2\sigma_i L_i = \kappa_i f_i$, including its half-Hessian and opposite endpoint signs. The trace Israel perturbation is already (I4), not a second independent scalar wall condition. Nonconstant embedding harmonics satisfying (I12) obey $\Box_\gamma \xi_i = -4k\xi_i$, but that necessary scalar equation is weaker than (I12). Consequently this sector distinction does not discard every generic mode with $\mu = -4k$. Constant harmonics also require separate gauge analysis; none is included by assuming that it represents a generic propagating mode. The distinction at shifted zero is nonvacuous: on a flat-slicing de Sitter patch, $Q = e^{-4\sqrt{k}\,t}$ has $\Box_\gamma Q = -4kQ$ but a nonzero traceless Hessian, since its temporal/spatial Hessian difference is proportional to $20kQ$. It is a local generic example, not a claim of spacetime square integrability. The absence of a physical radial mode at $\lambda = 0$ follows from the wall and energy argument, not from excluding all harmonics at that mass.

The first variation of proper separation is

$$\delta D(x) = \int_0^D \nu_{\rm met}(s, x)\, ds + \xi_b(x) - \xi_a(x). \tag{I14}$$

The lapse and endpoint changes cancel under a transverse coordinate transformation. For the retained generic modes, $\delta D/Q = -2\int \psi\, ds$. A nonzero overlap establishes a separation component; positive norm alone does not establish this overlap.

### *J.3 Kinetic coefficient and boundary terms*

After the normal total derivatives have been canceled by the Gibbons–Hawking terms, the four-dimensional derivative terms at quadratic order are

$$S_{2,\partial x} = 2\int ds\, d^4x\sqrt{-\gamma}\, a^2 \left[-\tfrac{1}{2}(Df)^2 + 3\{D\nu_{\rm met}\cdot D\psi + (D\psi)^2\}\right]. \tag{I15}$$

To verify the gravitational coefficient, expand $N\sqrt{-h}R[h]/2$, with $N = 1 + \nu_{\rm met}$, $h = a^2 e^{2\psi}\gamma$, and $R[e^{2\psi}\gamma] = e^{-2\psi}[R[\gamma] - 6\Box\psi - 6(D\psi)^2]$. The factor multiplying the bracket is $a^2(1 + \nu_{\rm met})e^{2\psi}/2$. Four-dimensional integration by parts gives

$3D\nu_{\rm met}\cdot D\psi+3(D\psi)^2$. This calculation assumes compactly supported variations or vanishing four-dimensional boundary flux. The scalar kinetic term contributes $-a^2(Df)^2/2$. The extrinsic-curvature terms in this gauge contain normal derivatives; $R[\gamma]=12k$ changes potential terms. Pure wall potentials have no tangential kinetic terms when $\xi_i=0$. No wall kinetic term has been dropped from the declared action. Substitution of $\nu_{\rm met}=-2\psi$ into (I15) gives the coefficient of $-\frac{1}{2}\int\sqrt{-\gamma}(DQ)^2$:

$$Z=2\int_0^D a^2(f^2+6\psi^2)\,ds. \tag{I16}$$

The initial factor two is the cover multiplicity. The canonical amplitude is $Q_c=\sqrt{Z}Q$. This derivation uses no inverse scalar gradient and remains regular at a turn. Negative $\mu$ with positive $Z$ is a tachyonic instability in this sector. It is not a negative-kinetic-energy mode. This is the pullback of the original-action kinetic form to admissible profiles, with vanishing four-dimensional surface flux. It does not identify an unrestricted off-shell master action or a complete operator domain. Appendix L supplies the finite-norm weak/classical mode-domain statement needed for the separated count.

## Appendix K. Regular scalar reduction and its physical domain

This section derives the regular transformation used in Proposition 1. All derivatives are in proper distance. The auxiliary $v$ used here is strictly positive at both walls; the lower-vanishing phase-anchor solution in Appendix N is a different function. We prove equivalence of classical solutions and traces, not equivalence of full closed Hilbert-space domains.

### *K.1 Starting identities*

Use proper distance $s$, $a>0$, $H=a'/a$, $T=\Phi'$, and $\lambda=\mu+4k$. Define $L=g+2T\psi$ and

$$\mathcal{Q}=V''+4k/a^2+4T^2/3.$$

The original constrained system is equivalently

$$\psi'=-2H\psi-Tf/3,\quad f'=L-2T\psi,\quad L'=-4HL+(\mathcal{Q}-\lambda/a^2)f-2T'\psi, \tag{J1}$$

$$TL-T'f=3\lambda\psi/a^2. \tag{J2}$$

The background implies

$$T''+4HT'=\mathcal{Q}T. \tag{J3}$$

For example, differentiating $L=g+2T\psi$ and using $T'=V'-4HT$ gives the last equation of (J1) with no division. These identities hold at scalar turns.

*K.2 Positive auxiliary function*

Choose any solution v of

$$v'' + 2Hv' - 2T^2 v/3 = 0, \qquad v(0) > 0, \quad v'(0) > 0. \tag{J4}$$

It is globally finite on the compact interval because this is a regular linear equation. While v>0,

$$(a^2 v')' = 2a^2 T^2 v/3 \geq 0.$$

Thus $v'$ stays positive, so $v$ cannot have a first loss of positivity. In particular

$$\mathfrak{h} = \frac{v'}{2v} > 0, \quad \mathfrak{h}' = -2H\mathfrak{h} - 2\mathfrak{h}^2 + T^2/3 \tag{J5}$$

is regular and bounded away from zero on this fixed compact interval. No sign of $H$ is required.

Set

$$C = \frac{T^2}{3\mathfrak{h}}, \qquad B = T' - \frac{T^3}{3\mathfrak{h}} = T' - CT, \qquad R = 2H - C, \tag{J6}$$

$$U = \mathcal{Q} - \frac{4TT'}{3\mathfrak{h}} - \frac{2HT^2}{\mathfrak{h}} - \frac{2T^2}{3} + \frac{2T^4}{9\mathfrak{h}^2}. \tag{J7}$$

Direct use of (J3),(J5) gives the division-free identity

$$B' + (2H + R)B = TU. \tag{J8}$$

The displayed smooth formula (J7), rather than the formal quotient of (J8) by $T$, defines $U$ at turns.

*K.3 Transformation and inverse*

Define

$$F = f + \frac{T}{\mathfrak{h}}\psi. \tag{J9}$$

Equations (J1),(J5) imply

$$F' = L - Cf + \frac{B}{\mathfrak{h}}\psi, \quad TF' - BF = 3\lambda\psi/a^2, \quad \psi' = -R\psi - TF/3. \tag{J10}$$

Differentiate the middle equality, substitute the last equality and (J8), and simplify. The result is

$$F'' + 4HF' + (\lambda/a^2 - U)F = 0, \quad \text{or} \quad -(a^4 F')' + a^4 UF = \lambda a^2 F. \tag{J11}$$

No curvature approximation or substitution into a flat factorization has been made. The coefficients p=aˆ4 and w=aˆ2 are strictly positive; the potential aˆ4U is regular.

For $\lambda \neq 0$ the inverse is

$$\psi = \frac{a^2}{3\lambda}(TF' - BF), \quad f = F - \frac{T}{\mathfrak{h}}\psi, \quad g = f'. \tag{J12}$$

For completeness, differentiating the first expression and using (J8),(J11) gives exactly $\psi' = -R\psi - TF/3$, without dividing by $T$. The definition of f then gives (J10) and (J2), with $L = f' + 2T\psi$. Differentiation of (J2) supplies the remaining equation (J1) away from zeros of $T$; continuity supplies it at those zeros. Under the endpoint hypotheses $T_a < 0 < T_b$, $T_a', T_b' > 0$ every zero of $T$ is simple and there are finitely many, so this extension has no exceptional interval. Consequently (J9),(J12) give a linear bijection between the two-dimensional classical constrained solution space and the two-dimensional classical solution space of (J11), for each $\lambda \neq 0$, including negative $\lambda$ and any nonzero wall pole below.

A pointwise determinant check makes the absence of lost local data explicit. For $\lambda \neq 0$, put $\alpha = a^2/(3\lambda\mathfrak{h})$. Using the constraint to eliminate $\psi$, the map between constrained initial-data pairs factors as

$$\begin{pmatrix} F \\ F' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -C & 1 \end{pmatrix} \begin{pmatrix} 1 - \alpha TT' & \alpha T^2 \\ -\alpha (T')^2 & 1 + \alpha TT' \end{pmatrix} \begin{pmatrix} f \\ L \end{pmatrix}. \tag{J13}$$

The first matrix has determinant one, and the second has determinant

$$(1 - \alpha TT')(1 + \alpha TT') + \alpha^2 T^2 (T')^2 = 1.$$

Thus the full pointwise map is invertible, including at $T = 0$ or $H = 0$. Its coefficients still depend singularly on $\lambda$ at zero; this check neither extends the inverse to that parameter nor identifies unrestricted function-space norms.

This is a solution-domain equivalence. It does not assert a uniform bounded invertible transformation on the entire unconstrained $H^1$ space; the inverse contains F' and $1/\lambda$. On compact spectral sets away from zero, it is an ordinary regular differential map on the displayed classical solution spaces. Its dependence on $\lambda$ is analytic there.

*K.4 Exact transformed walls and their poles*

At a finite wall write $L_i = r_i f_i$, with

$$r_a = \kappa_a/2, \quad r_b = -\kappa_b/2, \quad E_i = T_i' - r_i T_i. \tag{J14}$$

The stated endpoint hypotheses give $E_a, E_b > 0$, including zero Hessians. All quantities in the following wall formulas are evaluated at that wall. Substitution in (J10) yields the **undivided** condition

$$(3\mathfrak{h}\lambda - a^2 ET)F' = \{3\mathfrak{h}(r - C)\lambda - a^2 EB\}F. \tag{J15}$$

For $\lambda \neq 0$ this is equivalent to the original wall condition. One can see the equivalence directly: its left side minus its right side equals $3\mathfrak{h}\lambda$ times $(L - rf)$ after (J12) is used. Thus no quotient is needed at a pole.

Away from the pole, (J15) has logarithmic derivative

$$F'/F = \mathscr{R}_i(\lambda) = r - C - \frac{a^2E^2}{3\mathfrak{h}(\lambda - \lambda_{p,i})}, \quad \lambda_{p,i} = \frac{a^2ET}{3\mathfrak{h}}. \tag{J16}$$

Both residues are negative and both derivatives are positive:

$$\mathscr{R}_i'(\lambda) = \frac{a^2E^2}{3\mathfrak{h}(\lambda - \lambda_{p,i})^2} > 0. \tag{J17}$$

The lower pole is negative because $T_a < 0$; the upper pole is positive because $T_b > 0$. At either pole $\lambda_p \neq 0$ the coefficient of $F$ in (J15) is $-a^2E^2 \neq 0$, so the correct wall condition is $F_i = 0$, with $F_i'$ free. It is a regular boundary chart change, not a deleted eigenvalue.

To translate to boundary conditions $(A\lambda + B_0)F = (C_0\lambda + D)pF'$, $p = a^4$, choose

$$A = 3\mathfrak{h}(r - C), \quad B_0 = -a^2EB, \quad C_0 = 3\mathfrak{h}/a^4, \quad D = -a^2ET/a^4.$$

Then

$$AD - B_0C_0 = \frac{3\mathfrak{h}E^2}{a^2} > 0. \tag{J18}$$

The raw determinants have the same positive sign at both endpoints. A cited theorem that inserts an outward-normal sign into its determinant must be translated accordingly; this calculation must not be read as automatic satisfaction of its hypotheses.

At a Dirichlet scalar wall $f = 0$, the transformed undivided condition is

$$a^2T^2F' = (a^2TB + 3\mathfrak{h}\lambda)F, \tag{J19}$$

or $F'/F = B/T + 3\mathfrak{h}\lambda/(a^2T^2)$. Endpoints have $T \neq 0$. Thus transformed Dirichlet walls remain spectral Robin walls, with positive slope. This formula is obtained directly and includes $\lambda = 0$ as a transformed condition; $\lambda = 0$ still requires the separate analysis below. It is also the limit of (J15) as the corresponding nonnegative Hessian tends to infinity.

*K.5 Exact norm identity; why the endpoint space is indefinite*

For real solutions, direct differentiation using (J5) and $\psi'$ gives

$$a^2F^2 = a^2(f^2 + 6\psi^2) - 3\left(\frac{a^2\psi^2}{\mathfrak{h}}\right)'. \tag{J20}$$

Therefore the original regular norm obeys

$$\frac{Z}{2} = \int_0^D a^2F^2 ds + 3\left[\frac{a^2\psi^2}{\mathfrak{h}}\right]_a^b. \tag{J21}$$

At a finite wall away from its pole,

$$\psi_i = -\frac{a_i^2 E_i F_i}{3(\lambda - \lambda_{p,i})}, \quad \frac{3a_i^2\psi_i^2}{\mathfrak{h}_i} = a_i^4 \mathscr{R}_i'(\lambda) F_i^2. \tag{J22}$$

The same last equality holds at a Dirichlet wall with the affine derivative from (J19). At finite poles one must use (J21) with the regular inverse (J12), not the undefined quotient (J22).

Hence the transformed endpoint norm on eigenfunctions has a positive upper-wall term and a negative lower-wall term:

$$N[F] = \int_0^D a^2F^2 ds + a_b^4 \mathscr{R}_b' F_b^2 - a_a^4 \mathscr{R}_a' F_a^2 = Z/2 > 0 \tag{J23}$$

for a nonzero physical eigenprofile with $\lambda \neq 0$. Polarization of (J20) gives the corresponding mixed-product identity. At distinct spectral parameters the boundary rational functions give divided differences rather than the equal-parameter derivatives in (J23).

The signs matter: the natural endpoint linearization has one negative boundary direction. Positivity on the actual nonzero physical eigenfunctions is not positivity of that entire enlarged space. A standard positive-measure Sturm–Liouville theorem cannot be imported by suppressing this negative endpoint contribution.

One can display regular endpoint coordinates even at the finite poles. Set $\vartheta_i = \sqrt{3}\, a_i\psi_i/\sqrt{\mathfrak{h}_i}$ and $d_i = a_i^3 E_i/\sqrt{3\mathfrak{h}_i} > 0$. Then the finite-wall trace relations are

$$p_i F_i' = p_i(r_i - C_i)F_i + d_i\vartheta_i, \quad \lambda\vartheta_i = \lambda_{p,i}\vartheta_i - d_i F_i, \tag{J24}$$

and the product is $\int a^2F^2\, ds + \vartheta_b^2 - \vartheta_a^2$. At $\lambda = \lambda_p$ these equations impose $F_i = 0$ and retain $\vartheta_i$ and the derivative, so no physical wall pole is discarded. At a Dirichlet scalar wall set $e_i = \sqrt{3\mathfrak{h}_i}\, a_i/T_i$; then $\vartheta_i = e_i F_i$ and $p_i F_i' = p_i(B_i/T_i)F_i + e_i\lambda\vartheta_i$. These trace-coordinate relations alone do not prove an operator-domain theorem. Appendix M supplies an auxiliary form-domain realization; a basis theorem for arbitrary original Einstein–scalar Cauchy data is not inferred.

### *K.6 The exceptional zero is artificial and has negative norm*

At $\lambda = 0$, the original constrained solutions have

$$f = -uT, \quad L = -uT', \quad \psi = u'/2, \quad u'' + 2Hu' - 2T^2u/3 = 0. \tag{J25}$$

To justify coverage, on intervals where $T \neq 0$ the constraint gives this expression; the regular system continues it through every simple turn. These solutions form the full two-dimensional constrained space. At either finite wall, $L - rf = -Eu$, so $E > 0$ forces

$u_i = 0$. A Dirichlet wall likewise forces $u_i = 0$ since $T_i \neq 0$. Multiplying the equation for $u$ by $a^2 u$ and integrating under $u(0) = u(D) = 0$ yields

$$\int_0^D a^2\{(u')^2 + 2T^2u^2/3\}\, ds = 0.$$

Thus there is no nonzero physical $\lambda = 0$ eigenfunction for any of these wall pairs.

The transformation (J9) at $\lambda = 0$ has kernel $u$ proportional to the chosen positive auxiliary $v$. Its image is spanned by

$$F_0 = \frac{T}{a^2 v'}. \tag{J26}$$

Indeed, the Wronskian $vu' - v'u$ is a constant times $a^{-2}$, and $F = T(u'/(2\mathfrak{h}) - u)$. Equation (J11) holds for this image; equivalently it follows directly by substitution. At either endpoint

$$F_0'/F_0 = T'/T - T^2/(3\mathfrak{h}) = B/T.$$

This is exactly (J15) at zero, regardless of finite stiffness, and also (J19) at zero. The transformed separated problem therefore always has this spurious zero eigenfunction. Since its separated left condition is nontrivial, its eigenspace has dimension one. It has exactly the same interior zeros as $T$.

At zero the finite-wall norm coefficients reduce to the Dirichlet values, independent of stiffness:

$$a_i^4 \mathscr{R}_i'(0) = 3a_i^2 \mathfrak{h}_i / T_i^2.$$

Define $D = 3/(2a^2 vv') = 3\mathfrak{h}/(a^2 v'^2)$. Equation (J4) gives

$$D' = -a^2 F_0^2 - \frac{3}{2a^2 v^2}.$$

Consequently its transformed norm is strictly negative:

$$N[F_0] = \int_0^D a^2 F_0^2 ds + [D]_a^b = -\frac{3}{2}\int_0^D \frac{ds}{a^2 v^2} < 0. \tag{J27}$$

This does not contradict (J21): no original profile satisfies both physical walls at $\lambda = 0$, and the inverse (J12) is unavailable there. Applying the equal-parameter transformed boundary formula to the artificial zero does not manufacture a physical inverse.

### *K.7 Compact wall faces and the count offset*

For a single formula on the nonnegative wall square, set $t_i = 1/(1 + \kappa_i)$, $j_a = -(1 - t_a)/2$, $j_b = (1 - t_b)/2$, and $\widetilde{E}_i = t_i T_i' + j_i T_i$. The physical wall is $t_i L_i + j_i f_i = 0$. In the endpoint-only theorem $\widetilde{E}_i > 0$ uniformly, including both faces. Substitution of (J12) gives

$$(3\mathfrak{h}_i \lambda t_i - a_i^2 \widetilde{E}_i T_i) F_i' + \{3\mathfrak{h}_i \lambda (t_i C_i + j_i) + a_i^2 \widetilde{E}_i B_i\} F_i = 0. \tag{J28}$$

At $t_i = 1$ this is zero Hessian; at $t_i = 0$ it is (J19). For every $\lambda \neq 0$, its residual is exactly $3\mathfrak{h}_i \lambda (t_i L_i + j_i f_i)$. Thus even simultaneous Dirichlet limits retain the classical trace

correspondence, and no division by a wall parameter is required. Appendices L/N treat simple-root continuation and the uniform root-free strip in the physical variables, without a pointwise limit of rational slopes.

The same-sign determinants belong to established eigenparameter-dependent boundary theory [14, 15]; a positive-measure boundary theorem cannot be applied by dropping the negative endpoint direction. Appendix M constructs the required positive complement explicitly and proves the restricted count. Appendix N supplies the separate remaining-band exclusion. The exact characteristic factor in Appendix L preserves every nonzero physical multiplicity and identifies only one simple artificial zero. No original Cauchy-data spectral expansion follows from this solution-space transformation.

## Appendix L. Weak mode domain and the physical characteristic determinant

This section proves the exhaustive neutral mode-domain statement used in Proposition 1. It concerns the separated generic longitudinal scalar sector, with the action-derived equations and traces of Appendix J. A finite kinetic norm is a radial condition on $f, \psi$; it is not a claim that an arbitrary finite signed-action metric perturbation admits this gauge or belongs to this sector. The endpoint hypotheses $T_a < 0 < T_b$, $T_a', T_b' > 0$ are assumed for the full compact wall square below.

### *L.1 Global weak equations force classical regularity*

Let $I = (0, D)$. Require

$$\int_I a^2(|f|^2 + 6|\psi|^2)\, ds < \infty$$

and impose, as distributions on all of $I$,

$$\psi' = -2H\psi - Tf/3, \quad f'' + 4Hf' = A_\mu f + E\psi, \quad Tf' - T'f + (2T^2 - 3\lambda/a^2)\psi = 0, \tag{K1}$$

where $A_\mu = V_{,\Phi\Phi} - \mu/a^2 + 2T^2$, $E = 12HT - 4V_{,\Phi}$, and $\lambda = \mu + 4k$. These coefficients are regular and bounded. The scalar equation must hold through every scalar turn, not merely on the complementary subintervals.

Positive warp on the compact interval makes the norm condition equivalent to $f, \psi \in L^2(I)$. Momentum then gives $\psi \in H^1(I)$. Put $p = a^4$ and $R = p(A_\mu f + E\psi) \in L^2$. The scalar equation states $(pf')' = R$. Its primitive $G(s) = \int_0^s R(r)dr$ belongs to $H^1$, and the distribution $pf' - G$ has zero derivative, so it is a constant on the connected interval. Consequently $pf' \in H^1$; multiplication by $1/p$ gives $f' \in H^1$, hence $f \in H^2$. No assumption $f' \in L^2$ was needed in advance.

In one dimension $H^1$ functions have continuous representatives. Thus $\psi, f, g = f'$ are continuous. The right-hand sides of manuscript (10) are now continuous, giving $\psi, g \in C^1$ and $f \in C^2$, with one-sided derivatives at both walls. The constraint is continuous and zero as a distribution, hence pointwise zero. Conversely every classical retained solution has finite norm and satisfies (K1). The wall traces are therefore ordinary classical traces.

This proves that no additional finite-norm weak mode or interface datum occurs at $T = 0$, without dividing by $T$ or $H$. To see why the global scalar equation is necessary,

join two constrained classical solutions at a simple turn with the same $f_*, \psi_*$ but different $g_*$. The constraint there fixes $f_* = -3\lambda\psi_*/(a_*^2 T_*')$ and does not fix $g_*$. The glued pair has finite norm and satisfies the other equations and the scalar equation off the turn, but $f''$ has a nonzero delta term equal to the jump in $g$. It is not a solution of (K1). The retained scalar equation excludes this gluing; finite norm alone would not.

For $\lambda \neq 0$, the scalar-pencil weak solutions are classical by the same primitive argument applied to $-(a^4F')' + a^4UF = \lambda a^2 F$. Thus Appendix K also gives a bijection of the physical and scalar-pencil weak eigensolution spaces with their undivided wall traces. This remains a solution-space statement, not a bounded equivalence on unrestricted Sobolev spaces.

*L.2 Uniform boundary rank and the normalized physical launch*

Set

$$t_i = (1+\kappa_i)^{-1}, \quad j_a = -(1-t_a)/2, \quad j_b = (1-t_b)/2, \quad \widetilde{E}_i = t_i T_i' + j_i T_i. \tag{K2}$$

Zero Hessian means $t_i = 1$, and physical Dirichlet means $t_i = 0$. The endpoint hypotheses give

$$\widetilde{E}_a \geq \min(T_a', -T_a/2) > 0, \qquad \widetilde{E}_b \geq \min(T_b', T_b/2) > 0. \tag{K3}$$

The wall row is $t_i L_i + j_i f_i = 0$. Its norm is at least $1/\sqrt{5}$, since $t^2 + (1-t)^2/4 = (5t-1)^2/20 + 1/5$. The constraint and wall rows on $(f, L)$ are $(-T', T)$ and $(j, t)$; their determinant is $-\widetilde{E}$. Thus no boundary rank is lost, even at a compactified face.

Every nonzero lower-admissible solution has $\psi_a \neq 0$: if it vanished, those two independent equations would force $f_a = L_a = 0$, and IVP uniqueness would force the full profile to vanish. Normalize the whole lower-admissible line by

$$\psi_a = 1, \quad f_a = -3\lambda t_a/(a_a^2 \widetilde{E}_a), \quad L_a = 3\lambda j_a/(a_a^2 \widetilde{E}_a), \quad g_a = L_a - 2T_a. \tag{K4}$$

It satisfies the constraint and lower wall for every complex $\lambda$, including zero. It is affine in $\lambda$ and uniformly regular on the whole wall square. Every admissible eigenfunction is a scalar multiple of this launch. In particular the radial eigenspace is one-dimensional; this does not count the multiplicity of four-dimensional harmonics.

*L.3 Entire characteristic determinant, reality and simplicity*

Propagate (K4) with manuscript (10) and define

$$\mathscr{D}(\lambda, t_a, t_b) = t_b L_b + j_b f_b. \tag{K5}$$

For a compact complex spectral disk, the IVP matrix is uniformly bounded on the finite interval. Its iterated-integral series has the factorial bound $(MD)^r/r!$; uniform convergence makes the solution entire in $\lambda$. The normalized launch has the same

property. Thus $\mathscr{D}$ is entire in $\lambda$, and real analytic in the wall coordinates in a neighborhood of each compactified face. This does not assert analytic dependence on an arbitrary smooth background parameter.

By Appendix L.1–L.2, $\mathscr{D} = 0$ is equivalent to an admissible finite-norm weak/classical radial mode. Pairing a complex root with its conjugate in the regular Green identity gives

$$(\mu - \overline{\mu}) \int_0^D a^2(|f|^2 + 6|\psi|^2)\, ds = 0.$$

The wall fluxes vanish at finite, zero-Hessian and Dirichlet walls. The integral is positive, so every root is real. In particular $\mathscr{D}(i) \neq 0$, showing it is not identically zero. Its roots are discrete and finite in every bounded spectral interval.

At a real root with $\lambda \neq 0$, the upper wall line has the form

$$(f_b, L_b) = \zeta(t_b, -j_b), \qquad \zeta \neq 0.$$

If $\zeta = 0$, the constraint gives $\psi_b = 0$, contradicting the normalized nonzero launch by IVP uniqueness. Differentiating the Green identity in the second mass, with the fixed lower wall, gives

$$a_b^4(f_b L_{\lambda,b} - L_b f_{\lambda,b}) = -Z/2, \qquad \mathscr{D}_\lambda = -\frac{Z}{2a_b^4\zeta} \neq 0. \tag{K6}$$

Thus each root is a simple characteristic zero as well as having a one-dimensional radial eigenspace, including on all compactified faces. On a compact root set separated from shifted zero, continuity also bounds this derivative away from zero. This directly identifies characteristic multiplicity of the analytic boundary-value problem. Appendix M constructs an auxiliary operator with matching nonzero eigenspaces; no original physical Cauchy-data completion is assumed here.

*L.4 Shifted zero and the exact transformed determinant factor*

At $\lambda = 0$ the regular solutions have $f = -Tu$, $L = -T'u$, $\psi = u'/2$, with

$$(a^2u')' = \tfrac{2}{3}a^2T^2u.$$

Each physical wall is $-\widetilde{E}_i u_i = 0$. Therefore both endpoint values vanish, and integration against $\overline{u}$ gives

$$\int_0^D a^2(|u'|^2 + \tfrac{2}{3}T^2|u|^2)\, ds = 0.$$

There is no nonzero physical root. For the normalized launch (K4), $u_a = 0, u_a' = 2$, independently of the wall coordinates. Positivity of the flux makes $u_b > 0$, and

$$\mathscr{D}(0, t_a, t_b) = -\widetilde{E}_b u_b < 0 \tag{K7}$$

uniformly on the square. The physical determinant is entire and nonzero at shifted zero; its exceptional-root issue is confined to the scalar transformation.

Let $C = T^2/(3\mathfrak{h})$, $B = T' - CT$ as in Appendix K. The compact scalar boundary row (J28) is

$$W_i[F] = \alpha_i F_i' + \beta_i F_i, \quad \alpha_i = 3\mathfrak{h}_i\lambda t_i - a_i^2\widetilde{E}_i T_i, \quad \beta_i = 3\mathfrak{h}_i\lambda(t_i C_i + j_i) + a_i^2\widetilde{E}_i B_i. \tag{K8}$$

The determinant of its $\lambda$-coefficient vector and constant vector, in the order $(\alpha, \beta)$, is $3\mathfrak{h}_i a_i^2 \widetilde{E}_i^2 > 0$. Thus its two coefficients never vanish simultaneously at any complex $\lambda$, and it always specifies one scalar initial-data line.

The forward identities yield the exact off-constraint relation

$$W_i[F] - 3\mathfrak{h}_i\lambda(t_i L_i + j_i f_i) = -a_i^2\widetilde{E}_i\{T_i L_i - T_i' f_i - 3\lambda\psi_i/a_i^2\}. \tag{K9}$$

This identity does not divide by $\lambda$. Use the forward image of (K4) to normalize the scalar lower launch. It is entire and never zero: for $\lambda \neq 0$ use (J13), and at zero use $F_a = T_a/\mathfrak{h}_a \neq 0$. It spans the entire lower scalar boundary line for every parameter. Its upper characteristic residual therefore satisfies

$$\widetilde{\mathscr{D}}(\lambda, t_a, t_b) = 3\mathfrak{h}_b\lambda\,\mathscr{D}(\lambda, t_a, t_b). \tag{K10}$$

All nonzero physical and transformed characteristic multiplicities agree. By (K7), the transformed zero is artificial and simple. The negative-signature calculation in K.6 is consistent with, and independent of, this characteristic proof of its simplicity. Neither proof extends the physical inverse through zero.

*L.5 Exactly what the count establishes*

The characteristic determinant exhausts the specified finite-norm weak/classical separated radial mode problem. The restricted-form proof and the remaining-band phase argument count every determinant root in their stated intervals; Appendix L.1–L.4 rules out hidden weak/interface modes, extra boundary-rank modes and a missing exceptional root. Combined with the background and phase premises specified in Appendix O, this justifies the complete central negative-eigenvalue count within that domain.

Appendix M provides an auxiliary positive-space operator realization for the separated transformed pencil. It does not identify a complete space of arbitrary original metric/scalar initial data with that operator domain. No eigenfunction expansion, dynamical-dominance or evolution theorem for such arbitrary physical data is supplied. No such inference is used here. Exceptional harmonics, tensor/vector sectors and nonlinear evolution remain outside the count.

## Appendix M. The restricted index as a standard form-theory consequence

This section proves the count below $\lambda = 0$ after the regular physical reconstruction and exceptional-root calculation. Ordinary Sturm oscillation [11] and standard closed-form representation, compactness and variational results [12] supply the general spectral facts. The required application hypotheses, endpoint signs, kernel and physical spectrum are established explicitly below. The separate remaining-band phase argument is in Appendix N.

*M.1 Fixed Robin comparison and its negative index*

Write $p = a^4$, $w = a^2$, $q = pU$. At each finite wall use the notation

$$A_i = r_i - C_i, \quad z_i = \frac{a_i^2 E_i T_i}{3\mathfrak{h}_i}, \quad c_i = \frac{a_i^2 E_i^2}{3\mathfrak{h}_i} > 0, \quad d_i = \sqrt{p_i c_i}, \quad e_i = \frac{d_i}{z_i} = \frac{\sqrt{3\mathfrak{h}_i}\, a_i}{T_i}. \tag{L1}$$

Here $C_i = T_i^2/(3\mathfrak{h}_i)$, $E_i = T_i' - r_i T_i > 0$, and $p_i = a_i^4$; the pole $z_i$ is distinct from the bulk coefficient $p$. Endpoint signs imply $z_a < 0 < z_b$. The exact transformed boundary equations are

$$p_i F_i' = p_i A_i F_i + d_i \vartheta_i, \qquad \lambda \vartheta_i = z_i \vartheta_i - d_i F_i. \tag{L2}$$

They retain the spectral poles without a quotient. At zero,

$$R_i^0 := A_i + c_i/z_i = B_i/T_i, \qquad e_i = d_i/z_i, \tag{L3}$$

both independent of wall stiffness.

Consider the ordinary, fixed separated Robin problem with the same bulk expression and $F_i' = R_i^0 F_i$. Its form on $H^1(0, D)$ is

$$s_R[F] = \int_0^D (p|F'|^2 + q|F|^2) ds + p_a R_a^0 |F_a|^2 - p_b R_b^0 |F_b|^2. \tag{L4}$$

The smooth function $F_0 = T/(a^2 v')$ is its zero eigenfunction and has exactly the $n$ simple interior zeros of $T$. The regular separated Sturm theorem therefore places zero at ordered index $n$, starting at index zero [11, Section 6.5]. This ordinary Robin problem has exactly $n$ negative eigenvalues and a one-dimensional kernel. Its form consequently has negative index $n$. These Robin eigenvalues are comparison data, not asserted to be the physical eigenvalues.

*M.2 Closed augmented form and exact index preservation*

Let

$$\mathcal{H} = L^2((0, D), w\, ds) \oplus \mathbb{C}^2, \quad Y = (F, \vartheta_a, \vartheta_b), \qquad [Y, Y] = \|F\|_w^2 - |\vartheta_a|^2 + |\vartheta_b|^2. \tag{L5}$$

Thus $[Y, Z] = \langle JY, Z\rangle_{\mathcal{H}}$, with $J$ negative on the lower endpoint coordinate only. For finite walls take $\mathcal{V} = H^1(0, D) \oplus \mathbb{C}^2$ and define the Hermitian form

$$s[Y] = s_R[F] + (-z_a)|\vartheta_a - e_a F_a|^2 + z_b|\vartheta_b - e_b F_b|^2. \tag{L6}$$

The two added coefficients are positive. Expanding them gives the boundary part

$$p_a A_a |F_a|^2 - p_b A_b |F_b|^2 - z_a|\vartheta_a|^2 + 2d_a \operatorname{Re}(F_a \overline{\vartheta_a}) + z_b|\vartheta_b|^2 - 2d_b \operatorname{Re}(F_b \overline{\vartheta_b}),$$

which verifies the endpoint orientation directly.

The form is densely defined and lower bounded. The one-dimensional trace estimate

$$|F_a|^2 + |F_b|^2 \le \epsilon \|F'\|_2^2 + C_\epsilon \|F\|_2^2$$

absorbs the possibly negative Robin terms into the uniformly positive principal term. For sufficiently large $c$, the squared form norm $s[Y] + c\|Y\|_{\mathcal{H}}^2$ is equivalent to the squared complete $H^1 \oplus \mathbb{C}^2$ norm. The upper estimate uses bounded traces; the lower estimate uses the trace bound and the nonnegative squares. Hence $s$ is closed. The representation theorem gives its self-adjoint lower-bounded operator $S$ [12, Theorem 10.7]. The form-domain embedding into $\mathcal{H}$ is compact, by one-dimensional compactness and the finite endpoint coordinates; equivalently the resolvent is compact [12, Proposition 10.6]. Explicitly, the coercive estimate bounds $(S + c)^{-1}$ into $\mathcal{V}$, whose embedding into $\mathcal{H}$ is compact.

The negative index is unchanged. Any negative subspace of $s$ projects injectively to a negative subspace of $s_R$, because the squares are nonnegative. Conversely, lift every negative Robin subspace by $\vartheta_i = e_i F_i$, making both squares zero. Thus

$$\operatorname{ind}_- s = \operatorname{ind}_- s_R = n. \tag{L7}$$

In the polarized kernel equation $s(Y, Z) = 0$ for every $Z \in \mathcal{V}$, the independent endpoint variations force $\vartheta_i = e_i F_i$, and the remaining equation is the ordinary Robin zero equation. Therefore

$$\ker S = \operatorname{span}\{Y_0\}, \qquad Y_0 = (F_0, e_a F_{0,a}, e_b F_{0,b}), \qquad [Y_0, Y_0] = -\frac{3}{2}\int_0^D \frac{ds}{a^2 v^2} < 0. \tag{L8}$$

The last equality is the independent K.6 signature calculation. The vector $Y_0$ and the metric $J$ do not change with stiffness. This is the artificial scalar root, not a physical zero mode.

*M.3 Removing the negative kernel and recovering the physical spectrum*

Set $\mathcal{M} = \{Y : [Y, Y_0] = 0\}$. This is uniformly positive in the metric (L5), not merely nonnegative. To see this explicitly, write the ambient Hilbert space as its positive coordinates plus the one negative coordinate, and let $Y_0 = (u_0, \alpha)$, with $r = \|u_0\|/|\alpha| < 1$. Orthogonality gives $|\vartheta_a| \le r\|u\|$, so

$$[Y, Y] \ge \frac{1 - r^2}{1 + r^2}\|Y\|_{\mathcal{H}}^2, \qquad Y \in \mathcal{M}. \tag{L9}$$

Projection along $Y_0$ is bounded and preserves the form domain, since $Y_0$ belongs to it. Its range on $\mathcal{V}$ is dense in $\mathcal{M}$, and $\mathcal{V} = \operatorname{span}\{Y_0\} \oplus (\mathcal{V} \cap \mathcal{M})$. Because $s(Y_0, Z) = 0$, removing this kernel leaves the negative index unchanged. On the positive Hilbert space $(\mathcal{M}, [\cdot, \cdot])$, the restricted form is closed, lower bounded and compactly embedded, by (L9) and the preceding coercive estimate. Its self-adjoint form representative $T_{\mathcal{M}}$ has compact resolvent and exactly $n$ negative eigenvalues by the variational principle [12, Theorem

12.1]. It has no zero eigenvalue: a restricted zero solution extends to the full kernel, whose intersection with $\mathcal{M}$ is zero.

For finite walls the full weak equation

$$s(Y,Z) = \lambda[Y,Z] \quad \text{for all } Z \in \mathcal{V} \tag{L10}$$

gives the bulk scalar equation and exactly (L2), by interior, endpoint-coordinate and endpoint-trace variations. Interior regularity makes the solutions classical. For $\lambda \neq 0$, testing against $Y_0$ forces $Y \in \mathcal{M}$. Conversely a restricted eigenfunction satisfies (L10) against all $\mathcal{V}$, since both sides vanish against $Y_0$. Consequently the nonzero scalar-pencil eigenvalues are exactly those of $T_{\mathcal{M}}$. Appendix K and (K10) identify them, including characteristic multiplicity, with the physical nonzero roots. Thus exactly $n$ physical eigenvalues have $\lambda < 0$, independently of the physical-phase proof.

This construction is an auxiliary operator realization of the separated transformed pencil. Its spectral theorem is not an expansion theorem for arbitrary original Einstein–scalar initial perturbations: the inverse map contains the spectral parameter, and no unrestricted physical Cauchy-data space has been identified with $\mathcal{M}$.

*M.4 Dirichlet faces and stiffness comparison*

At a physical Dirichlet endpoint use the exact trace condition

$$\vartheta_i = e_i F_i, \qquad p_i F_i' = p_i R_i^0 F_i + \lambda e_i^2 F_i. \tag{L11}$$

Keep the same $\mathcal{H}, J, Y_0, \mathcal{M}$. Impose $\vartheta_i = e_i F_i$ in the form domain at each Dirichlet wall and omit its square in (L6); finite endpoints retain their free coordinate and square. The resulting domain is closed in the $H^1 \oplus \mathbb{C}^2$ topology and dense in $\mathcal{H}$. For density, approximate an arbitrary bulk $L^2$ function by smooth interior functions, then add arbitrarily narrow $H^1$ boundary layers realizing any prescribed endpoint $\vartheta_i/e_i$ with arbitrarily small bulk $L^2$ norm. No bounded derivative norm is required for that density approximation.

The coercive estimate and compact embedding still hold. Projection along $Y_0$ preserves these constrained domains. The negative-index comparison is unchanged: every negative Robin subspace can still be lifted by $\vartheta_i = e_i F_i$ at all endpoints. The kernel remains (L8). In (L10), a constrained endpoint variation has $\zeta_i = e_i G_i$; the boundary term gives exactly (L11), with the same oriented signs at either wall. Hence the proof covers either or both affine transformed Dirichlet faces directly. It does not send a finite pole to infinity inside a quotient chart.

The same comparison also explains the standard stiffness behavior. At finite walls $\rho_a = -z_a$, $\rho_b = z_b$ increase with the respective nonnegative Hessians, while $e_i, Y_0, \mathcal{M}$ remain fixed. The variational principle gives nondecreasing ordered eigenvalues, with Dirichlet ceilings. For a physical eigenmode, $\vartheta_i - e_i F_i = -e_i f_i$ and $\partial_{\kappa_i} \rho_i \, e_i^2 = a_i^4/2$; its norm is $[Y,Y] = Z/2$. The simple-root derivative is therefore exactly $a_i^4 f_i^2 / Z$, as used in Appendix N. These are standard variational consequences after the reconstruction, not separate novelty claims.

*M.5 Consequence for contribution and limits*

The restricted count is a standard-corollary chain: a regular curved reduction and exact physical trace reconstruction give a fixed Robin zero mode with $n$ nodes; ordinary Sturm theory gives form index $n$; positive endpoint squares preserve it; removing the strictly negative-metric artificial kernel produces a positive spectral problem with that same negative count. The action-specific transformation, physical admissibility and exceptional-root work are the nontrivial translation required to use the established machinery. The count follows from the explicit domain and form arguments above.

This proof concerns $\lambda < 0$, equivalently $\mu < -4k$. Positive $\lambda$ can still correspond to $-4k < \mu < 0$; the original background's complete tachyon classification continues to require the validated remaining-band phase enclosure and wall injection. The contribution is this physical reconstruction and certified application within established spectral theory.

## Appendix N. Phase monotonicity and the complete neutral count

*N.1 Green identity and nonvanishing*

Let $(\psi_i, f_i, g_i)$ satisfy the regular constrained system at mass $\mu_i$, and set $L_i = g_i + 2T\psi_i$. Direct substitution gives

$$\left[a^4(f_iL_j - L_if_j)\right]' = (\mu_i - \mu_j)a^2(f_if_j + 6\psi_i\psi_j). \tag{M1}$$

For verification put $U = \psi_if_j - f_i\psi_j$, $W = f_ig_j - g_if_j$, $\Delta\mu = \mu_i - \mu_j$. Then $W' = -4HW + \Delta\mu f_if_j/a^2 + 4(T' + HT)U$ and $TU' = (T' - 2HT)U - 3\Delta\mu\psi_i\psi_j/a^2$. These identities give (M1) without dividing by $T$. The lower flux vanishes for a fixed lower physical wall. Differentiating in the second mass and setting the masses equal yields

$$\partial_\mu\theta(s,\mu) = \frac{Z_s}{2a(s)^4(f^2 + L^2)} > 0, \quad \theta = \arg(L + if), \quad Z_s = 2\int_0^s a^2(f^2 + 6\psi^2)\,dr. \tag{M2}$$

For $\lambda \neq 0$, $(f, L)$ cannot vanish anywhere: the constraint gives $\psi = 0$, then IVP uniqueness makes the entire profile zero. Thus (M2) defines a continuous spectral phase lift, not only a derivative evaluated at an eigenvalue. Complex conjugate pairing in (M1) gives reality. Appendix L supplies the exhaustive characteristic and its simple roots on the full compact wall square.

*N.2 Wall response and compact continuation*

For finite Hessians the same Green identity with one wall varied gives

$$\mu_{\kappa_i} = \frac{a_i^4 f_i^2}{Z} > 0. \tag{M3}$$

Its sign is the same at both ends because the two physical wall orientations are opposite. A finite-wall $f_i$ at a nonzero shifted root cannot vanish without making all profiles zero.

Use $t_i = (1+\kappa_i)^{-1} \in [0,1]$, $j_a = -(1-t_a)/2$, $j_b = (1-t_b)/2$, $\widetilde{E}_i = t_i T_i' + j_i T_i > 0$. The entire lower launch in (K4) has $f_a = -3\lambda t_a/(a_a^2 \widetilde{E}_a)$. At an upper root write $(f_b, L_b) = \zeta_b(t_b, -j_b)$, $\zeta_b \neq 0$. The response extends to all faces as

$$\mu_{t_a} = -\frac{9\lambda^2}{Z\widetilde{E}_a^2} < 0, \qquad \mu_{t_b} = -\frac{a_b^4 \zeta_b^2}{Z} < 0. \tag{M4}$$

This follows from (M3) in the interior and analyticity of the simple normalized root at a face. Appendix L proves the physical determinant is strictly negative at $\lambda = 0$, uniformly on the compact square. Its continuity on a compact spectral neighborhood gives a common two-sided root-free strip around that point. No numerical value of that strip is required.

Given a remaining-band tachyon $-4k < \mu < 0$ at any target wall, increase both $t_i$ toward the soft corner $(1,1)$. Its mass decreases by (M4). It cannot escape through zero mass, or through the uniform strip at $-4k$. At a putative finite continuation endpoint, compactness of the mass interval and IVP continuity give a limiting root; simplicity extends it by the implicit-function theorem. Distinct roots cannot merge because the limiting characteristic derivative is nonzero. Therefore

$$N_{(-4k,0)}(t_a, t_b) \leq N_{(-4k,0)}(1,1). \tag{M5}$$

This is an injection, not a general equality. If a non-soft target had a massless root, the same path has negative initial mass derivative (at least one coordinate increases). It immediately becomes a remaining-band tachyon and would also reach the soft corner. Thus an empty soft-corner band and absence of a soft-corner zero exclude every extra tachyon and every massless mode throughout the square. The restricted count of Appendix M is one, including Dirichlet faces, so these exclusions complete the count.

### *N.3 Physical zero mass and its anchored phase*

At zero Hessians the normalized launch is

$$\psi_a = 1, \quad f_a = -3\lambda/T_a', \quad g_a = -2T_a, \quad L_a = 0, \qquad a_a = 1. \tag{M6}$$

At $\lambda = 0$, put $f = -Tu_0$, $L = -T'u_0$, $\psi = u_0'/2$. The auxiliary equation is $(a^2 u_0')' = 2a^2 T^2 u_0/3$, $u_0(a) = 0$, $u_0'(a) = 2$. Hence $u_0, u_0' > 0$ inside the collar. This is not the everywhere-positive auxiliary $v$ used in Appendix K's transformation. With exactly one simple zero of $T$, $f$ is first positive and then negative; at that zero $L = -T'u_0 < 0$. Its continuous spatial lift passes through $\pi$, and the upper signs $T_b, T_b' > 0$ place its endpoint strictly in $(\pi, 3\pi/2)$. There is no hidden extra winding, since $f$ has only that one interior zero.

For positive $\lambda$, (M6) has lower phase $3\pi/2$. This choice is tied to the preceding auxiliary lift. Indeed the regular equations give $L' = -4HL + (V'' - \mu/a^2 + 4T^2/3)f - 2T'\psi$, so at the lower corner $(s,\lambda) = (0,0)$, $L' = -2T_a' < 0$. It remains negative in a small corner neighborhood. The phase therefore joins through the left half-plane to the positive-$\lambda$

lower launch; resetting its principal value to $-\pi/2$ would change the winding by one. Nonvanishing then extends that lift continuously up to physical $\mu = 0$, which means $\lambda = 4k$.

Upper zero-Hessian roots satisfy $L_b = 0$; their angles above the auxiliary endpoint are $3\pi/2 + j\pi$, $j \geq 0$. Strict monotonicity (M2) counts crossings up to physical zero mass. Hence

$$\theta_b(\mu = 0) < 3\pi/2 \quad \Longrightarrow \quad N_{(-4k,0)}(1,1) = 0 \text{ and no soft-corner massless mode.} \quad \text{(M7)}$$

An equality would be a physical massless root, not a tachyon. An interval merely touching the threshold is inconclusive. The exact enclosure in manuscript (14), justified as a computer-assisted premise in Appendix O, is strictly below the threshold. Combining (M5)–(M7) with the one-root restricted count proves Proposition 1. It counts radial eigenvalues in the generic coupled sector, not harmonic multiplicities or arbitrary Cauchy-data expansions.

*N.4 Why the selected exact background has one turn*

For $\mathcal{Q} = V'' + 4k/a^2 + 4T^2/3 > 0$, the background identity is $(a^4T')' = a^4\mathcal{Q}T$. If $T$ had two distinct zeros, multiply this equation by $T$ and integrate between them to obtain $-\int a^4(T')^2 = \int a^4\mathcal{Q}T^2$, impossible for a nonzero solution. A zero with $T' = 0$ would force the solution of this regular linear equation to vanish identically. The opposite endpoint signs therefore imply exactly one simple zero. For the exact quadratic potential the coefficient is strictly positive. The retained numerical geometry also supplies the required endpoint signs and positive endpoint $T'$. No secondary concave/quartic example is used.

## Appendix O. Mathematical justification of the numerical premises

*O.1 Exact problem and dependency chain*

For the exact potentials in manuscript Appendix A, define the original upper-wall residuals

$$F_1(s, A) = H(s, A) - \tau_b(\Phi(s, A))/6, \qquad F_2(s, A) = T(s, A) + \tau_{b,\Phi}(\Phi(s, A))/2. \quad \text{(N0)}$$

The local branch is the first transverse zero of $F_1$, followed by a zero of $F_2$ there. The background wall coefficient remains 20. Numerical truth of the enclosures is a stated outward-execution premise; the following arguments explain why the required predicates imply the result, without re-running a producer.

Proposition 1 combines two certificates with the analytical counting argument. Their dependency is

The exact action and launch determine a background root and its geometry. That linked background supplies the regular zero-Hessian phase enclosure, which combines with Appendix N to complete Proposition 1.

manuscript Appendix A fixes all six rational action parameters. The proper-distance lower launch for any scalar value $A$ is

$$\begin{aligned} W_a = 0, \quad \Phi_a = A, \quad T_a = \gamma(A - v_0)/2, \\ H_a = -[\lambda_a + \gamma(A - v_0)^2/2]/6, \\ k(A) = H_a^2 - 1 - T_a^2/12 + m^2 A^2/12. \end{aligned} \tag{N1}$$

where $W = \log a$. The exact central launch interval is the one in manuscript Appendix A; the endpoint interval is $I_D = [32727/16384, 32729/16384]$. Those rational intervals, rather than a fitted central trajectory, are the spectral inputs. Interval evaluation may lose correlations between $A$ and $k(A)$; outward bounds enlarge the enclosed set and still include the required correlated trajectory.

The background wall potentials remain the original potentials when this launch is constructed. Setting the **perturbative** wall Hessians to zero means adding background-preserving quadratic wall terms, not replacing $\gamma = 20$ in (N1). For $\mu = 0$, the phase calculation explicitly replaces the default finite-Hessian perturbative launch by

$$T_a' = m^2 A - 4H_a T_a, \qquad \psi_a = 1, \quad f_a = -12k/T_a', \quad g_a = -2T_a. \tag{N2}$$

The certificate checks $k > 0$, $T_a' > 0$, and $f_a < 0$. Equation (N2) satisfies the exact constraint and $L_a = g_a + 2T_a\psi_a = 0$. Its lifted phase is $3\pi/2$, by the positive-shift corner argument in Appendix N.3. Physical zero mass here means $\lambda = 4k$, not the auxiliary endpoint $\lambda = 0$.

*O.2 Why the enclosed background exists*

Let $F_1, F_2$ be the residuals (N0). On a regular background trajectory, validated event bounds identify the first zero $D(A)$ of $F_1$: all earlier cells are excluded, the residual changes sign across the event bracket, and $\partial_s F_1 < 0$ on it. The retained event bracket lies strictly inside the integration interval $(0, 2)$, so the preceding positive and following negative sign regions are nonempty. Those sign regions are part of the event proof; a vacuous "all cells" test on an empty region would not establish a crossing. The implicit-function theorem defines a differentiable event map there. Propagating the launch sensitivity $y_A$ together with the background gives

$$D'(A) = -\frac{\partial_A F_1}{\partial_s F_1}, \qquad q'(A) = \partial_A F_2 + \partial_s F_2 D'(A), \qquad q(A) = F_2(D(A), A). \tag{N3}$$

The derivatives include the dependence of the launch and $k(A)$. On a frozen launch box $I$, interval Newton evaluates

$$N(I) = A_0 - \frac{q(A_0)}{q'(I)}, \qquad A_0 \in I, \quad 0 \notin q'(I), \quad N(I) \subset \operatorname{int} I. \tag{N4}$$

These interval-Newton conditions [16] establish existence and uniqueness of the shooting zero in $I$. The high-resolution producer's exact Newton image is the interval $I_A$ printed

in manuscript Appendix A, equation (31); its event enclosure is $I_D$. The independent background implementation establishes a root of the same shooting function in the common frozen box. Identification uses that uniqueness result, not overlap of approximate roots. Positive warp, positive curvature, event transversality and the opposite endpoint scalar-derivative signs are the retained geometric predicates. The quadratic potential is convex, so Appendix N.4 then supplies the one-turn and $T' > 0$ consequences.

Only the local root and geometry are needed here; the separate larger global covers and fixed-distance exclusions are not premises of Proposition 1.

The local background existence, tight launch/event enclosures and required geometry were regenerated and internally reviewed in the identified 18 October 2026 source executions, separately from the phase run. The original high-resolution implementation supplies the exact narrow image and event interval, and the separately written implementation supplies local root and geometry enclosures on their common frozen launch box. The phase run still does not prove background existence: it consumes those enclosures. The larger global covers and simultaneous-wall Jacobian were not regenerated by this local chain and retain their earlier certificate status. Appendix H states the separate assurance boundaries.

*O.3 Interval flow enclosure*

The spectral calculation propagates the background and regular scalar equations together. The primary state is $Y = (W, H, \Phi, T, \mathcal{E}, \psi, f, g)$, where $\mathcal{E} = e^{-W}$ obeys $\mathcal{E}' = -H\mathcal{E}$, $\mathcal{E}_a = 1$; occurrences of $a^{-2}$ become $\mathcal{E}^2$. The second implementation propagates $(W, H, \Phi, T, \psi, f, g)$ and evaluates $e^{-2W}$ directly. Both use manuscript equations (9) and (10), fixed $\mu = 0$, and the same exact inputs. They are separately written recurrence implementations, but share the Arb arithmetic library [17] and background/theorem premises.

Write either smooth autonomous system as $Y' = F(Y; p)$, where $p$ denotes fixed parameters in outward intervals. On each cell of width $h$, an interval starting state $Y_j$ is enclosed in a padded whole-cell box $B_j$. The accepted test is strict componentwise inclusion

$$\operatorname{hull}\{Y_j,\ Y_j + [0, h]F(B_j; p)\} \subset \operatorname{int} B_j. \tag{N5}$$

Local existence and uniqueness follow from the smooth right-hand side. More explicitly, before a hypothetical first exit the integral equation places every exact trajectory in $Y_j + [0, h]F(B_j; p)$, strictly inside $B_j$, contradicting that exit. A solution confined to a compact box in the smooth ODE domain extends to time $h$. This argument is uniform for all enclosed initial states and fixed enclosed parameters; it does not require a computed contraction constant. No numerical Picard contraction constant is claimed: the implementation tests strict self-containment, not a contraction estimate. Failed inclusions stop the calculation.

Define normalized autonomous Taylor coefficients recursively by $c_0(Y; p) = Y$ and

$c_{r+1} = (D_Y c_r)F/(r+1)$. Interval recurrences enclose these coefficients. The propagated endpoint and any clipped subinterval use

$$Y(s_j + t) \in \sum_{r=0}^{4} c_r(Y_j; p)t^r + c_5(B_j; p)t^5, \qquad 0 \le t \le h. \tag{N6}$$

Indeed, for each component $i$, Taylor's theorem gives a remainder $c_{5,i}(Y(s_j + \xi_i); p)t^5$, with $0 \le \xi_i \le t$. The components need not share one remainder point: each lies in the same whole-cell box, so outward evaluation of $c_5(B_j; p)$ encloses all of them. The recurrence computes the normalized jets $Y^{(r)}/r!$. The fifth coefficient evaluated on $B_j$ therefore bounds the normalized fifth derivative at every possible remainder point. Thus a degree-four polynomial plus a fifth-order remainder encloses the trajectory, rather than treating a truncated numerical series as exact. Polynomial evaluation is by interval Horner arithmetic. The primary cell width is 2/8192; the second is 2/16384. Both use 768-bit ball arithmetic, giving 8,183 and 16,365 cells respectively through the upper end of $I_D$.

Arithmetic operations and elementary functions use outward Arb balls. Inputs are formed from exact rational literals, not rounded decimal parameters. Each cell also checks finite state bounds, positive warp, and zero containment of the first-integral and linear-constraint residuals. The last two are consistency gates; they do not prove exact constraint satisfaction by themselves. Exact launch identities and analytical propagation of the constraints supply that assertion.

### *O.4 Whole-cell phase charts and the lift*

On every whole-cell box compute intervals for $f$ and $L = g + 2T\psi$. At least one of the following strict half-plane conditions must hold. It supplies the indicated interval angle:

Table 7 lists the admitted local phase charts.

**Table 7. Whole-cell conditions and their corresponding local phase charts.**

| Whole-cell condition | Local angle chart |
|---|---|
| $L > 0$ | $\arctan(f/L)$ |
| $f > 0$ | $\pi/2 - \arctan(L/f)$ |
| $L < 0$ | $\pi + \arctan(f/L)$ |
| $f < 0$ | $3\pi/2 - \arctan(L/f)$ |

All inequalities hold for the complete interval, so the selected denominator excludes zero. The entire exact path in that cell lies in one open half-plane: it cannot contain the origin or make an unobserved full winding. If no chart is admitted, the calculation stops rather than choosing a principal angle at the endpoint.

Starting from $3\pi/2$, add an integer multiple of $2\pi$ to the new chart so that it agrees with the previous lifted endpoint. Floating arithmetic proposes this integer, but does not certify it. Acceptance requires outward-ball verification that the shifted chart is at distance strictly less than $\pi$ from the preceding reference midpoint and overlaps the

preceding endpoint interval. The new endpoint is evaluated with (N6), assigned its own half-plane chart and checked for the same unambiguous lift and overlap with the whole-cell chart. These checks, together with whole-cell nonvanishing, propagate the chosen continuous phase, rather than relying on sampled unwrapping.

The branch-selection argument can be stated without relying on overlap alone. Let $P$ be the preceding enclosure of the true lifted shared-endpoint angle, with radius less than $\pi$, and let $A$ enclose the next half-plane chart. For a proposed integer $n$, require the rigorous inclusion

$$A + 2n\pi \subset (\operatorname{mid} P - \pi, \operatorname{mid} P + \pi). \tag{N7}$$

The preceding true angle and the shifted chart's representative at that same endpoint are now in one open interval of length $2\pi$, and differ by an integer multiple of $2\pi$; they must coincide. The continuous half-plane branch therefore propagates the correct lift throughout the cell. The accepted interval also has radius less than $\pi$, so the argument iterates. The initial $3\pi/2$ enclosure satisfies the radius condition. The source's strict distance-from-reference-midpoint bound, verified using an outward angle ball and a lower bound for $\pi$, implements (N7); a floating proposal for $n$ cannot replace it. The same reasoning applies to endpoint and clipped-event charts, using the accepted cell chart as reference. This proves the branch choice under the stated outward arithmetic premises; coarse saved-trace overlap alone does not establish it.

The saved compact traces export angle bounds divided by $\pi$ using exact rational floor/ceiling at denominator $2^{40}$. This outward trace coarsening does not replace the higher-precision producer checks. Rechecking saved trace overlap authenticates a necessary consistency property; the traces alone do not reconstruct (N5), (N6) or the elementary-function bounds.

*O.5 Uncertain wall location and exact terminal comparison*

For every grid cell intersecting $I_D = [D_-, D_+]$, evaluate (N6) on

$$[\max(0, D_- - s_j),\ \min(h, D_+ - s_j)]. \tag{N8}$$

The phase on each such clipped event interval is lifted into its whole-cell chart. The saved global endpoints are checked as exact rationals: the first starts at $D_-$, every next starts where its predecessor ends, and the last ends at $D_+$. Each engine has two event pieces. The hull of these lifted phase enclosures covers the true wall at the unknown $D_*$; this is stronger than evaluating the phase at a numerical estimate of the wall.

Both historical and freshly regenerated final intervals for $\theta_b(\mu = 0)/\pi$ equal $[577838035/536870912, 577858823/536870912]$. They lie strictly inside $[1076307/10^6, 1076346/10^6]$. Their approximate finer endpoints are 1.076307212933898 and 1.0763459336012602. The coarser rational bound is sufficient: its upper endpoint is less than $3/2$ by $423654/10^6 > 0$. No floating-point threshold decision is needed. The auxiliary phase at $\mu = -4k$ lies in $(\pi, 3\pi/2)$, and its spectral derivative is positive. Therefore the interval from the auxiliary endpoint to physical zero mass crosses no

admissible upper-wall angle. Appendix N.2 transfers this empty remaining band from the zero-Hessian corner to the full nonnegative family.

*O.6 The selected baseline mass certificate and charged nonresonance*

The selected completion uses a separate physical mass bracket, with original perturbative Hessians fixed at $\kappa_a = \kappa_b = 20$. Let $\mu_- = -33/25000$, $\mu_+ = -131/100000$. Normalize the regular constrained problem by

$$\psi_a = 1, \qquad f_a = -\frac{3(\mu + 4k)}{T_a' - 10T_a}, \qquad g_a = 10f_a - 2T_a, \qquad a_a = 1,$$
$$\mathcal{D}_{20}(\mu) = g_b + 2T_b\psi_b + 10f_b = L_b + 10f_b. \tag{N9}$$

The lower denominator is positive because $T_a' > 0 > T_a$. These launch values satisfy the exact constraint and $L_a = 10f_a$; the upper residual is precisely the oppositely oriented wall law $L_b = -10f_b$. The initial values and coefficients are analytic in $\mu$, so this physical residual is continuous across the bracket. No oscillator is included in this baseline problem; its mass is the comparison branch used after the negative-axis oscillator elimination.

The original packet is `lin-central-validated-mode-r1-20261005T172800Z`. Its `BINDINGS.json` and completed `output/RESULT.json` contain the six exact action parameters of manuscript Appendix A and exactly the same $I_A, I_D$. Each engine propagates the whole launch interval and clips every event cell to cover the entire uncertain wall interval. Strict Picard and polynomial Taylor/remainder predicates are those of O.3; the primary uses $E = e^{-W}$ as an independent polynomial state, and the second implementation uses its separately written recurrence. The active runner assigns each endpoint mass to these equations, rather than only storing it in a configuration field.

The completed exact-rational endpoint enclosures are strictly contained in the following deliberately coarse intervals:

Table 8 gives conservative exact enclosures for the endpoint residuals.

**Table 8. Opposite-sign endpoint residual enclosures for the selected mass certificate.**

| Implementation | $\mathcal{D}_{20}(\mu_-)$ | $\mathcal{D}_{20}(\mu_+)$ |
|---|---|---|
| Primary | $[29, 30]$ | $[-9, -8]$ |
| Second implementation | $[29, 30]$ | $[-9, -8]$ |

Both event covers run from 32727/16384 to 32729/16384, with the shared interior division 4091/2048. Each sign thus holds at the exact physical wall, not just at a selected numerical endpoint. The background-root uniqueness in O.2 identifies the launch and geometry used here with the present exact background; no agreement of approximate profiles is substituted for that linkage.

Direct rational interval evaluation of (N1) on the exact launch box gives

$$\frac{3}{10000} < k < \frac{131}{400000}, \qquad \mu_+ < -4k. \tag{N10}$$

The intermediate-value theorem gives a physical root between the two masses. The restricted count in Appendix M, already established independently of this mass bracket, makes it the unique baseline root below $-4k$. Consequently

$$-\frac{33}{25000} < \mu_{20,20} < -\frac{131}{100000}. \tag{N11}$$

This identifies an existing root; it does not use charged existence to prove the nonresonance needed for that existence. For $\Gamma_i \geq 20$, stiffness comparison gives $\alpha(\Gamma_a, \Gamma_b) \geq \mu_{20,20}$. Since $\mu_n \leq -18k < -27/5000 < \mu_-$, the analytic branch's scalar problems have no kernel. For the growing solution, (N10)–(N11) imply manuscript (24); the shifted comparison (S8) then excludes every positive-order shifted scalar kernel. The other graph, metric, oscillator and phase divisors are handled separately in Appendices C/D.

The local `numerical_evidence/MASS_CERTIFICATE_MAP.json` binds selected copies of the historical bindings, full-packet manifest, both active interval engines, runner and exact endpoint results to their originals. The original manifest contains the broader packet; the selected copies are not represented as a portable full replay. The binding records the executable, imported extensions/libraries, sources, 768-bit Arb precision and single-thread setting used for the historical route. These records identify the execution premise; they do not establish arithmetic correctness or continuous runtime surveillance by their hashes alone. Both implementations use Arb and shared input mathematics, so their agreement is not independent arithmetic. The signs and linkage can be checked from saved records, but the complete interval trajectories remain inherited outward-execution premises. No mass integration is rerun here.

*O.7 Referee inspection and saved-record verification*

The accompanying anonymous evidence package contains the linked background/root and phase source/record chains, final internal acceptance dispositions, selected-mass records and kinetic certificate cells. Its `README.md` identifies one read-only, standard-library verification entry point. That checker verifies package identities, exact terminal signs and phase comparisons, uncertain-event coverage, and the kinetic saved-cell hull. It imports no producer and performs no integration. The mass contract and freeze record are included for inspection. Historical producer scripts still have their original environment and predecessor assumptions; inclusion is not qualification of portable regeneration. Location metadata are anonymised, with separate original and derivative hashes. Unmodified arrays and source files remain distinguishable from inspection derivatives. The recovered exact-Fraction verifier and its separately qualified portable launcher now provide a stronger saved-cell route, distinct from that package's inventory checker. Its dependency manifest, new execution and rejection controls are documented in the pipeline supplement. Referee-package preparation and external anonymous delivery remain distinct from qualification of the identified public source tree.

## Appendix P. Neutral Einstein-fiber compatibility

Fix the exact root $(a, \Phi_0, D, k)$ of manuscript (9), with $a > 0$. Let $\gamma$ be a smooth time-oriented Lorentzian four-metric with $\mathrm{Ric}(\gamma) = 3k\gamma$. On its product with the reflected collar, set $g_5 = ds^2 + a^2\gamma$, $\Phi = \Phi_0$, $\chi_i = 0$, constant $\theta_i$ and $J_i = 0$. These fields solve the same action's bulk and wall equations. This is a standard warped-product construction [25, Section 3]; the verification below keeps the present scalar potential and centered interface action fixed.

Indeed, in the bulk,

$$\begin{aligned} \mathrm{Ric}_{ss} &= -4a''/a, & \mathrm{Ric}_{s\alpha} &= 0, \\ \mathrm{Ric}_{\alpha\beta} &= [3k - aa'' - 3(a')^2]\gamma_{\alpha\beta}, \\ \Box_5\Phi_0 &= T' + 4HT. \end{aligned} \tag{O1}$$

Equation (9) gives $\mathrm{Ric}_{AB} = \Phi_{0,A}\Phi_{0,B} + \frac{2}{3}(V - 6)g_{AB}$ and $\Box_5\Phi_0 = V_{,\Phi}$. At each wall, $K_{\alpha\beta} = \sigma_i H_i h_{\alpha\beta}$ and $2n\Phi_0 = 2\sigma_i T_i$ are precisely the neutral junctions. The centered interaction and its first derivatives vanish; constant phases and centered oscillators satisfy their equations. Zero current uses the canonical neutral continuation, not a nondegenerate fluid-velocity chart.

For a closed spacelike fiber two-surface, choose future null normals with $\gamma(L_+, L_-) = -2$. The orthogonally reflected lift has normals $\ell_\pm = a^{-1}L_\pm$, and

$$dA_3 = a^2 ds\, dA_2, \qquad \theta^{(5)}_\pm = a^{-1}\theta^{(4)}_\pm. \tag{O2}$$

The compact-direction expansion trace vanishes and the one-sided values match at the interfaces. A two-sphere lifts to $S^2 \times S^1$ on the doubled collar. The spacetime is piecewise classical with Israel matching, not globally $C^2$. No smooth-spacetime null-energy singularity theorem is imported across the negative-tension wall. This observation establishes neither formation, an event horizon, stability of this sector nor nonlinear evolution of the scalar growing mode. The charged estimates remain on their original background.

## Appendix Q. Convergent neutral growing family and observable remainder

*Q.1 Fixed chart, normalization and harmonic recursion*

This appendix concerns action (1) with the selected completion (21), $\kappa_a = \kappa_b = 20$, zero currents and constant phases. The background and oscillator-completed negative mode are the same exact objects used in Sections 3–4 and Appendix O.6. Set $k = h^2$, $\mu_* = -r(r+3h)$, and normalize the real first mode by $p_1(a) = 1$. The enclosures retain the parameter correlations; independent endpoints of the intervals for $h, r$ are bounds, not separately chosen exact solutions. The normalization, wall potentials and homogeneous ansatz are fixed throughout.

Use the pre-pullback metric of (22) with $p, b, f$ expanded in $\eta = \epsilon e^{rt}$. Pull the moving interval to $I = [0, D_0]$ with

$$\begin{gathered} s = u + X, \quad X = \xi_a + \frac{u}{D_0}(\xi_b - \xi_a), \quad j = 1 + X_u, \\ \mathcal{D}_0 = r\eta\partial_\eta, \quad \mathcal{D} = \mathcal{D}_0 - \frac{\mathcal{D}_0 X}{j}\partial_u, \quad \mathcal{L} = j^{-1}\partial_u. \end{gathered} \tag{NL5}$$

Here $\xi_i(0) = 0$, and the first graph coefficients vanish. Background fields are composed with $u + X$; their Eulerian time derivative $\mathcal{D}$ is identically zero. In particular, an additional temporal derivative of a displaced background must not be added. All trace derivatives at a moving wall are total derivatives. The action equations and local oriented junction laws are those of Appendix C, with zero current sources and temporal rate $nr$, not its redshift rate $-6nh$.

The first coefficient is the selected regular neutral mode, including both oscillators. At $n \geq 2$, the known lower coefficients supply compatible sources and the operator has parameter $\mu_n = -nr(nr+3h)$. Its scalar constraint denominator is

$$\Delta_n = 2T^2 + 3[-\mu_n - 4h^2]a^{-2}, \qquad \Delta_n \geq 3n^2r^2a^{-2} > 0. \tag{NL6}$$

For this inequality, $r > h$ and $n \geq 2$ give $n^2r^2 + 3nrh - 4h^2 \geq n^2r^2$. The wall oscillator divisors are $2 - \mu_n/a_i^2 > 0$. The graph, scalar and oscillator solves therefore retain their original coupling. No homogeneous solution is added at higher order. The first, resonant mode is prescribed separately; a nonresonant estimate is never asserted at $n = 1$.

*Q.2 Uniform real and complex inverse estimates*

The real scalar equation has the form $f'' + A_n f' - V_n f = F_{Q,n}$, with outward Robin data and $A_n = 4H + B_0T/\Delta_n$, $B_0 = 12HT - 4m^2\Phi_0$. Here $F_{C,n}$ denotes the load $F_n$ of (Q3), and the positive-leading convention is explicitly $F_{Q,n} = -F_{C,n} = S_{g,n} + B_0S_{c,n}/\Delta_n$. Hence $|F_{Q,n}| = |F_{C,n}|$, and the absolute inverse estimates are unchanged. Write $E = a^{-1}$ and $f = E^2y$. The transformed coefficients are

$$\widetilde{A}_n = \frac{B_0T}{\Delta_n}, \qquad \widetilde{V}_n = m^2 - \mu_nE^2 + \frac{4}{3}T^2 - 2h^2E^2 + 4H^2 + \frac{B_0(T' + 2HT)}{\Delta_n}. \tag{NL7}$$

The Robin coefficients change to $\beta_a + 2H_a$ and $\beta_b - 2H_b$. Whole-cell bounds give a finite cutoff $N$, positive $c$, positive transformed wall coefficients and uniform bounds $\widetilde{V}_n \geq cn^2$, $|\widetilde{A}_n| \leq A_*$, $|\widetilde{V}_n| \leq v_* n^2$ for $n \geq N$. For interior load $F_0$ and maximum boundary load $G_0$, a barrier is

$$w = \frac{F_0}{cn^2} + \frac{2G_0}{\nu n}\left(e^{-\nu n(u-a)} + e^{-\nu n(b-u)}\right). \tag{NL8}$$

Choose $\nu > 0$ and the cutoff so $c - \nu^2 - A_*\nu/N \geq 0$, $\nu N D_0 \geq 1$, $N D_0 \geq 2$, and $1 - A_*/(2N) > 0$. The bulk operator applied to $w$ dominates $F_0$, and its outward derivatives dominate $G_0$; the nonnegative Robin terms help. The maximum principle bounds $n^2\|y\|_\infty$. A one-sided Taylor estimate over length $1/n$, followed by the differential equation and absorption, bounds $n\|y'\|_\infty$ and $\|y''\|_\infty$. The finite orders below $N$ receive separate positive-potential and boundary checks. Uniqueness makes the regular two-point boundary map invertible. Transfer through $f = E^2 y$ supplies uniform constants for the original scalar. These are bounds for compatible reduced forcing, not arbitrary independent Einstein residuals.

The exact background is holomorphic on a common stadium about $I$. One sufficient construction uses coordinate variation $d = 2^{-16}$ and disks of radius $d/128$, on which the polynomial background vector field has norm at most 64 and Jacobian row norm at most 32. Complex Picard contraction glues these disks by uniqueness. On the resulting collar, the scaled denominator $\Delta_n/n^2$ has real lower bound $3/16384$; its controlled complex variation is less than half that bound, uniformly in $n \geq 2$.

On a smaller stadium of radius $\sigma$, set $d_\sigma(z) = 1 - \operatorname{dist}(z, I)/\sigma$ and $\|v\|_p = \sup d_\sigma^p |v|$. A primitive from index $p+2$ to $p+1$ costs $\sigma/(p+1)$; a double primitive to $p$ costs $\sigma^2/[p(p+1)]$. Continuing the real solution and choosing $\sigma$ so the resulting positive two-by-two coupling matrix is a contraction bounds the scalar value and gradient at indices $2n, 2n+1$; the ODE bounds its second derivative at index $2n+2$. These estimates are first taken on closed inner stadia, so finiteness is not assumed before absorption.

The metric is continued using its first-order mixed constraint, $p' = -2Hp - Tf/3 + S_p$, and the algebraic constraint supplies its real anchor. A further Nagumo estimate gives its second derivative. This avoids differentiating an uncontrolled algebraic source twice. Interior forcing and boundary loads can be estimated separately by linearity, then recombined before reconstructing the physical solution. The selected refinements retain their distinct real and complex constants and all finite harmonic cases; they do not identify auxiliary scalar responses with separate Einstein solutions.

*Q.3 Seven sources, component majorant and the numerical radius*

For a bulk coefficient define

$$\mathcal{V}_n(v) = n^2\|v\|_{2n} + n\|v'\|_{2n+1} + \|v''\|_{2n+2}. \tag{NL9}$$

Let $B_n = \mathcal{V}_n(p) + \mathcal{V}_n(b) + \mathcal{V}_n(f)$, $X_n = n^3 \sum_i |\xi_{i,n}|$, and $C_n = n^3 \sum_i |\chi_{i,n}|$. The positive component ray is $(B_n, X_n, C_n) \preceq (1, \ell_g, \ell_c) w_n$, with $\ell_g = 1/256$, $\ell_c = 1$. These weights

condition a proof bound; they are not a decomposition of physical energy or force. The summed aggregate is bounded by $(513/256)w_n$.

Coefficient extraction from the exact pulled equations gives four bulk loads and three interface loads. The trace equation determines $b + 2p$; the mixed constraint, Klein–Gordon equation and radial constraint determine the compatible scalar and metric sources. At each wall the tracefree Israel row determines the graph response, while the scalar normal and oscillator equations determine the other two loads. The independent wall oscillator enters the nonlinear graph source through its kinetic stress and the oscillator equation through its full trace and friction; its linear scalar-junction coupling remains in the response operator. Removing its explicit higher-order source from a row where that coupling is exactly linear is not physical decoupling.

Each raw load admits a positive rational envelope $\mathcal{H}_j(B, X, C)$. Products add radial norm indices. A bulk jet with $a$ radial and $b$ temporal derivatives, $a + b \le 2$, costs at most $n^{a+b-2}B_n$ at radial index $2n + a$. A graph or oscillator time jet costs $n^{b-3}X_n$ or $n^{b-3}C_n$. The transport terms in (NL5) always attach their extra radial derivatives to a positive-order graph coefficient; its order pays the additional index. At walls, count primitive positive-order factors before composing products and reciprocals. For orders $i_1 + \cdots + i_m = n$,

$$\frac{n}{i_1 \cdots i_m} \le m. \tag{NL10}$$

Consequently the outer harmonic payment on each interface source is bounded by the primitive degree $m$, including graph acceleration and the moving-normal factors. Constants do not count as positive-order factors. All input series are truncated below $n$, so each envelope's own constant and linear terms have zero order-$n$ contribution. Removing those two jets does not cancel positive upper bounds between different source maps.

For explicit coarse envelopes, let $y \ge 0$, $g$ be the gap between the prior and inner radial collars, and define

$$\begin{gathered} G = (1-2y)^{-1}, \quad Z = 64(1-4y/g)^{-1}, \quad A_1 = (1-y)^{-1}, \quad A_2 = (1-2y)^{-1}, \\ w = 4yG, \quad w_u = 2yG, \quad w_t = 4yG + 8y^2G^2, \\ d = y(1+w), \quad d_2 = y(1 + 2w + ww_u + w_t + w^2), \quad l = Gy, \quad l_2 = G^2y. \end{gathered} \tag{NL18}$$

Here $d, d_2, l, l_2$ bound the pulled temporal first and second jets and radial first and second jets. Each separately composed background factor is bounded by $Z$, so its square is bounded by $Z^2$. The elementary exponential bounds use $1/j! \le 1$. Applying absolute values to the four bulk rows gives

$$\begin{aligned}
\mathcal{B}_T &= 3d_2 + 3d + 6d^2,\\
\mathcal{B}_M &= Gy(1 + w_u + w) + \tfrac{4}{3}d(Z + l),\\
\mathcal{B}_K &= Z^2A_2[d_2 + (3 + 3d)d]\\
&\quad + A_2[Z + l_2 + (4Z + 5l)(Z + l)] + Z + y,\\
\mathcal{B}_R &= \tfrac{13}{2}A_2(Z + l)^2 + 6 + \tfrac{1}{2}(Z + y)^2\\
&\quad + Z^2A_2[6 + 3(d_2 + 3d) + \tfrac{7}{2}d^2].
\end{aligned} \tag{NL19}$$

For the wall rows set

$$N_- = ZA_1, \quad v_+ = 4yZA_1^2, \quad v_{t,+} = ZA_1^2[4y + (2y + 4Zy)4y], \qquad B_+ = (1 - v_+^2)^{-1}. \tag{NL20}$$

The inverse lapse, velocity, total velocity derivative and boost reciprocals are bounded by these positive functions. The corresponding graph/Israel, scalar-normal and oscillator source envelopes are

$$\begin{aligned}
\mathcal{B}_I &= N_-B_+[v_{t,+}B_+ + v_+(2d + 1)] + \tfrac{1}{2}N_-^2B_+y^2,\\
\mathcal{B}_J &= B_+[A_1(Z + l) + v_+N_-d] + \tfrac{1}{2}[2 + 21(Z + 1 + y) + y],\\
\mathcal{B}_O &= N_-^2B_+[y + (3 + 8Zy + 2y + v_+v_{t,+}B_+)y] + 2y + Z + 1 + y.
\end{aligned} \tag{NL21}$$

The 21 in the scalar-normal row includes the original Hessian 20 and the centered interaction. The oscillator boost friction and kinetic graph stress are retained. In (NL19)–(NL21) the harmless constant and linear terms are removed from each envelope separately, only after truncating all inputs below the target order. Choose a positive evaluation point smaller than $\min(g/16, 1/4096)$, with all denominators positive and $v_+ < 1/2$. Cauchy's coefficient estimate bounds each degree by its envelope value divided by that point to the degree; (NL10) pays the wall factor. The finite uniform response constant of Q.2 then supplies a finite scalar majorant. This establishes the analytic source-control mechanism without assuming the sharpened numerical constants. The selected numerical radius below additionally uses the tighter, component-specific domination premise; it is not computed by substituting its $K$ into these deliberately coarse envelopes.

For example, if $v = e^{b-p}E(S)S_t$, then

$$v_t = e^{b-p}E(S)\{S_{tt} + [b_t^{\mathrm{wall}} - p_t^{\mathrm{wall}} - H(S)S_t]S_t\}, \tag{NL11}$$

where derivatives of wall traces are total derivatives. The graph source retains $\chi_t^2/(2\ell^2)$, the boost factors retain $(1 - v^2)^{-1}$, and the oscillator friction retains $vv_t/(1 - v^2)$. Background compositions use their analytic Taylor envelopes. In the tightened source budget, the Klein–Gordon composition of $T'$ uses its separately enclosed first jets and the common Cauchy tail, rather than substituting a bound for $T$. Thus the source refinement preserves derivative order and the full pulled transport.

Apply the positive inverse-response matrix of Q.2 to these seven loads, including the graph and oscillator solves. If the resulting component envelope on the chosen ray has

degree-$m$ coefficient at most $mM/a_0^m$, choose $K \geq \max(1, 1/a_0, 2M/a_0^2)$. Then it is coefficientwise bounded, after division by each ray weight, by $Kw^2/(1-Kw)^2$. This follows from $m \leq 2(m-1)$ and $Ka_0 \geq 1$. The first-mode analytic continuation is a separate linear initial-value calculation, with bulk seed at most 1215, zero graph seed and summed oscillator seed at most 1/8. It yields

$$w = s_1\zeta + \frac{Kw^2}{(1-Kw)^2}, \qquad s_1 = 1215, \quad K = 109802048057794951. \tag{NL12}$$

Here $\zeta$ is an independent complex amplitude for the analytic majorant; coefficient comparison is made at $\zeta = |\eta|$. The numerical source-domination premise is that the retained exact seven-row/component calculation establishes these inequalities at $a_0 = 21/2305843009213693952$, with the stated weights, inverse constants and seed. Its selected result is identified in Q.6. The coarser bounds used to explain Q.2 alone are not asserted to reproduce this sharpened $K$.

Let $R_0 = 1/(96K)$. On $|\eta| \leq R_0/(2s_1)$, the right side of (NL12) maps $|w| \leq R_0$ strictly into itself: the nonlinear part is less than $R_0/2$, and its derivative in $w$ is less than one. The contraction gives an analytic majorant. Restricting the radial collar to its inner half costs $4^n$ for field values at index $2n$, and $2\,4^n$ and $4\,4^n$ for first and second radial derivatives at indices $2n+1$ and $2n+2$. Dividing the amplitude radius by four pays the exponential factor; the fixed derivative factors do not change that radius or the field-value bound $Y$. Thus

$$\rho = \frac{R_0}{8s_1}, \qquad Y = \frac{513}{256}R_0 = \frac{171}{899498377689456238592}. \tag{NL13}$$

This gives exactly (NL2). On every strict inner disk geometric decay absorbs every fixed polynomial harmonic payment needed for the equations. The fields, their required derivatives and analytic compositions therefore converge. The estimates also keep $j \neq 0$, the intervals separated and the wall speeds below one. All assertions refer to the selected homogeneous analytic family.

*Q.4 Recovery of the remaining wall equation*

Convergence of a solved subset is insufficient. For a general homogeneous metric $\mathrm{diag}(-N^2, R^2, R^2, R^2, B^2)$, wall $s = S(t)$, speed $v = BS_t/N$ and boost $\Gamma = (1-v^2)^{-1/2}$, the exact extrinsic components satisfy

$$K^{\tau}{}_{\tau} = \sigma\Gamma\left[\frac{\Gamma^2 v_t}{N} + \frac{N_s}{NB} + \frac{vB_t}{NB}\right], \quad K^{x}{}_{x} = \sigma\Gamma\left[\frac{R_s}{BR} + \frac{vR_t}{NR}\right]. \tag{NL14}$$

The derivative $v_t$ is along the wall. Form the junction residual $\mathcal{J}^a{}_b = K^a{}_b - K\delta^a_b + S^a{}_b/2$, scalar normal residual $\mathcal{J}_\Phi = 2n\Phi - (\tau + W)_{,\Phi}$, bulk residual $E_{nU} = \mathrm{Ric}_{nU} - (n\Phi)(U\Phi)$, and oscillator residual $\mathcal{O} = \ddot{\chi} + 3H_{\mathrm{wall}}\dot{\chi} + W_{,\chi}$. Codazzi and the off-shell material exchange identity give

$$D_a\mathcal{J}^a{}_\tau = E_{nU} + \frac{1}{2}\mathcal{J}_\Phi\dot{\Phi} - \frac{1}{2}\dot{\chi}\,\mathcal{O}. \tag{NL15}$$

The sign and boost factors follow either by direct Christoffel/Ricci evaluation or by the two-dimensional base and warped spatial factor. Specifically, the geometric divergence is $-3\dot{K}^x{}_x + 3H_{\rm wall}(K^\tau{}_\tau - K^x{}_x)$, whose acceleration terms cancel. For $\rho = \dot{\chi}^2/2 + \tau + W$, direct differentiation gives $\dot{\rho} + 3H_{\rm wall}(\rho + P) = \dot{\chi}\mathcal{O} + (\tau + W)_{,\Phi}\dot{\Phi}$, producing (NL15) with the stated junction convention.

Once the bulk, scalar and oscillator rows vanish, the right side is zero. Inductively the lower junction coefficients vanish, and the tracefree row makes the remaining order-$n$ junction residual a pure identity tensor. Its leading divergence is $nr/\ell_{i,0}$ times its scalar coefficient, where $\ell_{i,0} > 0$ is the neutral wall lapse; all lower-order residual terms already vanish. Since $nr \neq 0$, that residual vanishes independently at each wall. The zero harmonic is the already fixed background and is not removed by this argument. With convergence, all residual Taylor coefficients vanish and hence the summed fields satisfy every equation.

*Q.5 Separation, actual clock and both signs*

The same pullback gives $D(\eta) = \int_I e^{b(\eta,u)} j(\eta,u)du$. On the analytic disk the aggregate bound implies $|D - D_0| \le M_D = 9Y/(1-Y)$. Thus Cauchy gives a tail after order $N$ bounded by $M_D q^{N+1}/(1-q)$, $q = |\eta|/\rho < 1$; Euler derivatives have the corresponding differentiated geometric tail. The actual lower-wall lapse is $\ell_a = N_a\sqrt{1 - v_a^2} > 0$, and

$$F(\eta) := \frac{dx}{d\tau_a} - rx = \frac{r\eta D'(\eta)}{\ell_a(\eta)} - r[D(\eta) - D_0]. \tag{NL16}$$

The normalization gives $\ell_a = 1 + \eta + O(\eta^2)$. Hence $F_0 = F_1 = 0$, $F_2 = r(d_2 - d_1)$, and the two retained coefficient enclosures give $-50 < F_2 < -21$. This is the quadratic coefficient in $\eta$; it is not the coefficient after reparameterization by $x$. For inverse-lapse coefficients $e_j$, with $e_0 = 1$,

$$F_n = r\left[(n-1)d_n + \sum_{j=1}^{n-1} j d_j e_{n-j}\right]. \tag{NL17}$$

The finite coefficients through nine, positive distance and scalar majorants through degree 24, and a separately bounded inverse lapse control the full remaining series. At $q_* = 3/4$, the positive tails are evaluated anew, with their supersolution and contraction predicates checked there; a half-radius estimate is not extrapolated upward. Tails beginning at degree 25 scale downward by $(q/q_*)^{25}$ for $q \le q_*$.

The full clock budget proves $|F - F_2\eta^2| \le 10\eta^2$; the endpoint ratio to this budget is approximately 0.001683018132. The other three remainder ratios are approximately $4.45249 \times 10^{-22}$ for separation relative to $25|\eta|$, $1.77384 \times 10^{-5}$ for the wall scalar relative to $|\eta|/40000$, and $7.31751 \times 10^{-5}$ for its Euler derivative relative to the same bound. Displayed decimals summarize exact rational comparisons. All four ratios

increase with $q$, so the endpoint inequalities hold on the whole smaller interval. A fifth gate, $25|\eta|/64 - 60\eta^2 > 0$, protects opening. These yield (NL3), with no higher-order sign inferred from a polynomial alone.

The first wall coefficient obeys $1/20000 < \chi_{a,1} < 1/10000$. The scalar and Euler remainders bounded above therefore give (NL4). At $\eta_*$, this supplies a fixed nonzero scalar threshold. Analytic convergence makes the restrictions at $\eta = \epsilon \to 0$ converge to background data in pulled bulk $C^2$, first-time-velocity $C^1$, and the finite-dimensional wall/oscillator position and velocity variables. These data are compatible because they restrict exact solutions. The positive lapse has a uniform upper and lower bound on the protected interval, so each logarithmic coordinate departure time corresponds to finite proper time. This proves the constructed-family witness without assuming general-data well-posedness, a full unstable manifold or evolution beyond $\eta_*$.

*Q.6 Exact additional premises and evidence access*

Besides the background and selected-mode premises, Theorem 3 uses the whole-cell inverse/seed enclosures, seven-source coefficient domination and final observable comparisons described above. The source-row inventory, derivative payments, positive component response and majorant contraction are mathematical requirements, not conclusions of a file hash or producer flag. The records identify where their numerical constants and source implementations are checked. Earlier weaker or unsuccessful packets are not silently rescored.

The archive namespace is `data/raw/l001/experiments/` in the public pipeline. The following terminal and result records identify the selected route; the revision's private source map records their full SHA-256 identities and the immutable pipeline commit for editorial delivery.

| Premise | Record directory | What it supplies |
|---|---|---|
| Uniform real and complex response | `20261026_GROWING_NEUTRAL_ALL_ORDER_INVERSE`, `20261026_NEUTRAL_ANALYTIC_INVERSE` and the component refinements | Finite-order coverage, high-order barrier and analytic continuation |
| Seed and selected positive source bound | `20261027_BACKGROUND_SEED_CERTIFICATE_R1`, `20261027_FINAL_FIXED_GRID_CERTIFICATE` | The selected seed, ray, $K$, $Y$ and physical radius; the refined lane is used |
| Missing wall-row recovery | `20261026_NEUTRAL_WALL_IDENTITY` | The off-shell identity (NL15) with the actual orientations and matter |
| Finite ninth coefficient | `20261027_NINTH_REAL_SOURCE_ENCLOSURE` | The ninth proper-separation coefficient in both retained radial implementations |

| Premise | Record directory | What it supplies |
|---|---|---|
| Positive tails at $q_* = 3/4$ | `20261027_NINTH_THREE_QUARTER_TAIL`, `20261027_NINTH_THREE_QUARTER_INVERSE_TAIL` | Newly evaluated distance, scalar and inverse-lapse tail bounds on that domain |
| Five-gate observable conclusion | `20261027_ACTUAL_NINTH_FULL_BUDGET` | Endpoint admission with unchanged analytic radius and the complete observable budget |

The actual seed 1215 is supplied by the background-resolved refinement named in the table; the earlier `20261027_COMPONENT_SEED_CERTIFICATE` supplies 12190. The selected final driver binds the refinement. Its sharper inverse-response weights come from `20261027_MASS_RETAINING_F_CERTIFICATE`, with the distinct real and complex denominator bounds traced through `20261027_DENOMINATOR_FACTORIAL`.

Stage-specific internal AI audits retain their declared scopes; the final full-budget audit binds the accepted result and records 7,264 checks. A subsequent bounded internal audit reconstructed the seven sharpened source estimates, compatible real/complex inverse, actual seed and selected majorant, followed by a fresh independent challenge. It found no failed estimate requiring a radius change under the declared exact-equation, enclosure and execution premises. It did not rederive the full action variation, regenerate interval trajectories or repeat the final observable-tail audit. Shared inputs and arithmetic do not establish arithmetic independence, and internal agent scrutiny is not external human peer review. Figure 3 renders saved bounds; archival verification establishes byte identity. Neither replaces the mathematical estimates or the retained execution/outward-arithmetic premises. Anonymous delivery of the necessary evidence remains a separate completion item.

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